

Bus 310 The Corporation

Numbered Transcript

BUS 310 • Dr. Yaron Brook

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Lecture 1: Corporate Context

At a glance — Brook sets the corporate context and the prosperity of modern life.

- 1.1 We live in amazing times. Our standard of living, quality of life, materially is the best ever. And we're unbelievably richer than pretty much any other generation that comes before us. We have an incredible abundance of stuff. I mean, this computer that is more powerful than supercomputers—the crazy supercomputers of just 40, 50 years ago. I mean, look at how many cameras we have straight at me right now. I mean, look this facility. This is all amazing and technology that really—maybe we could imagine if we were a science fiction writer, but most people could not imagine, and certainly not imagine that we would have it so soon. Today.
- 1.2 And yet all of this stuff that we have—I'm looking at the air conditioning vents and air conditioning itself—all of it is a product of a particular business form. All of this stuff is a product of corporations, of businesses. The world would not be as wealthy, as prosperous. We wouldn't have the material abundance that we indeed have without business. And the reality is that in America, as in most of the wealthy world, business has always—or over the last 150 years, when this wealth has been created—has manifested itself in a particular form, and that is the corporate form, this idea of corporations.
- 1.3 And yet, our culture resents businesses, generally. Big businesses, in particular. Notice that most of the stuff—this is the biggest company in the world. I'm not allowed to say what it is, but my computer over here—the biggest company in the world. The cameras are all a big company. The air conditioning is probably one of the big corporations out there. Big businesses—hate it—we despise it, we ridicule it, we go after it. People have very little understanding of what it takes to run a business, or what's involved in creating a business and then running it and being successful at it. They have no real idea what a corporation is.
- 1.4 How is it different than a partnership? How is it different than a lot of other business forms? They don't know how they function. They don't know why they're successful and how they achieve success. What do they run for? There's a lot of confusion about what is the purpose of business. And of course, "what is the relationship between business and the state and government?" is a huge topic of confusion, of misunderstanding, and of evolution, because that relationship constantly is altering and changing.
- 1.5 Now, we're going to talk about all these things in this course. We're going to talk about what a corporation is, how it functions, why it's structured the way it is, how they became so successful, what their relationship with government is, what is the role of regulation, antitrust, industrial planning in America. We're going to be focused on the American Corporation. There's some differences between the way corporations have

evolved and become successful here, than let's say, in Japan or in Germany or other places around Europe. We followed very much our own path for better and for worse in the United States relative to these other countries.

- 1.6 This is an issue that has constant relevance. There's constantly things going on in our world around us that makes this question about corporation, the world of government, how they function, what they are super important to policy. I was just reading this morning a story: about a year and a half ago, Nippon Steel made a bid to buy U.S. Steel. U.S. Steel is a very old American business. It's one of the first big businesses in America. It was put together by Carnegie in the early 20th century—Carnegie and J. P. Morgan. It's like a classic big American business.
- 1.7 And the steel industry has a history—we'll touch on it a little bit during the course—history of up and down over the decades. And generally, American Steel has not been particularly competitive, particularly not U.S. Steel, since the 1970s, since it really started facing competition from Japan and Europe. And indeed, U.S. Steel in the U.S. is struggling. So Nippon Steel, a very successful Japanese company, offered to buy them—\$14.6 billion.
- 1.8 Shareholders of U.S. Steel: "Yes, this is fantastic. We love it. They're giving us a premium over what our stocks earned. We're going to make some good money here." Even the union at U.S. Steel loved it because it's an infusion of capital. Nippon is really good at running steel companies. This is re-energizing for U.S. Steel, this injection.
- 1.9 But it's a foreign company buying an American company, and that sets off certain alarm bells in certain quarters. It's steel, which is like— we have a long history of steel as being American. That sets off certain alarm bells in other quarters. And it's national security. We need a lot of steel to build tanks and airplanes. And yet the amount of steel that goes into the defense industry is very small, and U.S. Steel is not a factor there. Other companies take care of it. But everybody—national security—they raise that banner very, very quickly. And the Union of Steelworkers— not the Union Chapter at U.S. Steel, but the Union of Steelworkers— they don't want a foreign company in the U.S. Now, their workers are working for a foreign company. So, they're against.
- 1.10 So all this is a situation in which the government steps in and says, "Wait a minute, we need to authorize this." Generally, the US government authorizes big mergers. We don't have the freedom to merge whatever companies we want. And Biden decided, no. Even though there's a commission that analyzes the national security risks involved in this, the commission came back with neither here nor there. They didn't want to commit, which was politically very good for them.
- 1.11 Biden said, "No, I'm against this deal." On the campaign trail, Donald Trump said, "I'm against this deal." Why? It's a foreign company—unions— always national security. The government is always going to use the national security terminology. Anyway, just last

week, it was announced that Nippon Steel is indeed buying U.S. Steel. The deal is going to go through. Trump has approved it. But there's a caveat.

- 1.12 Nippon Steel has agreed to give the US government something called a golden share. That is, they've given them veto power over most corporate decisions related to U.S. Steel. For really the first time in American history, the government has a golden share in a private company owned by a foreign company where it can now decide who are going to be the directors— at least the outside directors—the so-called independent directors are going to be basically approved and appointed by the US government, by the President.
- 1.13 They can decide on every single major business decision that affects U.S. Steel. This is something completely new. How does this fit into the history of the corporation? Now we have a U.S.—not owned, because they're not going to get any revenue from U.S. Steel—but controlled by the President. For a while, there was some thought, "Okay, this will phase out after three, four years." The U.S. government has in the past tried to nationalize industries or taking big stakes in companies, but usually they get rid of it very quickly. In the U.S., generally, our system is such that we believe in private property and the government shouldn't own businesses.
- 1.14 No, no. This golden share is there forever in perpetuity. This is a deal that's being negotiated. It's being closed. It's being done. This changes again the nature of corporate governance, the nature of the corporation in America today. Now we have this overhang of, "Well, what does the government want next?" This relationship between the corporate America—between business—and government has shifted just this week. And we will see all the shifts or many of the shifts that have happened over the last 150 or so years during this course.
- 1.15 So in our culture today, corporations are often the villains. You could check out Netflix or Prime or whatever, and go check out the movies, and you won't find a corporate hero, but you'll find lots and lots and lot of corporate villains. Every dystopian story, almost every dystopian story, that we have out there—who ultimately controls the world? Some corporation. It's always a corporation. You remember WALL-E? You remember WALL-E, the Disney movie, where you had this dystopian future? Who did that? A corporation. This is true, generally, in movies. The villain is always some corporate boss pursuing greedy prophets. And in pursuit of greedy prophets, grows, grows, grows, and ultimately takes over the world.
- 1.16 And that's the motivation. That's what they want to do. They're the bad guys. Our culture is ripe with this. Again, you find me a movie or a novel where a corporation represents goodness—where a corporate leader is a hero. It almost doesn't exist. It's very, very rare. There's some exceptions, but it's very unusual and very rare.
- 1.17 And indeed, today—it used to be, at least the perception used to be—that at least I'd say in the latter part of the 20th century, the perception was Democrats hated

corporations because they are generally on the side of socialism. They're generally perceived to be on the side of that the state should control business and capitalism, free markets, which corporations represent to a large extent. That is inefficient, ineffective. So we want much more regulations, controls.

- 1.18** And it's always been perceived as the Democrats are the ones who hate big business. And indeed, if you look at statistics in the past, Democrats have always resented big banks, big corporations, big business. But 40% of Democrats say they don't trust them. It used to be the Republicans actually had a relatively favorable view of big business and big banks. But that has really changed. Today, Democrats and Republicans agree: big banks, big business—something's very wrong with them. They have generally negative views of both. We now have a consensus across the political spectrum; about 60% of both Democrats and Republicans agree that there's something very wrong and we need more control.
- 1.19** They have a deep distrust of American business, particularly big business. Everybody loves small business. Everybody loves the little guy. Little guys are great until they grow big. The whole point of having a small business is to do what? For the most part, if you're ambitious, it's to make it grow big. We like them when they're small. It's like kids, right? They're cute when they're small. It's when they grow up, they become a problem.
- 1.20** In corporations, we despise them when they grow up, when they become big, when they become successful. How do you become big? Mostly by success. We don't trust them. We resent them. It's why antitrust is so popular. We want to break them up. We want to make them small again. Generally today, there's a bipartisan suspicion of big business across the board. It really goes deep.
- 1.21** There's a lot of people who question things like: aren't corporations just a creation of the state? Did government make corporations? And therefore, shouldn't we have a lot more control over them if they are creations of the state? Don't they have to register—get a corporate charter—and therefore, they're a government-created entity? And if such, the government has a role in regulating it.
- 1.22** And you see that widely on the left. And again, you see that—certainly Lina Khan, who was the head of the FTC under Biden, who is important because she was not just a regulator, but she was an intellectual. She came from a law school. She wrote a lot of papers. She explained her positions. She really viewed corporations as creatures of the state. The state should have a lot of involvement and should dictate how they should be run and what size they should be. You saw that in the old days. Those of you might remember Ralph Nader who used to go after corporations constantly—we want them controlled, we want them broken up, want them regulated, we want them to go away.
- 1.23** Of course, over the last 10 years—and maybe this is some of the reason Conservatives have changed their views, or Republicans, at least, have changed their views vis-a-vis

business—is you've seen, again, business accept this idea that they're creatures of, if you will, the state, society, and maybe they shouldn't just pursue profit. Maybe they shouldn't be so greedy. Maybe they should pursue something called social responsibility or the public interest or the common good. And as a consequence, they've adopted things like DEI or ESG. DEI being diversity, equity, inclusion, and ESG being environmentalism, social, and governance, which, again, are slogans typically associated with the left. Once business adopted them, as, "Yeah, we're socially responsible. These are the things that make us socially responsible." Now the right has turned against them.

- 1.24 But that's because they embrace this idea, as we'll see, that they are not just there, in a sense, to make money, and we'll get into what making money means, but that they're there to serve some social purpose. But the reality is that Conservatives have always had a tricky relationship with business. They've never been real fans. They've never gotten on board completely.
- 1.25 There was recently an article saying, this is the title of the article: "Big Business and Conservatives Are Headed for Divorce." So now the Conservatives, Republicans are completely against. From the article, "The new Conservatism Is hostile to globalization, immigration, big tech, media companies, and the idea that market forces should determine the common good." So they might defer on why they resent big business, but again, it's a universal trend now.
- 1.26 Here's Irving Kristol. I don't know if you remember—you guys are too young. Irving Kristol was a leading conservative in the 1990s. He was really the founder of neoconservatism, one branch of conservatism, not very popular right now, but in those days, very popular. This is what he wrote: "But one must concede that both the founding fathers and Adam Smith would have been perplexed by the kind of capitalism we have in 1978. They could not have interpreted the domination of economic activity by large corporate bureaucracies as representing in any sense the working of a system of natural liberty."
- 1.27 So he is objecting, again, to the large corporate entity. He goes on, "The large publicly owned corporation of today, which strives for immortality, which is committed to no line of business, but rather, like an investment banker, seeks best return on investment, which is governed by an anonymous oligarchy, would have troubled and puzzled them, just as it troubles and puzzles me. And they would have asked themselves the same question we have been asking ourselves for almost a century now: Who owns this Leviathan? Who governs it? And by what right and according to what principle?"
- 1.28 Those are the kind of questions we're going to try to answer in this course. Who governs them? By what right? And what principle? He says, "No other institution in American history, not even slavery, has ever been so consistently unpopular, as has the

large corporation with the American public." Not even slavery has been this unpopular. "It was controversial from the outset, and it has remained controversial to this day."

- 1.29** So he's right. Back in the 19th century, what did they call the successful businessmen of the 19th century? The big businesses. What were they called—the people who ran the big businesses of the 19th century? Robber barons. Why is that such a bad term? A robber? It means you don't create wealth, you steal it. And baron? A baron is an aristocrat, somebody who doesn't, again, create their status. Baron is very un-American. It's very European.
- 1.30** Now, there used to be robber barons. They were robbers and they were barons. Really, the whole aristocracy in Europe, how do you become an aristocrat? Five, six, seven, thousand years ago, how do you become an aristocrat? By stealing. There's no production. There's no actual production of values. There's no actual production of wealth. It's stolen wealth. In Europe, this association of wealth with stealing makes a lot of sense because they've never had really—up until about 150, 200 years ago—they never had a system in which wealth was created. Wealth was always redistributed or stolen. Robber barons reflects aristocracy of Europe.
- 1.31** And that is now attributed to the Carnegie's and the Mellon's and the Rockefellers and the J. P. Morgans. People who—I think as we'll see to some extent, but as if you know history—actually built, created, the wealth that they had. Very different than a baron, very different than the robber.
- 1.32** He goes on to say, "If large corporations are to avoid having government appointed directors on their boards"—golden share, government appointed directors on their boards; they're getting it now—"they will have to take the initiative and try to preempt that possibility by themselves appointing distinguished outside directors, directors from outside the business community. I love that. What you want running a company are directors who know nothing about business so that they can monitor and supervise CEOs whose business is business. We want somebody completely ignorant of the business of business. That makes them good directors. We'll see, that is the dominant view today.
- 1.33** There's political hostility. Again, more so, I think, today than in the 1970s, although there's some similarities. Globalization has affected our views, our view on globalization, our views on immigration. There was just now Trump—what was it last week? Of course, this will be broadcast sometime in the future. So sometime in June of 2025—Trump said, "ICE is not going to go into farming and certain businesses, hotels and restaurants, in order to deport immigrants because those businesses really need those workers."
- 1.34** Huge backlash against that. "Wait a minute. Why are you discriminating? If they're illegal immigrants, they're illegal immigrants, why do we care if it's business or not?" Now, just on Monday, it was rescinded, and now they're going after all of them. There's

a resentment towards immigration, and then there's a resentment towards business that employ immigrants because, "Hey, they're making a profit off of illegal immigrants. They should be paying employees a lot more money in order to hire Americans," is the view.

- 1.35** There's resentment against automation, technology. It always has been. It'll destroy jobs. Automation. What will we do once machines take all our jobs? We won't have anything to do. There's always been and always is, resentment against CEO pay. They make too much money. They make all these figures 200 and something times what the lowest employee in a company makes. And that's always a conflict that exists in America. It has its cycles depending on what makes the news. About 10, 12 years ago, it was the big thing CEO pay, how much money they make? Less so today.
- 1.36** A lot of people resent the idea—if you own stock in a company, how much say do you actually have in running the company? Are you really an owner if you have no control? It's this idea that the owners of the company don't really control the company. Who controls the company? The managers, the CEO, and yet there's this separation. So are we really doing stuff that's good for owners? CEO pay is a good example of that. Maybe the CEO is making a lot more money than he should. Shareholders shouldn't allow it. It's an interesting case.
- 1.37** If you remember, again, this is just not that long ago, Elon Musk, it's probably a year ago, Elon Musk got a pay package about five years ago from the board of Tesla, which was worth something like \$50 billion in stock. If he reached certain thresholds, he was going to get \$50 billion, like the biggest pay package, the biggest pay day anybody has ever had in the history of any place. And some minority shareholders sued and said, "Wait a minute. This board is a board appointed by Elon Musk. They don't represent our interests. We would never approve a \$50 billion payday. This is wrong."
- 1.38** It actually went to a Delaware Court—we'll get to Delaware later, why Delaware court—It went to a Delaware court where Tesla is incorporated. And the court said, "Yeah, that's nuts. That's unreasonable." And they voided the whole pay package. Later on, it went to shareholder vote, and shareholders actually voted for it. But there's still this issue of, majority shareholders versus minority shareholders? Is this okay? Again, the courts have ruled, "No, he can't get that pay package." So these issues around both CEO pay and around whether shareholders are properly represented is an issue that comes up when it comes to corporations.
- 1.39** And then there's the argument that corporations are inefficient, that they focus on profit at the expense of everybody else and everything else. This idea of small is beautiful. Small is beautiful, big is bad. And of course, tangled in all of that is envy, envy of people's success. Certainly, if you look at some of these very, very wealthy entrepreneurs in the tech industry, people hate their guts. They make their lives better. I think of somebody like Jeff Bezos. Changed the world. Who doesn't use Amazon? Everybody uses Amazon. Our life is dramatically better because of Amazon. How would

we have survived COVID lockdowns, ridiculous lockdowns without Amazon? And we love to hate Jeff Bezos.

- 1.40 A lot of that is envy. It's also this moral notion that his self-interest, his pursuing his profits, his pursuing his own ideal, his pursuing a lot of money, his pursuing success, and these days, the good life—there's something wrong with that. He should be much more, a little bit of sacrifice, a little bit of sharing with the rest of humanity his wealth and success. And there's a lot of ignorance and unthinking that leads to resentment, these kinds of resentments.
- 1.41 All right, so we're going to try to cover a lot of these. We're going to cover a lot of these issues and bring them up and try to see how they should be addressed. They're really out there in the world, very little defense of business, very little defense of corporations. They keep chugging along because everybody knows we need them. Everybody knows our jobs depend on them. Everybody knows our standard of living depends on them. We love to hate them.
- 1.42 And then the government comes in once in a while and shakes them all up or breaks them up or just harasses them for a while and then goes away and we continue going because what's the alternative? None of us actually want to embrace the alternative. So we live with our resentment and our hate, and then we express it in our movies and our books. We cheer for the guy who's going after the corporate leader.
- 1.43 We need to remember, though, how successful they have been, how crucial they are to our lives. And in spite of the crises that have happened—I don't know, we'll talk about the Great Depression and World Wars and the inflation of the 1970s and the great financial crisis—and in spite of some legitimate horror stories of real fraud, of people doing really bad things—in spite of all that, somehow, here we are in pretty good shape. At least again, materially doing fantastic.
- 1.44 And even this idea of materially, I say materially, but even if you think about non-material spiritual stuff—let's take a simple example of something like music. Music is a spiritual experience. Our lives are better off for being able to enjoy music. Do you think you have more ability to enjoy music today than you did 20 years ago, 100 years ago, 400 years ago? I mean, it's not even close. If you have one of these, you basically have access to every piece of music ever produced at a marginal cost of zero.
- 1.45 You pay a monthly fee, listen to anything, anytime, anywhere. And that, again, is not an accident. That is something created by a business. That changed music forever. And it's a sequence of events. Ultimately, it's resulting in Apple creating iTunes, creating those, remember those iPods originally, and of course, today in an iPhone. And pursuing what, by the way? Why did it do all these things? In pursuit of innovation, success, vision, beauty, but ultimately in pursuit of money. Greedy Apple made it possible for us to elevate ourselves spiritually by having more access to better music all the time.

- 1.46 All right. So in this course, we're going to talk about what corporations are, how they function, why they function the way they do. We're going to talk a lot about the history—because this is about history—about the history of corporations. A lot about the interaction between corporations and the rest of the world, particularly government. So we'll talk about antitrust. We'll talk about regulations, some regulations. You could have a whole course on it easily. Antitrust, you could have a whole course on.
- 1.47 So we'll condense all these. We'll talk about some, about industrial planning, the attempts of government to try to control winners and losers and structures of businesses. We'll talk about how all this affected the evolution of businesses to where we are today. We'll talk about the Big Five. What are the Big Five? The Big Five tech companies that everybody's so obsessed about today. How did we get to the Big Five? How did we get here? We'll talk about some of the dynamics that brought us here. In doing so, we're going to cover a history that basically starts somewhere in the middle of the 19th century, but really in terms of the corporation gets going at the end of the 19th century, early 20th century. And the 20th century is just fascinating, I think, in terms of the ups and downs, the changes, the evolution that has happened to the corporation and to our economy more broadly, and the two are heavily related, as we will see.
- 1.48 So let's start. Let's start really by jumping into what these things are that we call corporations. What is a corporation? A corporation is a particular type of business organization. It's a particular type of business organization that is created in order to pursue a business goal. And behind that business goal is profit via economic activity. Now, not all corporations are business organizations. Not all corporations seek profits. You can have nonprofits. Nonprofits are also corporations. They're also structured, as we'll see, as corporations. Today, there's even something called a B Corp. A B Corp is a corporation that lets you know in advance before you invest money in it, that it is not focused on making money as the predominant—it'll also make money, but it has a social goal, whatever that is. It's telling you up front.
- 1.49 Now, nonprofits is a big topic. We're not going to talk about it. It's just not relevant to what we're talking about in this course. We're going to be talking about business organizations. The B Corp, it's a small—it's not very popular, it's not very big. I suspect for a variety of reasons, I think you'll glean through the course, that it's never going to be very big. To a large extent, what drives business, what drives corporation is profit. It's a single-minded, if you will, pursuit of economic success, of business success. Once you pollute that with other goals, it's going to be a mess, and it's unlikely to be successful.
- 1.50 I'll give you one example. It's not exactly a B Corporation, but it's similar. Everybody know OpenAI? It's in the news, it's everywhere. This is one of the biggest successes. Recently, ChatGPT, this is huge. It's going to change the world. Well, the interesting thing about OpenAI, and there's a lawsuit going on right now between the founders of

OpenAI, Elon Musk and the other guy's name just slipped my mind. You want to say it? Sam Altman. Thank you. Sam Altman and Elon Musk. And when it was set up, it was set up as a nonprofit. OpenAI was a nonprofit with the idea that whatever it generated would benefit humanity. And then nobody would disproportionately benefit from this—that it was shared by everybody. And that's how it was set up.

- 1.51 But interestingly enough, it actually got over time—and a lot of money came in. Elon Musk, I think a country put in \$10 million or something, a lot of money went in and it grew. And what was about two years ago, three years ago, they had this product launch, which was a real breakthrough, ChatGPT, which was a step forward in a dramatic way. Microsoft had invested money in a a weird situation. Microsoft was investing in a nonprofit under the idea, I guess, that they would get part of the technology. But what OpenAI has now done is it spun off the ChatGPT portion, which is really the whole company, into a for-profit entity. Now, originally, they said this for-profit entity would just fund the nonprofit, and the nonprofit entity would distribute the money for good causes, and it would be responsible for the governance of the for-profit in making sure that it served the general public interest, the common good, however you want to define it.
- 1.52 And a lot of people invested in the for-profit spinoff of venture capitalists, again, Microsoft. It was always a little suspect because these people are investing in this subsidiary as if it's really making money. But the not-for-profit expects it not to make money because it has this other goal. And there's a huge rift, governance-wise, contradiction between what the nonprofit wants and what the subsidiary wants.
- 1.53 Basically, it was six months ago or so, they decided to spin off the subsidiary as its own for-profit company and completely sever the relationship with the nonprofit. And the nonprofit will get the money. They will, in a sense, sell it, and they will get the money, and they will do their philanthropy with that. But from now on, ChatGPT will be a for-profit. And this is what the lawsuit—Elon Musk is saying, "Wait a minute, that wasn't the original intent." Now, he, of course, has an incentive because he's created his own AI company, XAI, which competes with ChatGPT. So he would rather they stay a nonprofit, which he knows is not going to be as good a competitor as a for-profit OpenAI.
- 1.54 Anyway, you can see the tangled mess. But the reality is that at the end of the day, in order to be successful, in order for ChatGPT to actually win, if you will, it's got to be a for-profit. And the nonprofit knows it, and therefore they're spinning it up. So we're not going to talk too much about nonprofits. We're going to focus in on the full profit, on the typical business corporation that we love to hate and that we're familiar with in all of our daily lives.
- 1.55 So what is it that makes a corporation a corporation versus a more typical business setup? If you go back in history, the most typical business setup is, in a sense, a self-proprietorship. Somebody just opening a business— shoe horses: you make the shoes

from the metal, you do it all. It's your one-man operation. Maybe you have a couple of employees, but basically it's your business. All the income flows through you. You get to make all the decisions. You control it and you own it, and that's it.

- 1.56** Now, you can't grow a business like that very big. There's only so much one person can handle. There's no real access to capital. There are no outside investors. It's just you. A step in growing is to create, again, very common, a partnership. Bring in other people who can help manage, who can also bring in capital, and now you can have a larger business.
- 1.57** And indeed, typical business organizations historically were partnerships. Anything of size was a partnership. These were distinct legal entities, but where the owners, the partners, are all liable for everything that happens in the business, at least in a traditional partnership, that is the case. They're liable for it. If you sue the business, you're really suing the partners, you're not suing the business. If the partners sue, they're suing as individuals. There really isn't a business separate from the partners. It's one unit.
- 1.58** A corporation is something different. A corporation is a creation of distinct business entity. A business entity that, in a sense, has a life of its own. It can sue in the name of the entity. It is sued by other people directly. And it can bring in investors, large numbers of investors, from all kinds of places. Those investors are not now partners, so they're not expected to participate in the management and the running of the company. They're also not liable for the things that the company does. They are just investors in this legal entity. And as a consequence, it can grow much bigger, raise a lot more capital, and indeed can do somewhat riskier things because the investors can only lose what they invested in the company.
- 1.59** You can't go—this is called limited liability. Let's say a company takes on a huge amount of debt and then goes bankrupt. All the debt holders, all the people who are owed money can get is what is held under the corporate name, the assets that are held there by the corporation. If you own shares today in Apple, and Apple goes bankrupt, and it owes a huge amount of money, and they sell all the assets of Apple, and they still owe a huge amount of money, they can't come after you as a shareholder of Apple to take your money.
- 1.60** Now, if you're a partner, and the partnership gets sued, and they liquidate the partnership, and they pay it all out. If they still owe money, they'll come after you as individuals, at least in, again, a regular partnership. There are ways in which partnerships can shield themselves from that as well. So this idea of a limited liability is a huge benefit to the corporation. Huge benefit for a business. All you can lose is your initial investment. And that makes investing much more attractive.
- 1.61** It also makes it so, I don't have to do quite as much due diligence when I make an investment. If I'm making an investment and my entire net worth is at risk, I'm going to

know you really, really well. I'm going to make sure this business is a good business, and I'm going to make sure you don't take too many risks because I don't want you to take risks because my entire net worth is—and that is very constraining and very limiting. Think of all the real breakthroughs that have happened in business. They always involve risk.

- 1.62** How many companies try to establish their personal computer and how many of them failed, and how many succeeded? How many railroads succeeded versus how many failed? How many in any industry succeeded versus fail? And yet the reason they can succeed is because people are willing to invest in them because they know their entire wealth is not affected by it, just their investment. So limited liability is a huge benefit that corporations have. Now, you can create that in partnerships, but on a smaller scale with limited partners versus general partners. There are kind of legal ways to do this.
- 1.63** Corporations also are, theoretically at least, immortal. There's no lifespan. The lifespan is as long as they're successful, as long as nobody's bought them, as long as they haven't gone bankrupt. Because the owners are constantly changing. And therefore, when the owners die, the business doesn't close. The shares pass on or they've sold the share, somebody else owns it. So you can have a business like U.S. Steel that we talked about earlier, which was founded 150 years ago. Less than that, 120 years ago. And here it is, still around. And if not for Donald Trump, maybe it wouldn't be around anymore. It would be called Nippon Steel. But here it is. It might be owned by a Japanese company. Tomorrow, it might be owned by somebody else.
- 1.64** So you can get this idea of companies that can plan and can pursue strategy that is not limited by anybody's lifetime. It's actually now indefinite. The founder of Apple, Steve Jobs, died a few years ago. He's no longer around, and yet Apple continues. And if they do well, they'll continue for a long, long time. And if they don't do well, they won't.
- 1.65** This idea of an indefinite lifespan is, again, an attribute of corporation. Again, you can do that in partnerships. You can create partnerships that have that effect. But a lot of partnerships, it's a partnership. The partners die. It's done. It's over. Now, there are partnerships that survived long periods of time. Some of the investment banks, like Goldman Sachs, used to be a partnership. It lasted for a long time, that partnership. So you can structure legally a partnership into a different form. But even Goldman Sachs ultimately became a corporation because there are multiple advantages to being a corporation. It's much cheaper, easier, smoother to structure yourself as a corporation and get this idea of an indefinite lifespan and limited liability and all these other features that you could create elsewhere but are not necessary for that creation.
- 1.66** Part of the consequence of this is that one of the attributes of a corporation is that you have a separation of ownership from control. In most businesses, the owner runs the business. The owner controls the business. And if you think about most corporations, at the beginning, when they're small, that is probably the case. The entrepreneur, the

founder owns a big chunk of the business. He maybe sold a little bit off to investors, but he owns most of it and he's running it.

- 1.67** But as the business grows, and when ultimately they go public, owners typically hold a fraction of the shares in the company. Jeff Bezos doesn't own a big percentage of Amazon. He's the second the richest man in the world, even though he only owes a small percentage of Amazon because Amazon is so big. And at that point, he is the CEO — when he was CEO—he's running it even though he's not a big owner. So you get the separation of ownership from control. You get a division of labor. You get specialization.
- 1.68** I can now go out and hire a really good manager to run my corporation without me as an owner or as one of the shareholders being involved on a day-to-day basis. So you get a separation. Adam Smith talks about the huge benefits of specialization and division of labor. Here you get another way in which labor is divided. Ownership is divided from management, and that increases efficiencies for managers. You can now become an expert at being a manager. Go get an MBA. I don't know if I'd recommend that, but theoretically, at least, go get an MBA. That gives you the skill set, supposedly, of becoming a good manager. Or at least a diploma that says you have the skill set for being a good manager. I used to teach an MBA program, so I can criticize it.
- 1.69** Corporations have these huge advantages legally and structurally. They make things more efficient and more productive. They are better utilizers of this division of labor. They provide for this massive amount of ownership. If you think about us as owners of shares, I mean, all of us probably own shares in our retirement account, some of that account, brokerage account. We can now diversify our own holdings. But how many of us—you probably get proxy. You get a vote on stuff once a year. How many of you actually read those? I know what I do to them. In the trash as soon as I get them. We're not thinking about how to run the company. We don't manage this. If we don't like how the companies run, what do we do? We sell our shares.
- 1.70** Now, we'll get to the most sophisticated effects of that and what some people can do in terms of when they're not happy with running a company. But for most of us, this is a way in which all of us participate in the enormous success of business without being a part of that business, by just providing capital, without actually getting involved in how to run it. Massive benefits to investors. And again, the creation of this class, if you will, are people who now specialize in management. That's what they do. That's their shtick, is how to run a great business.
- 1.71** For the corporation side as well, they get the benefit. And if you take in my finance course on this platform, we talk about this. The managers get the benefit of now having the corporation listed on an exchange valued constantly by investors like us. That gives them incredible price information about how well they're doing. Stock markets are signals. Stock prices are signals to managers. You're doing well, you're doing poorly, you're just doing okay. You can compare their stock to their competitors, see who's

doing better. And managers are constantly looking at this stuff, not just out of a sense of vanity, but out of a sense of, they're important signals the market is sending me through the prices of my industry and my own company.

- 1.72 I like to say that sometime in the early part of the 20th century, the best company that made the best buggies in the world, probably started to see their stock price decline, and the managers went, "What's going on? We're super profitable. We're making really good buggies. Everybody's buying our buggies. What's going on?" And they popped their head up out of their business and looked around, and what did they see? An automobile industry over there, which the market had seen before them. The market had recognized, lowered their valuation because they had no future—they were all going to go out of business, and shifted the capital to the automobile industry. That's a signal.
- 1.73 Think about in our world of tech, the changing of the guard in terms of tech companies, and what's hot now, and what's hot yesterday, and how stock prices change, and how that is a signal for: what is the value? What is the prospects for the future?
- 1.74 These are the qualities or the way organizations or corporations are structured. There are lots of complications on top of this. There are lots of iterations that we can do. That's the basic idea. Separation of ownership and control. You get a new class of managers. You get this idea of indefinite time span, and you get limited liability, which is crucially important.
- 1.75 Now, a lot of people, a lot of critics of this form, worry a lot about separation of ownership control. We'll get to that. But before even that, the only reason this can exist, the only reason we have corporations, is because of the state, because government has endowed them with the ability to have this indefinite nature, endowed them with limited liability. You can't have limited liability unless the state recognizes that you have limited liability. This is all just a creation of the state.
- 1.76 Ralph Nader, famous consumer advocate, supposedly, "A corporate charter"—which is what the state gives any corporation. If you have a corporation, you have to go and register and you get a charter from the state. "A corporate charter is in effect an agreement whereby a government gives the corporate entity existence, and that entity in return, agrees to serve the public interest."
- 1.77 If you accept that definition, if you accept that that's true, then it's the state that tells corporations what they should pursue. It's government that tells them what they should pursue. Maybe the government decides, "Pursue profits," which is in the public interest, but then they might change their mind and say, "No, don't pursue profits. Pursue some other cause, ESG, DEI, climate change, national defense."
- 1.78 It's the state that dictates what that should be. But is it? Is it the case that the corporation is a product, a creature, as they call it, of the state? And the theory behind this is called the concession theory. If you look back at history, you can see its roots,

really, all over the place. I mean, it used to be before about 200 years ago, that if you wanted to start a business, any business, you had to get a concession. From whom? From the ruler, from the king. In reality, the king owned everything. Even in England, which is the most liberal of all kingdoms 200, 300, 400 years ago, the King owned all property. He owned all rights to trade, and ultimately, he owned all the profits.

1.79 So if you wanted to do something new, like in England, in the case, if you wanted to go and basically get a boat and sail to India and start a new spice trade with India, well, that is owned by the king, that trade. So if you want to do that, you have to get permission. You have to get a concession. You have to get agreement in order to do that. The king grants you the ability to do it. So the king grants concessions. He grants the right to create an entity that is engaged in that activity with perpetual existence. And this is the original corporate forms or some of the original corporate forms. And what you got originally were towns that were incorporated in this way. You had churches, guilds, universities, all created corporations.

1.80 And then you had trading organizations like the East India Company that got a concession from the king to engage and trade with India, with the East. And these trading companies, for example, the East India Company or the Virginia Company that established ultimately Virginia—Virginia was a commercial venture. Virginia was not a political venture. It was a commercial venture. The first settlers who came to Virginia came there as part of the Virginia Company in order to create economic activity to benefit this company, all granted to them by a concession from the king. And these are the first stock companies. People actually issued stock, just like today. They had limited liability, and they could raise capital from a lot of different people.

1.81 But it was at the approval of a king, and almost always, really always, it involved a monopoly. A monopoly not in the sense of today's monopoly—we'll get to what is meant by monopoly today, or is not meant, or the confusion about what monopoly means today—but what monopoly meant was the king will not allow anybody else to do what you're doing. If somebody else wants to trade with the East Indies, they go to jail. There's no competition, not allowed competition. We're going to use the state to prevent competition.

1.82 This was created, the first people to really embrace this with the Dutch East India Company, and then the British, and there were shares, and they were traded. The beginnings of the stock markets were a lot of these companies that were granted monopoly power over a particular segment, a particular industry, a particular aspect.

1.83 One of the first corporations in the U.S. was a corporation in Massachusetts, granted the right to build a bridge over the Charles River and to take a toll. And it got a monopoly. It was the only company allowed to build a bridge over the Charles River until people got upset by the fact that there was and they wanted more. And it actually ended up going to American Supreme Court, where they said, "No, you can't do that. You have to grant more—if other people want to build bridges, it's okay."

- 1.84** But initially, corporate charters were associated with monopoly. But if you go back in history, as early as the 13th century, there were already stocks, shares, different business entities that had a lot of the corporate characteristics. So in America, we didn't really have much economic policy driven from the federal government. So in the early part of the 19th century, the federal government was not very involved in the economy, but the state governments were very involved in the economy.
- 1.85** So one of the myths about the 19th century in America is we had laissez-faire: anything goes, we had real freedom, free markets. No, not at all. There were quite a bit of regulations and controls at the state level, not at the federal level. There was always a tension between the commerce clause that says you can't regulate trade between states and between the reality on the ground where there was a lot of regulation of trade between states at the state level.
- 1.86** And that tension continued most of the 19th century. And indeed, states would grant monopolies to build canals, to create businesses, all again in the corporate form and their ability to sell stock and have this indefinite life. They all demanded certain benefits to the state. The state would take basically money. A lot of the state revenues came from this granting of corporate monopolies, corporate rights to these different companies. This is how early on corporations were structured.
- 1.87** A change to that only came in the 19th century, in the middle of the 19th century, where businessmen didn't pay attention to states, managed to create business entities that were corporate-like without getting a grant from the state and without getting any special favors, but also coming in under the radar.
- 1.88** The first new corporate laws that were starting to approach modern corporate laws were in Connecticut in 1837. They created standardization and simple procedures in order to start a corporation. Britain passed similar laws in 1844 and 1862. But really, the big change happened in the late 19th century in New Jersey. New Jersey was the first state to basically grant corporate charters with almost no strings attached. You paid something to set up your corporation, and that was it. They left you alone.
- 1.89** One of the big innovations for New Jersey was the ability to have your company incorporated in New Jersey, but to own businesses all across the country. And then states started competing on who could be better for business because they wanted to attract businesses to come and incorporate there. And then you got a little bit of that competition going, and state chartering became a lot easier, simpler, and allowed for a lot more activities and monopoly aspect of it. And the "do what the state tells you to do," disappeared.
- 1.90** In Delaware, funnily enough, in 1913, before he became President, Woodrow Wilson was the governor of New Jersey, and he hated big business. He hated corporations, so they passed a law to undo a lot of the incorporation stuff that New Jersey had done before and to penalize businesses who actually incorporated New Jersey. Guess what

happened to the businesses? They all left, and they all incorporated in a small state. To this day, it dominates incorporation in the United States, 65% of all Fortune 500 companies are incorporated in one state. That is Delaware.

- 1.91** And it makes sense for a small state because they get the fees on the incorporation. Given that they're a small state, that constitutes a huge amount of their revenue, and they become very friendly towards businesses, very friendly towards corporations. And what Delaware has done over the last 100 and some odd years is specialize in corporate law. Delaware It has a body of law with regard to corporations that is more developed than any other state. They have judges that are specialists in corporate law. And as a consequence, and they're generally friendly towards managers.
- 1.92** So managers like to incorporate in Delaware. But shareholders don't penalize managers who go to Delaware. So it must be pretty friendly to shareholders as well. So generally, Delaware is considered the best place to incorporate, at least until this case with Elon Musk. And now everybody is saying, "Oh, we should incorporate in Texas." So we'll see if that actually happens. Saying you're going to move out of Delaware and move to Texas is one thing. Actually doing it is something else. We'll see how many of those Fortune 500 actually move to Texas. But as of last year, 65% of Fortune 500 or S&P 500 were there. It's predictable. Everybody knows exactly where it is and so on.
- 1.93** So given this history, we have to ask ourselves: are indeed corporations a creation of the state? Is it the state that gave them, made it possible for them to have this limited liability? It seems that way. Kings and then states and now Delaware. It seems that the state has provided them with all these benefits of incorporation. And if so, if they're a creation of the state, what duties in a sense, do they owe the state? Because that's the claim, they're creatures of the state, they must be socially responsible. They must take into account, quote, "the public interest." And with that question, we will start next class.

Lecture 2: Corporate Purpose

At a glance — Corporate purpose and the history of the corporation's creation.

- 2.1 So last class, we talked about the history of the creation of corporations a little bit. We talked about what makes corporations unique and what they are. And of course, we talked about the hostility towards corporations, and we'll keep coming back to that theme.
- 2.2 At the end of last class, we ended on the concession theory of the corporation. The concession theory of the corporation is basically the idea that corporations are creatures of the state. They were created by government. In a free market, they would not exist. They cannot exist. All their features from the immortality, in a sense, to their limited liability, all these features are features that can only be created through a special charter.
- 2.3 And while a lot of this idea of granting monopoly through the corporate charter went away in the middle of the 19th century. The idea is no, but even without the monopoly, everything else that was given to the corporation was given to them by the state. And of course, this has significant consequences because if you hold that view, then there's a sense in which the corporation owes something to the state. And there's a sense in which the state has a responsibility to control, regulate, and maybe put in provisions there that might not be consistent with what managers and what shareholders would like to do at the corporate level.
- 2.4 So is the corporation a creature of the state? And I'd argue that the answer is no, it is not. Corporations are creatures of individuals contracting with other individuals to create an association. An association of individuals for the purpose of seeking out a particular business opportunity, a particular profit opportunity. But that is open-ended because the corporation has such a long life. That profit opportunity can change and deviate and change over time.
- 2.5 It is an association of individuals, again, contractually related to one another, engaging in business activity, engaging in economic activity. And it's based on basic individual rights. The right to associate with whoever you want, to write contracts, the kind of which, within the context of a legal system, you should be free or the terms of which you should be free to negotiate and determine.
- 2.6 And these corporations existed even at the time when the state was granting corporate charters that were very restricted, businessmen would create businesses on the side. They basically had all the characteristics through contracts of the corporation without getting the state charter. The state charter is a convenience. It reduces transaction costs. It reduces uncertainty. It creates a standardized format by which corporations get incorporated, but it is not a requirement. It's not a necessity.

- 2.7** You can find corporations with limited liability and with shareholders and the ability to extend over time on a small scale, granted, even going back to the 14th century. Certainly in America, the 19th century, these corporations existed side by side by those who were incorporated. They were just inconvenienced by the fact that the incorporated, the ones who use state charter, had advantages with regard to the law.
- 2.8** So every one of the features that people say, "How do you get limited liability in the marketplace? How do we get this idea that shareholders are not liable in the personal assets, only liable for what they invested, only for—how do you get that in a contract?" Well, it's easy. You write it into the contract. If I've got a corporation, I've got shareholders, and I want to come to you, you're a banker, and I want to take a loan, one of the provisions of the loan is you cannot go after my shareholders. If I go bankrupt, you have access to all the assets of the company, but you cannot go after my shareholders. You can recreate that. Indeed, other forms of businesses like partnerships do create that.
- 2.9** Even within the world of partnerships, you have general partners who are fully liable and limited partners. They're limited, which means they have limited liability. Not a creation of the state, creation of contracts. It's a contract between the general partners and limited partners. When the general partners contract with bondholders or with banks, with creditors, in other words, then it is understood that the creditor cannot go after the limited partners if the company goes bankrupt.
- 2.10** So this is the thing that can be easily done in a legal framework, in a contractual framework without the state having special laws for it. It's more convenient if we say, instead of writing it into every contract, we just say, corporations have a limited liability, and therefore every time a bank deals with a corporation or every time a bondholder, or anybody who lends money to a corporation, a creditor to the corporation, deals with a corporate entity, they know. "Oh, okay. We don't necessarily have it in the contract, but it's part of the contract because I'm dealing with such an entity." But again, it's convenience, legal convenience, a legal benefit, but it does not make the essential entity a creation of the state.
- 2.11** So limited liability is something you can create independent. Even longevity, the fact that it lasts forever. It's in the nature of the ownership contract that we have. That is, when you buy shares in a corporation, you can sell them. And a new person who buys them has the same ownership rights that you did. Just having that contract makes it possible for the corporation to last indefinitely. The fact that the shareholders are not necessarily—are not expected to run the company, makes this division of labor part of the contract.
- 2.12** It's all just part of the contractual relationships between the different entities that make up a corporation. The state is a convenience. It's a recognition. It's not a little convenience. It's a significant convenience. But it's convenience that is part of the job

of the state. That is to make law in its role as protecting and identifying and recognizing property rights. It's not some special privileges that is granting the corporate entity.

- 2.13** So whatever rights a corporation has— once in a while, a case will come to the Supreme Court, where it's a question of, "Does a corporation have, for example, the right to free speech?" So this came up with regard to campaign finance. Can corporations use their money to finance candidates? We all can because that's part of our free speech. It's recognized as part of the way we express ourselves in terms of speech is through our money. Our money is an expression of our ideas. Our money is an expression of our speech, and we can contribute to any candidate. There's no— there are limits, but we can make those contributions.
- 2.14** Do corporations have the same rights? Do corporations have a right to property? Do corporations have rights? And the answer is yes. Not because there's some unique special creature or there's something separate from human beings, but because our rights flow to the corporation. We don't give up our rights as manifest in our ownership of this particular business. When we invest in this business, when we become owners of this business, the business reflects us as individuals, and therefore it has our rights. So it's not rights above and beyond the rights of individuals. It's rights derived from its individual owners.
- 2.15** The right to property is, again, through a contract. The right to speech is our right to speech. We form an association and we have an opinion. The government has no right to suppress that because it's us speaking as individuals through a particular entity.
- 2.16** So all the state is doing is documenting the existence of a corporation, creating a certain framework, a certain legal framework that standardizes the relationships. And that creates a body of law that makes it all simple and knowable. But all the special characteristics of corporations are indeed ultimately market innovations. The things that the market created, and the government stepped in and said, "No, we want a piece of this," or "We want to control it," or "We want to have now the right to grant it so that we can have some power over others."
- 2.17** The fact that government recognizes these innovations is corporate law, and it's all a consequence, really, of the property right of individuals, the right to contract, and the right to associate of individuals. If you think about— it unites the right of association, the right to contract, and the right to property— all come together. Us as individuals, all come together in this corporate entity. And again, if you look at history, there have been plenty of examples of entities created that have all the corporate features without any government involvement. So it doesn't require it in any sense.
- 2.18** Now, this has profound consequences because basically what it says is that the corporation then is an entity that is responsible to its owners. It's created by its owners and is responsible to them. We'll talk more in detail about this whole idea of social responsibility and what that means and what that implies in a few minutes.

- 2.19** Let's think first about how corporations actually get created. What's the process by which we create these entities? How did Apple come into being? How does any business come into being? Apple basically started out as a self-proprietorship or as a venture of two guys, Jobs and Wozniak, two hippie-inspired children of the '70s, long hair. Jobs did not believe in taking a shower in those days. And they were part of hobby groups that were putting computers together, assembling stuff. This is in the days where computers didn't even have a screen. It was flashing lights and cables coming in.
- 2.20** Wozniak was the tech guy, they put together this computer to demonstrate it. And Jobs was always the marketing and business side of the duo. Jobs had this idea, "Oh, we could create something here. We could actually make it a business." They ultimately found people who were willing to invest in them. And when they found people to invest in them, they incorporated.
- 2.21** So you first have to have an entrepreneur with an idea. So an entrepreneur gives birth to the corporation, not the state. It's an entrepreneur that makes the corporation possible, gives it a reality, and focuses it on a particular economic activity, a particular business activity. In this case, the idea was to create a personal computer. Not the first, but to create just a better personal computer.
- 2.22** And then as part of the corporation—corporations grow—it brings in investors. Investors are not going to invest in you unless you already formed the corporate entity. Because now they have a particular system of law, their particular charter, their particular structure that they can understand and they know what they're investing in. Now they have limited liability. They can sell their shares. They have transferability.
- 2.23** And of course, as part of the investors investing in Apple originally, they demanded that they bring in professional managers, and they brought in professional managers and consultants. And Apple just exploded ultimately and became not really what it is today. It took a few iterations to get here. It took some failure before it got to where we are today—the biggest corporation in the world. But it became a big deal in the early 1980s. It became the first really big, successful personal computer maker before it was overtaken by Compaq and Dell and a bunch of others.
- 2.24** And what's Apple today? Is it a computer maker? Phone maker? App maker? Music maker? What is it? Today, it's more of a consumer electronics company. We'll talk about the evolution of that industry later. But originally, you need an entrepreneur. You need investors. Those initial investors, often called seed investors. And then beyond seed investors, you have venture capital investors. And then ultimately, if you get enough investors and you build up the business and money is coming in, and even if money is not coming in, you can convince people money will come in. There is a future.
- 2.25** Then you can go public, and going public means now you can get lots of investors. You list on an exchange, people can buy and sell your stock, and when you IPO, initial public offering, it means a large amount of capital comes into your company. The beauty of a

corporation is that you can raise large amounts of capital, build big things. Without public markets, without going public, without that, capital is limited, or at least was historically limited. Today— we'll talk about private equity and other sources of large amounts of capital, but historically, at least, the only way to grow big was to access large pools of capital.

- 2.26 And why these large pools of capital? Just because you're getting shares from a lot of different people. Even if everybody could put in a little bit, you aggregate them. It's a lot of money. So you take on investors, you invest, you make money, and then over time, you might take on more investors, you might issue more shares, you might grow even bigger, or you might use the profits that you're making to grow bigger, or it might be a combination of both.
- 2.27 When you use your internal, when you use what's called retained earnings, the profits from the company itself, then we call that an internal capital. That's internal capital. When you go out to the markets, banks, bondholders, or the stock market, that's external capital.
- 2.28 And we'll see whether you go to internal capital, external capital is to some extent dependent on the state of the market and what is going on in the world. So there are periods in American history where companies rely on internal, on their own profits, and the periods in which they rely externally on capital markets, the periods in which companies are very, very profitable.
- 2.29 Apple doesn't need an external market. It makes so much money. It doesn't know what to do with all the cash that it generates. It sits on a huge amount of cash today. So it doesn't need to go and borrow money or it doesn't need to go and issue shares. But there are other periods, certainly when Apple was not quite as profitable, it had to go out and raise capital from external markets with the promise that one day it would be so profitable, it could pay it back or it could make you enough returns to justify the investment.
- 2.30 So all of this, from the entrepreneur starting the company to raising capital, all of these —and then to the continuous use of capital and constant investment and growth and innovation, that whole process is a constant process of marketplace, of going into the marketplace and dealing with others on a contractual basis all within the context of a legal system that would protect property rights, all within a context of a legal understanding of how these contracts all work and how the corporation works. So what the government does, what the state does—it protects these contracts. It doesn't create them. It doesn't make them.
- 2.31 Indeed, I would argue that to a large extent, what the state has actually done and what government has actually done is interfered in these contracts, prevented certain contracts from being written, prevented certain relationships from happening. And then through regulations and antitrust, stepped in and said, "No, you can't do that. You

can't grow that way. We have to break you up. We have to take assets from some people, give them to others, redistribute them." It's not that the state so much creates the corporation, it's in our mixed economy, in our modern world, the state regulates, controls, and limits the corporation and what it actually can do.

- 2.32** And much of that, much of that is done in the name of, in one way or another, social responsibility or the common good or the greater good. So there is an ongoing debate, a debate that's been going on since the beginning of, really, of American corporations, about what the purpose of the corporation is. What should guide it in making decisions? Is it to maximize shareholder well-being?
- 2.33** Is it to do what its owners wanted to do, which for the most part is make the money? That's why we invest. Most of us invest our money because we're saving for retirement. We want it to grow. In other words, we want the corporation to become more valuable, which is how we are going to fund our retirement. So what we're really focused on is the corporation making money for us.
- 2.34** Or do corporations have some social responsibility? Yes, make money. It's okay. Make a little bit. Make some. Not clear how much. They never tell you. But really what you should focus on is some social agenda. And again, the social agenda will change over time, but that is what the focus should be. And really, the social responsibility agenda, the whole thing, has its roots in this concession theory, has its roots in this idea that corporations are really creatures of the state, therefore they owe something back. But it's gone way beyond that. Today, everybody uses this.
- 2.35** Here's an example of a quote from a newspaper article. "Today, it is absurd to regard the corporation simply as an enterprise established for the sole purpose of allowing profit making. One simply has to ask, 'Why should citizens, through their government, grant special rights, powers, and privileges, and protections to any firm, except on the understanding that their activities are to fulfill their purpose?'—citizens' purpose?—"Corporations exist because we allow them to do so."
- 2.36** Top-down, authoritarian state allows you to exist. Therefore, you must serve the state. A lot of people have that attitude to individuals, not just to corporations, right? So corporations have to do what's socially responsible. Now, there are many problems with this, but start off with, what does that even mean? What is socially responsible? And who gets to decide? Clearly, we have different views about social responsibility depending on our personal philosophies, our personal politics, our personal view of society.
- 2.37** Does personal responsibility mean we should be democratic? That is, we should do the most good for the most people or what most people want us to do? Or how do you even decide? What determines social responsibility? For that matter, what determines the common good? What is the common? Where is the common? Point me to the common. How do I find it? And how do I figure out what's good? What is good for all of

us? We're individuals. We all have different values, different ideas. We're pursuing different goals. It's not clear that we have a common good.

- 2.38** And even with social responsibility—social, what does social mean? Responsible to whom? I mean, shareholders are a part of society. Should companies be responsible to them? But what is argued is, no, they should be responsible to the environment. Well, who gets to decide what's good for the environment or what's not? How do we figure that out? Consensus? There's almost never a consensus about anything. And if I have a property right, if this company is mine, if I'm a shareholder, then why are you imposing your values on me?
- 2.39** The whole idea of common good, social responsibility, the good of the state, they're all meaningless concepts at the end of the day. There is no good of the state. All the state is is a bunch of citizens. There is no social well-being. All there is is a bunch of individuals who constitute society. And we each have, again, different values.
- 2.40** One could argue that the best thing for society, what is really the common good, is to leave us each alone so we can live our lives as we see fit rather than imposing values on us.
- 2.41** I think that was, to a large extent, the Founding Fathers' view of the welfare of people, "Leave them alone. Protect their rights. Protect their individual rights. And other than that, leave them alone." Once you start imposing criteria of social responsibility—what is a CEO supposed to do? Is he supposed to weigh different pressure groups? How does he make a decision about any particular corporate activity?
- 2.42** Another way to express social responsibility is often to express stakeholder capitalism. So a corporation, a business, should take into account the effects of every action that it does on all of its stakeholders. And then try to do what's good for the stakeholders, where shareholders, owners, in other words, are just one of the many stakeholders a company has. I used to do this exercise with my students. I used to say, "Okay, let's make a list of all the stakeholders." So we made a list of all the stakeholders. So think about it. So shareholders, obviously, stakeholder, employees. What about suppliers? Are they a stakeholder? Yeah, suppliers are stakeholders. Who else? Bondholders, people who lend you money, you borrow from them. Banks are going to be stakeholders. Who else? The local community. Because if, let's say, you leave, you move your factory somewhere else, some of the store owners will lose their business. So they're stakeholders. They have a stake in how well you do. If you do well, they do well. If you do badly, they do badly.
- 2.43** And you make a list of all of these. What about employees in the future? Do we take them into account? If you're very profitable, you'll hire more people. And if you're not profitable, you won't hire those people. So do the future employees count as stakeholders? Do you care about the future? Is that a consideration? And you could go on and on and on and expand this view of stakeholders however broadly you want. And

you make a list of all the stakeholders. And you say, "Okay, we've got a corporate decision to make right now." I know this is politically incorrect these days, but we could make a lot more money if we moved our factory to Mexico. We move the factory to Mexico. We could import a lot of the stuff back in. Labor is cheaper there. There are maybe fewer regulations. And given what we make, it's easier for us to move to Mexico. There's more skilled labor there, the labor we need.

2.44 Okay, let's make the decision, make a stakeholder theory. Social responsibility. We're going to be a lot more profitable. So we know shareholders are better off if we move the factory to Mexico. What about bondholders? Are they better off or worse off? Well, it's complicated. Bondholders—the thing they like the most is stability and no risk. So they might be against moving to Mexico. Even though being in Mexico is more profitable, it's easier for them to get their money back, but it's also riskier; there's a little bit of risk. Something could go bad. So maybe some bondholders are going to be for, some bondholders are going to be against. It's not clear.

2.45 What about employees? Are they going to be for going to Mexico or against? Well, it depends. Some employees, we're going to be firing them. They're against. But maybe there's a group of employees we're still going to keep, and now they're going to be working for a more profitable business, and they're going to be for it.

2.46 And what about our suppliers? Well, some suppliers, maybe the suppliers very close to where we are today, won't like this because they'll probably lose our business. But some of our suppliers may love it because now we're more profitable and we buy more from them. And then the future suppliers, the suppliers we might get from Mexico or in the border states next to Mexico, that they might love it. The new suppliers. I don't know if we take them into account or not. And again, you could go on and on and on with this.

2.47 Even employees. Well, should we take into account the fact that we're going to hire employees in Mexico and they will like it? Are they stakeholders? Or is there a border, a restriction, on where stakeholders end? How do we make the calculation?

2.48 Now you say, "Okay, well, should we move the plant to Mexico or not?" What do I do? Add pluses and minuses? What ultimately happens? How do you make decisions when you've got 100 different groups and they all believe in something, they all want something different and there's no consensus? How do you ultimately make decisions? You're not going to have them vote. They're not equally important. So what ultimately is going to make a decision?

2.49 I mean, this is politics because this is how politics runs. This is how politics is constantly run. You're going to do something politically. You're going to cut taxes on tips. That's on the agenda right now in the big, beautiful bill. And you've got probably a hundred different pressure groups, each wanting something a little different. How do you make a decision? Well, the most obvious way to make a decision there is to appease the ones

who are the loudest. Whoever beats on your table the loudest, whoever gets in your face the most. In politics, whoever has the potential to hurt you electorally the most, you're going to appease them.

2.50 In a business, whoever advocates the strongest, whoever threatens the biggest strike, whoever can do you the most damage—you listen to them. And imagine running a business this way. Imagine a CEO who has to run every decision that he does. That's a big one, moving to Mexico. But every decision he has, he's got 100 stakeholders and you have to weigh them all and figure out—you can't function that way. And in the end, all you do is you find a group to appease, and you go ahead and appease them. And often what happens is, particularly on a decision like moving to Mexico or something like that, the government will step in and say, "No, you can't do that," or, "Yes, we encourage that, or something like that."

2.51 So you get an external force that is stronger that ultimately determines. But where's the business and where are the owners? Don't the owners—shouldn't they make the decision? Because imagine if the decision was just made by shareholders. So your job as a CEO is to maximize shareholder agreement; do what the owners say. Then it's easy. Long term profitability goes up, risk goes up a little bit, but not too much. Yeah, we're moving. Or not, if it turns out that it's the other way around. But the decision metrics is simple and straightforward.

2.52 Imagine if you add to the stakeholders the national interest, national security, and a bunch of other things, again, amorphous, that now you're letting the government have a voice in what you do. It's impossible to manage. Versus a system that basically says, "Okay, you've got one principle that guides you: share the wealth maximization of a shareholder, wellbeing maximization, or doing what your shareholders want you to do, to the extent that you know that it's efficient, it makes sense, it's simple, and it's consistent with property rights. It's consistent with the fact that they're owners versus everything else which negates property rights. It makes them not really owners. Custodians? I don't know what it makes them. But now you have other factors, other people you have to consider.

2.53 And what happens if you just maximize shareholder wealth? You chain your employees to your machines and you whip them three times a day to get the most work out of them because you don't care about employees. You only care about shareholders. Is that how it works? Well, no, because A, you've got contracts with these employees. B, you want them to be really productive because that's how you make money for the owners. If they walk out and leave or if they stop working hard for you, then you're not doing your job maximizing shareholder wealth.

2.54 The only way to maximize shareholder wealth is to treat your employees well based on how productive they are. So you stop paying suppliers because you don't care about suppliers. You only care about shareholder wealth. You don't care about stakeholders. Only care about shareholders. So you stop paying your suppliers. Well, of course not,

because then they'll stop supplying you. And if you're erratic about paying them, they won't trust you anymore and quality will go down and you won't be able to rely on them anymore. So who are you hurting? Your shareholders.

- 2.55** So by having one decision parameter: I'm going to maximize or benefit shareholders—that determines that you treat your employees well, that you're going to treat your suppliers well, that you're going to treat all of them appropriately within the context of trying to make money.
- 2.56** Now, this doesn't mean you buy every employee a Mercedes. Treat them well. It means you treat them appropriately. It doesn't mean that Walmart is going to treat its employees the same as Saks Fifth Avenue. Saks is going to pay them more than Walmart in the context of what they're each trying to achieve, the marketplace that they have, and how they become profitable. But if Walmart pays their employees less than what is reasonable and what they're actually producing, then those employees will leave and go work somewhere else, and the shareholders at Walmart will suffer.
- 2.57** So this idea of, "We're going to maximize shareholder wealth" is not a negation of, "Oh, we don't care about anybody else." Of course, you have to care about lots of people, all those stakeholders, because it's by caring about them that you maximize shareholder wealth. I mean, the most obvious is customers. "I don't care about customers because I care about shareholders." Well, no, of course not. You care about customers because you care about shareholders. Shareholders require you to care about your customers. So a corporation is obliged to maximize their well-being, maximize their wealth, maximize their returns to its shareholders.
- 2.58** In our modern times, over the last 10 years, the two big issues that have come up in terms of corporate social responsibility have been DEI and ESG. I mean clearly agendas, in this case of the left, around either social equity, around environmentalism or other social issues that they believe are true—and they put pressure on companies to deviate from shareholder maximization in order to appease the particular interests of these particular social interests, of this particular political agenda.
- 2.59** It's quite possible that with the shift away from DEI and ESG, you will see the right come in with its DEI and ESG equivalent. Their particular agenda that they will want corporations to move towards, to drive towards at the expense, against, of shareholders. But the reality is that we've achieved the prosperity and the success that we have achieved, we've achieved the great innovations that the corporation has—a great standard of living and wealth creation that we've achieved because corporations are focused on primarily wealth maximization, making shareholders money, focusing on driving towards profits.
- 2.60** And what are profits? What are profits? How do you make a profit? By producing something that ultimately people want and are willing to pay you for it more than it cost you to produce it. Why do I buy the iPhone for \$1,000? Because the iPhone is

worth what to me? More than a \$1,000. In this case, much, much, much, much more than a \$1,000. More than a thousand. So my life's better off, and Apple makes a profit. If you think about profit as a win-win relationship, every time there's a profit, every time a company reports a profit, it means somebody else has benefited, somebody's life is better off.

- 2.61** When companies, corporations, announce high profits, everybody goes, "Oh, my God, they made so much money. It's terrible." I go, "No, this is great. It means they've created enormous value. They've improved our lives because we've been willing to trade with them." And why are we willing to trade with them? Because it'll make our life better. Otherwise, we wouldn't do the trade. Nobody forces me to buy an iPhone. Nobody forces me to buy, I don't know, the particular kind of ice cream that I like. I buy it because it produces value for me.
- 2.62** I'm better off. And yeah, they made a profit. That's a sign that the trade was win-win and that they can produce more ice cream to benefit more people over time. So you can't make a profit without creating value in a free market where people are reasonably free and where people are making decisions for themselves and not imposed on them. You can't make a profit without making the lives of other people better, which is what it means to create values. Create values for your customers, create values for people out there.
- 2.63** Now, does that make society in some sense where we define society as a group of individuals better off? Yeah, they're all better off because now they've engaged in this win-win transaction. They are winners. So is the company. It's a profit, it's just a reflection of one side of a win-win relationship, which is fantastic. Now, we also have to remember that there's a property rights issue here. Shareholders actually own the company. It is an ownership. That's the contract they have with the entrepreneur and with bondholders and with everybody else in the corporation. They are the owners. Well, what does it mean to own something?
- 2.64** Ownership means the ability to use that property in a way that you see fit. It doesn't impose any responsibility from outside on you. Quite the contrary, it says you can make decisions about your property. If shareholders own the company, the company needs to be run for their benefit. That's property rights. If you put in some other criteria, then you're violating the property rights. You're violating the right, you're violating the whole concept of ownership as it relates to this.
- 2.65** At the end of the day, social responsibility is just a cover; the whole stakeholder and all of this stuff is just a cover for the people who don't want private property, who don't want companies to be able to be run based on how the owners want to run it. They don't want that freedom. They want centrally planned. They want to be able to guide the economy. They want, at the end of the day, to be able to guide our lives. It's a form, it's not a direct form, but it's a cover for socialists or statists or any type of statist who wants control over the economy and over our lives and over our businesses without

literally nationalizing it all. That would be very unpopular in America, although this golden share in U.S. Steel comes close to nationalization.

- 2.66 Generally, out-and-out nationalization is unacceptable. Nobody does it. So they find other ways to sneak in their social agenda, their political agenda. Corporate social responsibility, stakeholder theory are ways to do that, ways to get a political agenda in. DEI and ESG are great examples.
- 2.67 ESG is the idea that the environment is the most important thing. We must focus on the environment. Obviously, corporations don't focus on the environment, so we're going to put pressure on them. We're going to create an environment where they have to focus on the environment politically, and that way we'll get our environmental agenda passed rather than go and do it politically. That is, you can pass laws to get the same agenda passed, but that's a lot harder; that's a lot more difficult. And you have to convince the majority of the people to vote for that agenda, which people haven't in the past.
- 2.68 Same with DEI. We can't convince most of Americans to discriminate in a particular way we believe discrimination should happen. So we're going to find ways to guilt companies into doing it for us so we don't have to pass the political agenda. We'll manifest these ideas into corporate life without it being top-down, without going through the political process, which is messy and difficult and requires persuasion.
- 2.69 So, at the end of the day, the purpose of the corporation should be— and I think for most corporations it is, although they all have beautiful social responsibility slogans, every business has one—at the end of the day, the purpose of the corporation is to make money at the particular thing that they are doing. It's to create value for the shareholders, to create value for the owners, again, within the context of what it is that they do.
- 2.70 They might have inspiring slogans, messages that relate to that. Or as some corporations do, they might have what they consider inspiring messages that are unrelated to creating value, that are on the side, that contradict, in order to appease the various pressure groups that are trying to get them to be more, quote, "socially responsible."
- 2.71 Given that the company, the corporation, is geared towards, or should be geared towards, maximizing shareholder wealth, doing what the owners wanted to do, how do we make that happen? Because one of the things that was identified early on with regard to corporations, particularly big corporations that are publicly traded, is there was a tension and a difficulty in making sure that managers, the people who actually control the day-to-day activities of a company, actually do what's in the best interest of shareholders.
- 2.72 Because you could imagine managers having other incentives. Yes, they're there to run the business for shareholders and to make their money, but they might, for example,

they might want to be invited to really cool parties. And to be invited to really cool parties, you might want to do, I don't know, certain social responsibility stuff in order to appease the correct crowd. Or you might want to grow the business really, really big because if you're really, really big business, then people perceive you as really, really important, and you get invited to the right parties.

- 2.73** In other words, you might have personal incentives that are not consistent with making money for shareholders. Part of it might be determined by how you're compensated. If I compensate my CEO based on sales— the more sales he does, the more money he'll make—what do you think he's going to maximize? Sales. Is maximizing sales the same thing as maximizing profits? No. It might be, but it usually is not. So we could create the wrong incentives within a corporation.
- 2.74** If managers really control the corporation, and in some ways, if managers control their own salaries, because boards decide on the salary of a CEO, but who appoints the board? The CEO does. If you look at corporations, your shareholder vote, but shareholders almost always vote in the directors the CEO wants. And who's the CEO going to want to be directors, theoretically? His buddies, his golfing buddies.
- 2.75** And then he goes to them and says, "Look, this is the salary I want. This is the incentive structure I want. I want to be concentrated on sales. I like sales. The more sales I do, the more money." They're likely to say yes. At least the theory is they're likely to say yes. And as a consequence, he's going to maximize sales, and not care about shareholders.
- 2.76** You can think about a lot of scenarios where managers do things that are inconsistent with the long-term profitability of a business, inconsistent with the interests of shareholders. This is called the separation of ownership control, the problem of separation of ownership control. There was a very famous book written in the 1930s by two economists, Berle and Means. This is a real problem. We've given control to people whose incentives are not aligned with shareholders, and they're going to screw this up, and they're going to do bad things.
- 2.77** And indeed, this idea has really dominated the literature about corporations since then, even the whole discussion about CEO pay, the Elon Musk discussion— it was the directors he appointed. They're all his buddies. Are they going to vote no to his \$50 billion compensation? Of course not. So how do we align the interests? Adam Smith even recognized that this was an issue. The more dispersed the ownership is, the more shareholders there are, and the smaller the shareholders are, the less power they have, the less voice they have, and the more managers can get away with, again, at least in theory. So this is a real problem and can create real conflicts of interest within the company and between different interests within the company. So how do we resolve this?

- 2.78** Now, one thing to realize here before we get into how to resolve it, one thing to realize here is that this idea of separation of management control, as we said really last class, it's not so much—there's an issue there, but it really is part of the innovation. It's, at the end of the day, a plus if we can resolve the conflict, because by creating a separation of management control, we now can hire specialist managers, and all of us can participate. The pool of capital is much greater. So this is an example of division of labor. It's an example of how markets solve problems, not how they create them.
- 2.79** And then the question is: how do you solve this issue? Well, first, it makes an assumption about human nature, which is sometimes true, but not universally true. And the assumption about human nature is that people are only motivated by money, and that people will screw one another whenever they have an opportunity to do so.
- 2.80** If you incentivize Steve Jobs based on sales—I don't know if the board of directors somehow did that—would Steve Jobs maximize sales? Is he just an automaton, like a little rat that you wind up and just does it? No. Steve Jobs loved what he did. He valued the company. He wanted to see beautiful, amazing products. Part of that was to maximize the value he was creating. Steve Jobs was not motivated by any particular set of materialistic incentives. It was much more complicated than that.
- 2.81** And he had integrity. He wasn't going to abandon his integrity because there was some mismatch of incentives that was created in his compensation package. No matter how you incentivized Steve Jobs, I don't think it would have made much difference— as long as he could make money, much difference to what he did. And most businessmen are like that.
- 2.82** Most businessmen actually have personal integrity. They take on the job not just to maximize their income at a particular point in time. They take on the job because they love creating values. They love the process, and they are committed to maximizing the wealth of their shareholders. They're committed to maximizing profits.
- 2.83** So it makes an assumption, which I think is wrong. An assumption that is way too prevalent in our world: we're all flawed, we're all fallen, we all do nasty things if just given a little opportunity. I think if that were the case, again, we wouldn't have the amazing world we live in. We'd be screwing each other over all the time. There are plenty of opportunities to do that. The fact is that an overwhelming majority of people are pretty honest, pretty dedicated, and pretty dedicated to doing a good job, doing something of value.
- 2.84** We want to see, we want to get meaning from our life. A lot of the meaning we get from our life is the work that we do. So again, I think it's a reflection of a pretty horrible view of life and a view that there's always conflicts of interest. That me succeeding means you failing rather than, particularly in a corporation context, me succeeding means you succeeding.

- 2.85** In business, generally, the attitude should always be win-win. Me succeeding means my customers succeed. Me succeeding means my employees succeed. Me succeeding, my suppliers succeed. That's how you really grow a successful business. The attitude of a zero-sum world—do you know zero-sum? That is a never transaction. There's a winner and there's a loser. That is a loser way of viewing the world. It's a loser way of viewing the world from a material perspective—from how much money you'll make—but it's also a loser way of viewing the world from a spiritual perspective, from the self-esteem and the self-valuing that you have, if you view everything as winners and losers.
- 2.86** Now, business, but life, generally, not just business, life should be about maximizing your win-win relationships, maximizing the win-win opportunities that you have in life: to win, but with the other parties winning as well. That's the fundamental essence of trade. The fundamental essence of trade is win-win. And that's why trade is one of the most important activities that we engage in as human beings. Think about trade beyond just buying and selling. But everything we do, we're involved with other people. So that's one. It's a negative view of man.
- 2.87** The other thing is, if you start adding into the corporation these ideas of social responsibility and stakeholders, you create these win-lose situations. You create situations in which, as I said before, there is no one standard by which we measure our success. And now you're creating conflict. And now you're generating bad vibes, bad morality, bad ideas. What should CEOs do? What should managers do? Which group should they benefit? At whose expense? We're sacrificing people, one to the other, instead of trying to maximize shareholder wealth by making not everybody better off because some people are going to lose their jobs, some suppliers we're going to fire. This is not some Pollyanna-ish view of business. There's always change and there's always evolution. But overall, if you do your job well, people are better off, even if not every single individual is better off.
- 2.88** So how do we resolve the reality of this potential conflict. And by the way, once in a while, it does manifest itself. We'll talk about—people today think the American ideal was somehow in the 1950s and 1960s or the 1960s and even the 1970s. Some people are arguing, particularly today, again, both on the left and both on the right, there's this nostalgia for the 1960s. We'll talk about that. It was actually a pretty terrible time, particularly from a perspective of corporations.
- 2.89** One of the things that made it terrible was indeed, corporate managers behaved wrongly. There were a lot of incentives that got them to do that. But American business became big, bloated, and inefficient, and unproductive. We'll see how all that changed in the mid '70s and into the 1980s. But during that period, there was some of this behavior that we're talking about. People did—some CEOs bought a bunch of private jets, flew over the country in a private jet as their stock price was going down. "Screw shareholders, I'm flying private." It doesn't last very long because there are mechanisms, market mechanism, to correct for that.

- 2.90** So first, there is a board of directors. And while it's true that the CEO appoints those directors, again, those directors have a fiduciary duty to shareholders. They can be sued if they're acting on behalf of the CEO at the expense of shareholders. They also are typically themselves business people, often business people, and again, have that integrity and that responsibility to their agents of the shareholders, and they try to act that way. So portraying boards of directors as just puppets of the CEO might occasionally be true, but systematically is an insult to the people who serve on boards of directors, and just, again, an insult to human nature. I don't think human beings are like that.
- 2.91** There are various signs that suggest this. CEO pay is incredibly correlated with stock market performance of the company. That is shareholder wealth. CEOs don't last in their job for very long. When they don't do well, they're fired fairly quickly. CEO tenure is very low. It's not a job you go in for 20, 30 years, particularly over the last 30 years. Again, '50s and '60s were a little different for a variety of reasons. So board of directors are not that bad. They're actually quite good.
- 2.92** When we start intervening in the board of directors, it's when things go shaking. So for example, there's a big push to get independent directors on the board of directors— independent means unrelated to the CEO. But it also could mean, for example, directors who are not business people. We heard a quote around that last time. We want outsiders to come in. But is that what you really want? What is the ideal director? If we care about the alignment of shareholders with the CEO, what is the ideal director?
- 2.93** The ideal director is a shareholder. And then the director is aligned with the shareholders because he is a shareholder. If you maximize shareholder wealth, he also maximizes his own. But the idea of independent directors is often the idea that, "No, they shouldn't own any shares. They should be independent. They shouldn't have any interest," which is bizarre.
- 2.94** I remember Warren Buffett was on the board of Coca-Cola for many years. Warren Buffett owned a big chunk of Coca-Cola, and Warren Buffett is super smart. He's like the best director ever. If I had a business, I would love to have Warren Buffett both own a piece of the business and be on the board of directors. And if I'm a shareholder, I love the fact that Warren Buffett is on the board because he's smart and he's also a shareholder. So he's like me. Well, better than me, right? So he's going to represent my interest really, really well.
- 2.95** But CalPERS—CalPERS is the pension plan of the Teachers Union in California. It's the largest, or used to be the largest institutional investor in the United States. CalPERS said, "No, we don't want Warren Buffett on the board of Coca-Cola." They actually filed a proxy thing to get him off the board of Coca-Cola because he's an insider, because he owns too much shares. Because we believe in independent directors, people who don't have an interest, which is, again, a perversion. It's this idea, if you don't have an interest, you're going to do better defending shareholders than if you're a shareholder.

- 2.96** Again, the implicit assumption there is: self-interest is bad. If you're somehow motivated by your own self-interest—you're a shareholder, you want to maximize your wealth—that's bad. We want people who don't have a self-interested interest. They're taking care of shareholders out of the goodness of their hearts because they don't have any self interest. That we elevate. That we think is noble and good.
- 2.97** You want independent directors, but who are typically businessmen who know something about it, but you always want them to own shares. Ideally, not only does a new board member who comes on the board own shares, but he actually takes money out of his pocket when he joins the board and buys more shares to even more align himself with the investors. I think companies should—and many companies do this—require board members to own a significant portion of their own net worth in the company. That will align their interests perfectly.
- 2.98** Indeed, if you think about it, one way to align interests is to have the CEO to own shares. One of the reasons that CEOs started getting stock options—about 30, 40 years ago, we started giving CEOs stock options; we stopped, unfortunately, about 15 years ago—is because it aligned the interest. It gave them— if stock went up, they made a lot of money. If stocks went down, they didn't make anything. So their incentives were aligned with shareholder incentives.
- 2.99** Now, there are other ways in which we can align incentives. One, if the CEO starts doing a bad job, if the manager starts doing a bad job, what happens? People stop buying the product, profitability goes down, the company doesn't do well, and we get all the signals: time to kick him out. So the CEO is incentivized to maximize profits because that is going to ultimately determine his long-term future. Not necessarily because somebody is overseeing him, but because if he starts failing in the product market, the company will fail, and he'll lose his job. You assume he's motivated not to lose his job. So the ultimate arbitrator of this is the product market, the ability to sell and the ability to create values.
- 2.100** Capital markets. If a CEO does a bad job, what happens to stock price? It goes down. Now, stock prices could go down even if the CEO is doing a brilliant job because the market might have changed. You might be making the best buggy in the world, but if buggies have no future, your stock price is going to go down even if you're the best CEO in the world. But let's say you are slacking off as a CEO and stock price goes down. And then somebody else looks at that and says, "If I took over that company, I could make a lot of money because I could buy cheap, where it's priced now, and if I did a good job as a CEO, the price would go up."
- 2.101** So what you're creating by doing a bad job is the opportunity for others to come in, buy your stock, and replace you—walk into your office and fire you. Now, it doesn't work quite that simply. We'll talk about that in a future class. But the capital market is a great discipline. Stock prices going down is a signal you're not doing well. And if they go down enough, there's an incentive for somebody else to come in and take you over and

replace you as CEO. It's called the market for corporate control, the ability to change CEOs.

- 2.102** One of the reasons it's good to have a large shareholder like Warren Buffett is because that shareholder, whether he sits on the board or not, has a lot of influence on what you do and is monitoring the CEO to make sure he does what is in the interest of all shareholders. We've got board of directors. We can incentivize managers by giving them stock or stock options. We can incentivize directors by requiring them to be shareholders. And we've got this market for corporate control, where if you do a bad job, you will be replaced. And we'll see that happening. That's exactly—all the CEOs that screwed up in the '60s all lost their jobs by the '80s. And so that mechanism actually works. It actually happens.
- 2.103** One last point before we go on to a new topic, to the next class. One of the things that makes this problem more real is the fact that our regulatory environment, the regulations and controls that the government imposes on businesses, often exacerbate the problem by making it difficult for large blockholders to be board members. So for example, we make it very difficult for bankers to be on the board of directors of non-bank companies. At some point, we actually forced them all to get off the boards of railroads. We make it very difficult for other institutional investors, pension plans, insurance companies, and others to actually be board members. The regulatory environment does not view favorably large blockholders sitting on the board of directors, and yet these are the most qualified directors and the ones with the greatest incentive to actually monitor what managers do.
- 2.104** So we've actually created a regulatory environment that exacerbates whatever exists in terms of this separation of ownership and control. Add to that that we have made hostile takeovers—my ability to come into a company where the CEO is not doing a good job, fire him, and replace him by taking over the company—we've made hostile takeovers really, really, really difficult in this country. We had a lot of them in the '80s, early '90s, very few hostile takeovers since, partially as a consequence of the regulatory environment has shifted, where it's very difficult today to do a hostile takeover and therefore use the corporate market for control to discipline managers.
- 2.105** So corporations are a creation of the marketplace. Their purpose is to maximize shareholder wealth. And in spite of the criticism, markets are very good at aligning the interests of shareholders with managers. The more external intervention there is, the more distortions and perversions there are—but ultimately, again, we wouldn't have, and corporations wouldn't have been as successful as they are, if this alignment hadn't existed. If you've invested in S&P 500 over the last 50 years, 100 years, almost any period, you've done very, very well as a shareholder, which suggests managers are doing a pretty good job at maximizing your wealth, maximizing economic profits, maximizing the creation of values.

Lecture 3: Corporate Evolution

At a glance — Corporate evolution — what a corporation is and does.

- 3.1 We are at the point where we know what a corporation is, we know what it does, we know what its goals at least should be—maximize your shareholder wealth—and that the whole attack on it over social responsibility is completely misguided. And of course, the attack on it in terms of separation of ownership control just misses the point and is not really there. Although there is something there, and we'll keep seeing that crop up, and that issue will keep coming back, and we'll see how markets correct for it as it does crop up.
- 3.2 What we're going to do in the next few classes is really look at the evolution of American corporations over time. We're going to start with the 19th century, with the late 19th century, and go all the way, really, until today. We're going to do it not sequentially necessarily, but around particular themes, particular themes of how the corporation evolved, given particular pressures that are being put on it.
- 3.3 But before we get there, because I think this plays a big role in it, I want to talk about the difference, or I want to talk about political power, because as we'll see over and over again, politics plays a huge role in the evolution of business, in the evolution of the corporation. Politics is everywhere from the beginning all the way to today. It's guiding it, I would argue, strongly, not in a particularly positive way. I think in a way that hampers business, in a way that reduces our quality of life, standard of living, wellbeing, and wealth. But it's there. It's a reality.
- 3.4 And you cannot understand the way corporations have evolved. You can't understand even what's going on today. You can't understand even the existence of the Big Five high-tech companies without understanding the political dynamics that have played into this. And we're going to talk. We're going to have a section just talking about antitrust and how antitrust has evolved since the late 19th century to today and the different attitudes towards antitrust and how they have shaped different corporate behavior.
- 3.5 We're going to talk about regulations. We're going to talk about the attempts of government to do industrial planning and how that has shaped businesses and corporations. We're going to talk about the World Wars, World War I and World War II, and how the attempts of government to manipulate business during that period, what consequences that had. We're going to talk about all that. In addition, we're going to talk about the market and how the market responds and how the market reacts to all this.
- 3.6 But in order to understand all that, I think it's first important to understand what it is that government brings to the table. And in that sense, I think there's a massive confusion in the world between two types of power. Economic power: the power that

businesses have, the power that a corporation has. And political power: the power that governments have. There's a fundamental difference in the nature of these two forms of power that I think goes a long way to explaining the relationship between business and government and the role government plays.

- 3.7** So what is the essence of political power? What is it that makes political power different than really any other power? What does government bring to the table? What is government about? What is the essence of government? I think George Washington said in one of his inaugural addresses, he said, look, "Government is a gun." Government, fundamentally, government power, fundamentally, is the power of coercion. It's the power of force. It's the power to dictate terms. And if you don't abide by those terms: pick you up and put you in jail. Literal force. It's not "I disassociate with you." It's no, "I impose my will on you."
- 3.8** Laws, for better and for worse, are the imposition of rules on us. Rules that if we disobey, there are real physical consequences to them. In the sense of, at the end of the day: jail. Physical force. They might fine you, but that's just an intermediary step. What happens if you don't pay the fine? Ultimately, it's about using force to restrict your behavior, using force to restrict you.
- 3.9** Now that's required. We need that. There is a role for that force. It's a self-defense role. That is to protect us from crooks and criminals and fraudsters. We don't want to negotiate with fraudsters. We don't want to negotiate with murderers. We don't want to negotiate with people who can do us harm. The role of government is to come in and stop them and to separate them from us by putting them in jail, by reducing their impact on our life or eliminating the impact of their life.
- 3.10** Government is there, ideally, as an ideal, to protect us, to protect our rights. And what do our rights mean? Our rights mean our freedoms—our freedom to act, our freedom to live our lives as we see fit based on our own judgment, our own values, our own mind. Free of authority, free of coercion, free of force. Government is supposed to protect us so that we can live such lives— lives of freedom.
- 3.11** But of course, it doesn't only do that. We haven't had this ideal form of government, really, ever. Government also uses its force to protect what they would say is the "common good," the public interest, what society wants. And the important thing to remember in this context is it's doing so through force and through coercion. There are no options. You can't say, "No, I don't like that law. I'm not going to follow it." Unless you're willing to bear the consequences of not following it. So that's one type of power, one type of power relationship.
- 3.12** What is the essential characteristic of economic power, of the power of a business, the power of a corporation? What is the essential characteristic there? One is a relationship of coercion. What is the other? Well it's fundamentally a voluntary relationship. It

doesn't matter how big Amazon grows. It doesn't matter how enticing its products are, how cheap it offers them to me.

- 3.13** I literally don't have to buy stuff from Amazon. I can go to Walmart. I can go to my local store around the corner. I can go to the mall. And even if none of those exist, I can go off to start a farm and grow my own fruits and vegetables. Amazon has no ability to use force, to use a gun, to use coercion to make me deal with them. Even if somehow they got 100% monopoly, I could still go and make my own stuff. You can't stop me from doing that.
- 3.14** So the essence of economic power is voluntary interaction. And the way you grow big, the way you're successful economically is by offering your customers values, by making as many win-win relationships with your customers as possible. The more such win-win relationships, voluntary win-win relationships, the bigger you're going to be, the more successful you're going to be. There's no coercion. There's no force. There's no compulsion. A business has to convince me it's in my interest to deal with them.
- 3.15** And that's not just relating to customers. If I'm an employee, I don't have to work for them. I can leave. I can resign. They can fire me. It's a voluntary relationship. When it's not win-win, for whatever reason it's not win-win anymore, we can go our own ways. The company can't force me to stay. That's called slavery. I can't force, or shouldn't be able to—I think I can—I can't force the company to keep hiring me. In some circumstances in America today, you can do that; you join a particular union, you negotiate. There are ways in which to coerce the company into doing it and to use the force of government on your behalf. But pure economic power without political power is only the power to negotiate, to contract, to voluntarily exchange and trade.
- 3.16** Now, we live in a world where there is a mixture, where economic power often is not pure in a sense that corporations can use government to impose their will. They can lobby, they can help government change regulations, to make it difficult for their competitors to compete with themselves. There's a constant interaction between political power and economic power in the world in which we live.
- 3.17** In my ideal, in a true free market, those two are completely separate, and there's no interaction between them. We live our economic lives, and the government protects our rights, and that's it. It's simple. That's not the world we live in. We live in a world in which government brings the power of coercion into almost every transaction we engage in in the marketplace. There's some element of coercion or force associated with it.
- 3.18** And businesses are constantly engaged with government, either protecting themselves from the attempts of government to control them and regulate them, or in an effort to try to control and regulate their competitors in order to make their life easier. But there's a constant engagement going on. There's a constant back and forth going on. And we'll see that play out over the course of the last 150 years in America. And we'll

see how that interaction between the forces of government, the forces of coercion, and the forces of voluntary trade, how that interaction shapes the world in which we live, shapes at least the corporate world, the economic world in which we live.

- 3.19** We have to remember that politicians are pulled in a number of different directions. Ideally, they would be individuals of principle. They would live by principle, and they would be immune from pressure and influence. But particularly in politics, politics doesn't self-select for those kinds of people. Politics self-selects for people who are adroit at dealing with multiple constituencies and in a sense, selling their soul to the highest bidder.
- 3.20** And that's true on the entire political spectrum. It's not like this is a left or right issue. Politicians are not—I can't think of an exception in our modern world—people of principle. They are people of juggling. They're jugglers. They're constantly juggling different interests.
- 3.21** And a big part of what drives people to politics is exactly that: the ability to have power over others, the ability to manifest control. So that's the kind of people that get attracted to politics. In a world in which we live, you have to do that. Different people have different interests, and they're all voters, and they're all going to come to you with different things. Some people bring money, and some people don't. Some people bring votes. Some just appeal to your good nature, but everybody's trying to get something from you.
- 3.22** Because we live, today, in a democracy where it's all about who can muster the most votes, which creates a situation of pressure groups and where the government can intervene in almost every aspect of our lives. So they have immense power and a whole array of incentives. And lots of people putting pressure on them constantly. And again, we will see that as we go through various examples of that.
- 3.23** Corporate lobbying, cronyism, is definitely a problem. Once you give politicians or the political machinery power over business, business is going to want to defend itself. Business is going to have influence over those that influence it. You can't get politicians power over business and expect business to just say, "Okay, we're not going to intervene. Whatever they want, we'll do." It's not how the world works. It's not how the world has worked. So what you constantly get is an interaction between the two.
- 3.24** I often tell the story of Microsoft in the 1990s, when Microsoft was the largest corporation in the world, and it literally spent zero dollars on lobbying. Nothing. No presence in Washington, D.C. Nothing in D.C. No lawyers, no building, no activity. They were not interested. They were like the biggest company in the world, they didn't need politicians. And they were thriving in Seattle.
- 3.25** And they were literally brought in front of a Senate hearing with Orrin Hatch, who was a senator from Utah, Republican senator from Utah. At some point in this hearing, Orrin Hatch stood up, started yelling at them, "You need a presence in D.C. You need a

building here. You need a lawyer here. You need to lobby. You need to protect your interests right here in D.C.!" In other words, "You need to bribe me." That's another way of putting it. And, "We're important! D.C. is important. D.C. is more important than you. You need to come and—"

- 3.26** Microsoft literally at the end of the hearing said, "You leave us alone, we'll leave you alone. Because we're not interested. We have nothing to gain here. We're doing great by ourselves. Don't need you." Well, the problem is, for the business, is that Washington didn't leave them alone. Within a few months, there was an antitrust lawsuit against them. And ultimately, they were regulated, they were controlled, they were almost broken up.
- 3.27** And what is the consequence of that? The consequence of that is how much dollars do you think Microsoft spends today on lobbying? A lot. Tens of millions of dollars a year. They built a beautiful building in D.C., about equal distance from the White House and Congress. And they spend a lot of time with lawyers lobbying constantly. And is all of it just to defend themselves from regulations? No. A lot of it's also to manipulate, in order to protect themselves from competitors. That's how it always evolves.
- 3.28** Once you allow politicians to have influence, corporations will try to have influence in the other direction. That is the nature of the beast. And for the last 150 years, the more power we've given politicians over business, the more business has imposed power or tried to influence politics. And the whole evolution here is a constant interplay between those two periods in which government has a lot of power over business, then it gets reduced, then it comes back, and we move through these cycles.
- 3.29** Although overall, you have to say that the trend of history is more power over business, more controls, more influence is the general trend. The regulatory book in Washington D.C. Only grows, or almost only grows. There have been very few periods of real deregulation, real shrinkages of the power of government over business. So remember. Political power: force, coercion. Economic power: voluntary transaction. You can walk away from a corporation, you cannot walk away from your government.
- 3.30** Okay, so with that, let's talk about how business has evolved, how the corporation has evolved, has changed, structured itself over the last 100 or so years. We're going to look at a number of different drivers of business, of the corporation. We'll talk about the fundamental, which is market forces. How does the market shape business? And in particular, how does the market shape the structure of corporations? What kind of businesses do they get into? Whether they're vertically integrated, horizontally integrated, well diversified or specialized. All of this, how does the market determine how this happens? The structure of ownership, determined by the market and other forces.
- 3.31** We'll talk about macroeconomic factors. How does, for example, a great depression affect structure of business? What does it do to small businesses versus big businesses?

Was the New Deal good for big businesses, small businesses, some combination? What's going on there? The Great Recession, the financial crisis of 2008. Who benefited? Who lost? And how did that shape behavior moving forward?

- 3.32 A big one is antitrust. Obviously, antitrust has played a huge role in American business since the passing of the Sherman Act in 1890, which was the first real big federal regulatory, you can call it legal action, to regulate business; 1890. The Clayton Act: 1914. And all the way to today, with the different interpretations of different administrations and the different interpretations of different Supreme Courts.
- 3.33 Because one of the things you'll discover about the law is much of it is written in super ambiguous terms, particularly law related to business, is written in super ambiguous terms to give politicians power over when to apply it, when not to apply it, who to apply to, who not to apply to. And then it gives the courts a lot of leeway in terms of the different interpretations of what it actually means.
- 3.34 Industrial planning. The United States has gone through phases where it has embarked on attempts at industrial planning, certainly in World War I or World War II, certainly in the Great Depression. And then, of course, more recently with Biden, we had a number of bills from the Inflation Reduction Act to the CHIPS Act to try to say, "Okay, America needs a chip industry. So we, the government, are going to incentivize, subsidize, motivate a chip industry." Inflation Reduction Act was actually around, "We need batteries and we need solar panels, and we need these industries, and we need to build them in America."
- 3.35 So there was a huge subsidy and incentives around those kinds of industries. This is the government deciding winners and losers. Which industries are good? Which industry is not so good? Which industries do we want more of? Which industries we don't care that much about?
- 3.36 We'll talk a little bit about that, and you're going to see certainly under Trump, you're going to see more of that, not less of it. Trump is definitely, or the Republican Party as it's constituted today, is very much interested in boasting certain types of businesses and not other types of businesses. The whole tariff policy is a way, is a means, at least by some interpretation, a means to do industrial planning.
- 3.37 And then finally, regulations. We won't talk a lot about regulations because God, it's a big topic and there's no end to it, but we'll talk about a few biggies. And we'll talk quite a bit about deregulation, because deregulation that happened surprisingly—for most people, I think, really surprisingly, shockingly—in the 1970s under Jimmy Carter and in the 1980s, under Ronald Reagan. We'll talk about that deregulation, how it affected the structure of businesses and how business evolved as a consequence of that. But that was a big shift in the regulatory environment in America.
- 3.38 So let's start. And what we'll do today is talk about market forces, and we'll see what else we cover, but we'll definitely cover market forces. And next class we'll start talking

about some of the various government interventions, a class on antitrust, and a class on some of the others.

- 3.39** As we described, business is fundamentally started by entrepreneurs. What do entrepreneurs do? What is it that entrepreneurs do? What is an entrepreneur? Well, an entrepreneur is somebody who discovers or figures out a profit opportunity nobody else has seen. And that could be by discovering a new product that nobody else is producing and they think they can produce and that people will buy.
- 3.40** Now, if you're just discovering a product and profit is not on your mind, you're not going to do very well. Profit indicates that people are going to value it more than the cost to produce it. And if that's not the case, you're not going to survive. So profit is an inherent part of entrepreneurship. It could be you discover a new form of business, a new combination, a new way of doing business that is going to create profits. It could be that you create a new system of manufacturing; the same good, but you're just going to do it differently, in a way that nobody else has seen before that generates a profit. In other words, again, every time I say profit, you can substitute "value" for that because that's what profit indicates. It indicates an added value.
- 3.41** So all businesses start with somebody having an idea for something different, something better. Some are right, and they succeed. Many, many, many, many are wrong, and they fail. Most businesses of all types, most businesses, not just most, an overwhelming majority of businesses fail. Very few succeed. A venture capitalist investing in businesses has already selected the ones he thinks are going to be successful. How many of the investments he makes are indeed successful? Typically less than half. Half fail. And that's of the companies he chose to invest in.
- 3.42** Of the companies he actually looked at, 90% of which he didn't invest in, most of those fail. So a vast percentage of companies of entrepreneurs fail. And they never get to start a business, they never get to build a business. They never get to really have an impact. And of course, the reality is that almost every entrepreneur who is successful typically fails.
- 3.43** One of the things that I think makes the United States different than the rest of the world, particularly different than Europe, is our high tolerance for failure. In America, if you fail, "Eh, big deal. Go do something else. Do it again. Do it differently. Next time, you'll be successful." Everybody views it that way. Your family views it that way. The bankers view it that way. Venture capitalists view it that way. I know venture capitalists will say, "I won't invest in anybody who's never failed. I want people who failed. I want people who know what that's like, who've learned from it." Failure is built into the American entrepreneurial system.
- 3.44** In Europe, if you fail, your family looks down on you, the banker will never give you money, and no venture capitalist is ever going to invest with you. It's a different cultural attitude towards risk-taking, towards being an individual, between trying something

different and new, towards being an entrepreneur and towards failure, generally in life. I think one of the great virtues of America is our tolerance of failure, is our willingness to tolerate failure and accepting it as a learning opportunity.

- 3.45** In some cases, we purposefully fail in order to learn. I think the best example of this is SpaceX; will launch a rocket knowing it's going to blow up. They know. They know it's not going to succeed. But by launching it and having it blow up, they'll learn so much so that the next time, ultimately, it will be a success.
- 3.46** That attitude in many cultures does not exist. That willingness to take a loss in order to achieve a long-term gain does not exist. That thinking forward, thinking long term, investing in the long term, even at the expense of taking a hit in the short run. That's, I wouldn't say uniquely American, but that is one of the things that categorize Americans and differentiates it certainly from Europe or Japan or some of these other countries.
- 3.47** An entrepreneur starts a business, but there are a lot of different ways that you can organize your business. Even if you establish a corporation— which they all do because you want to limit the liability and all the other benefits we talked about—there's a lot of ways in which you could do it. You, for example, could subcontract. You could sign contracts with everybody who needs to provide you with the service. You could have no employees, or very few employees, and everything could be subcontracted, and just put together.
- 3.48** Even the putting together, where's the iPhone put together? China. Apple doesn't put the Apple together. Apple doesn't put the iPhone together. The iPhone is put together in China. Apple doesn't source any of the materials that go into this. Well, now they do the chip. But even the chip, who makes the chip? Literally makes the chip? TSMC in Taiwan. Apple doesn't make the chip.
- 3.49** So what does Apple contribute to this? Planning, design, marketing, sales. The biggest value added in this is not assembly. It's not the chip. It's not the glass, which is not made by Apple. I mean, nothing really here is literally made by Apple. Everything here is subcontracted except the engineering, except the design. Even the chip. Who designs the chip? TSMC might fabricate it, might make it, but who designs the chip? Who innovates? Who makes it possible for the chip to get faster and faster all the time, to put more transistors on it? Well, Apple in its design, and then the manufacturer, TSMC, has to be able to produce what Apple designs. You can design what you want. If you can't actually build it, it's useless.
- 3.50** So some companies basically, in a sense, outsource everything. And that makes sense in certain types of industries. If you're going into business and there's a robust market in every piece that you need in order to make the stuff, the new thing that you have invented, then why bring all that manufacturing in? Why bring all those people in when they're all good at doing what they are doing already? And you can benefit from the

price signals that a market creates. You can benefit from the indications of supply and demand that exist in the marketplace.

- 3.51** When would you bring it all in? You would bring it all internally when that is cheaper than outsourcing it. Every time we interact with other people, there is a transaction cost. There's a cost to transacting, to every transaction. There's a cost of bringing in suppliers and making sure they're aligned with us, making sure that they are going to deliver the product on time. There is a cost that everything is going to be to specs the way we want it. There's a cost that they don't leak the knowledge that we're providing them about the parts that we want to our competitors. There is a lot of issues with having somebody do it on the outside.
- 3.52** But there's also issues bringing somebody inside. Now, they're only working for me. Is their capacity optimized? Do we have internally the skills to match the external supplier? Can we do it as well inside as they can do it outside? There are costs to doing things inside the business or outside the business. When the costs are lower to do it inside the business, we'll do it inside. When the costs are lower to do it outside the business, it makes sense to outsource, it makes sense to do it outside.
- 3.53** For a long time in economics, there was no real understanding about why companies grew so big, why they brought everything in-house. What was going on here? Why were these corporations— why didn't we just outsource with everybody else? The marketplace works really, really well. We've got a whole theory about prices and information that we get in the marketplace. Why don't we do everything in the marketplace? Subcontract, everything.
- 3.54** And so an economist by the name of Ronald Coase in the 1930s came up with this idea of transaction costs, and that within the corporation, under some circumstances, the transaction costs are lower, and therefore we bring it in. And when they're not lower, when it's indeed cheaper to do it outside, we do it outside. Simple idea, but nobody had thought of it. Nobody had explained it that way before he actually won a Nobel Prize in Economics for that idea.
- 3.55** As the costs change, you'll see the company move products out and in, functions out and in over time. Apple is a good example. The first Macintoshes were all built in a production facility in California by Apple on an assembly line. But as it became clear that Apple did not really have a competitive advantage in doing that, and as other subcontractors developed the skills to build computers, slowly things got outsourced to those outside manufacturers.
- 3.56** They might be other manufacturers in the United States, or they might be manufacturers outside of the United States. But you can only have manufacturers outside the United States do this if they have the skill to do it, if they can do it well. It's not just about cheap. It's not just about labor costs. It's about quality, and reliability,

and predictability. So there's this internalizing and externalizing that is constantly going on.

- 3.57** For example, sometimes you want to do something and there's just no market out there to do it. I'll give you an example. In the early part of the 20th century, one of the big businesses that were coming out in Chicago, becoming a big deal, was meat packing. The production of meat: the slaughter of animals, production of meat, and then transporting that meat. And because of the railroads, you could actually transport meat. That was— what's the word—harvested, produced, in Chicago, all over the country through the railroad system.
- 3.58** But the reality was that nobody had ever produced and nobody had ever had a thought about producing a refrigerated, railroad car in order to move the meat around. And nobody had the knowledge of how to do that. So it turned out that the meat packer, a particular company called Swift, had the idea of, "We can now grow our market dramatically if we can refrigerate it." They developed refrigerated cars within their own company and then sold them to the railroads to pack with meat to deliver all over the country.
- 3.59** When Standard Oil wanted to move its oil, kerosene, the product of oil refineries, all around the country, all the shippers were moving it in barrels. They would load barrels up onto the train, and the train would take those barrels, and it was incredibly inefficient. It was very time-consuming. It was very difficult to do. And one of the innovations that Standard Oil came up with—because they were in that business because they were thinking about it, even though they were not in the rail car business, they weren't in the business of designing rail cars—but they came up with the idea of a tanker.
- 3.60** Nobody had done that, and nobody had ever done a tanker, right? Where you just pump the oil into it, it's a big container, and it sits right there on the train. It's easy to move, easy to unload. You don't have to unload. One of the advantages Standard Oil had, over all its competitors, is that it had tankers, and they were still using barrels. And it's one of the comparative advantages that it had. So it got into the business of building tankers.
- 3.61** Think about Amazon. Amazon was a bookseller, remember those days? You were too young. But they just sold books, right? And they slowly grew their business, and it's all online. And they wanted to turn it into a marketplace and invite other people to be able to sell stuff on Amazon. And they needed this vast array of servers to be able to manage all the transactions that were going on. It wasn't just enough to have a few computers, like when they started. They just had a few computers running all the transactions.
- 3.62** Now they need massive server farms. So they build these big warehouses with massive servers inside. And then they said, "We've got this. We're really good at building these

server farms. We could easily scale this. Then we could do what? We could sell space in these server farms to other companies." Now, what is that? What do we call that today? That is the cloud.

- 3.63 Amazon invented, created the cloud or made it viable. A lot of people thought of it before, but they made it viable. Not because it was Amazon's business; Amazon's business was selling stuff. But they couldn't find anybody who could build server farms as big as they needed. And then once they got into the business of building server farms themselves, why not build server farms for everybody else?
- 3.64 And Amazon has never really made money selling us stuff. I mean, the profit margins in retail are tiny. They have massive competition. If they raise prices too much, somebody else will create an online service that will beat them, and people have. And of course, there's Walmart, where you can walk in and get your stuff. And Walmart has an online presence as well. So margins on Amazon are tiny.
- 3.65 Where does Amazon make its money? And the first time it started making money is from cloud computing, unrelated directly to actually selling stuff. But there was an opportunity, and suddenly they went into a different business in order to fulfill that opportunity.
- 3.66 So good businessmen are constantly looking for opportunities to expand markets, to get into markets, or to create new markets in products that didn't exist before in which they have a comparative advantage. Amazon had a comparative advantage in building server farms. Now they're in that. Now, Google is also in that business today. Microsoft is also in that business today. There's competition. But it was Amazon that created that business, the cloud business. Those are examples, all the way from meatpacking to oil to Amazon.
- 3.67 So corporations form to create, take advantage of opportunities. They internalize functions when it's cheaper to do so than to outsource them. I mean, Amazon, if at the time—this is the mid 2000s— if at that time there was another company providing cloud services for cheap, Amazon would have outsourced that. It would have never internalized. The only reason they internalized it, because there wasn't such a service, they had to create it. And once they created it, they got an economy of scale and they made it cheap.
- 3.68 One of the reasons we saw a massive acceleration in creation of corporate America, corporations in America, and growth in corporations in America in the late 19th century, is that suddenly economies of scale mattered. Economies of scale means that as you build more and more of the same thing, your prices go down. You get better at it and you achieve economies from size, from the quantity that you do. You can buy stuff in quantity that is cheaper. You can create production lines where you're more efficient. Everything about it, as you grow bigger, your costs become lower.

- 3.69** Part of it is suddenly—if you think about business before then, all business was small business. There was no big business. I mean, the only big businesses might have been like the East India Company, or something like that, that did trade. But in terms of manufacturing, everything was local, everything was small. Ownership and control was the same. Managers owned the companies and they owned the factories, and they were in the factory every day, and everything was compact and available.
- 3.70** But then in the 19th century, you had a couple of revolutions happen, really, that enabled suddenly scale, which required big corporations. One was the discovery of vast quantities of coal and an understanding of how to use that coal to produce energy. So coal was a means by which you could suddenly scale production dramatically. Because the energy, what drives production? Energy. You need to spend energy to produce stuff. And coal is a cheap form of energy.
- 3.71** There was no really source of energy before that. Wood burning, very low efficiency. Windmills, you still go in the Netherlands, you can see those windmills turning around. It's just not very effective. Wind doesn't always blow, and it's not very productive, and it's not very efficient. But coal is incredibly efficient. It might be dirty, but it's incredibly efficient. So suddenly we had a cheap source of energy that could fuel big factories, big production.
- 3.72** Now, that is good, but it's not enough because even if you can produce a lot of stuff, how are you going to get it to where it needs to go? If I'm in, I don't know, if I'm producing steel in New Jersey, how am I going to get that steel to California? And if I'm not going to get that steel to California, what's literally going to happen is there's going to be a steel mill in California and a steel mill in Ohio and a steel mill in every place because getting steel from place to place is so hard. Imagine horses and buggies transporting stuff across the continent. Takes months. It's impossible.
- 3.73** The second revolution is the railroads. Suddenly, we can move stuff. Suddenly, we can move stuff. And these are, as we'll see later, these are incredibly disruptive. Railroads are incredibly disruptive. Think of a farmer who's out there in, I don't know, the middle of Missouri, the middle of Ohio, in the middle of Indiana, somewhere. And they're basically farming. They eat all the crop that they make. Maybe they have a little bit left over and they can take the buggy and ride it into town and sell a little bit of it. And all the products that they consume, they actually make themselves because everything's so far away. How are they going to get goods to themselves?
- 3.74** Imagine that farmer suddenly gets a rail line delivered right to his backyard. And suddenly he can export out of his farm large quantities, and bring in large quantities of things. The whole dynamic of his life changes. Everything about how he lives and how he produces is going to change. This period in the 19th century is incredible. Incredible changes in people's lifestyle, changes in wealth because they become a lot richer because they can produce a lot more at scale. But it's also very disruptive.

- 3.75** So railroads, suddenly you can move stuff anywhere. I can build one factory in one place and deliver to the entire United States. I can create a business that is not just local and small. I can create a business that has the marketplace of all of America. Just think then, in a sense, the next step of that is Amazon and its delivery system. But it's the same principle, it's just better technology. And that's the evolution of business. Technology, innovation, and scale.
- 3.76** At some point, we globalize so we can sell not just in America, but for that, I have to have real cheap shipping costs. Think about containers. Shipping containers, how they changed the world. Shipping containers changed the world in the 1950s. They made globalization possible because suddenly it's super cheap, really super cheap, to ship stuff anywhere in the world. Before that, it was too expensive. You only made for your own local market.
- 3.77** And something simple, like a shipping container, how did people ship before containers? Like in boxes and different crates and different things, different sizes. Loading a ship was very difficult and very expensive and very time consuming. Loading up the train to get it to the ship was very difficult, very time consuming. But now everything's standardized in a shipping container. You take it from the truck, to the train, to the ship, it gets off the ship, into a truck. It's all modular. And it changes the world.
- 3.78** That little invention called the shipping container changes the world. Think about that thousands and thousands of times over the last 150 years. Little changes like that, how they shape our lives, how they shape industry. So coal and railroads. Small business is out because now I have a national market. I can get my stuff any way I want to.
- 3.79** But that also creates complexity. I have to figure out where to deliver, what to where. I have to have high-quality managers thinking about these things, planning, strategizing. Can't just do it off the cuff. The railroads themselves have an incredible complexity. They're running rail against each other, but also in cooperation because the train has to go from my railroad onto your railroad, and there have to be contracts to regulate all of that.
- 3.80** I need a huge amount of capital to build these railroads, which means financial markets, which means bankers have to provide me with a lot of capital. I have to go sell stock on the stock market. So again, corporate entities, large corporate entities, some of the most dynamic corporate entities, some of the most dynamic competition during this period was among railroads. And there were lots of railroads.
- 3.81** Because it was so capital-intensive and you needed expertise over a particular region, every region had its [own] and railroads would compete in the same region. You should have railroads all over the country, lots of them, lots of complexity, lots of business acumen required, knowledge required, extensive geography, high specialization.

- 3.82** Really the first episodes of professional managers, of managers who manage. That's what they do in life, get an MBA—there weren't MBAs back then—but actually strategize and think about how to run a business. The first real examples of that are in the railroads. And these are complex businesses, hard to manage. And again, railroads play a huge role in how this country's business evolved.
- 3.83** For the first time, you get separation of ownership and control as well, because the people running the railroads are not going to be the same people providing the capital. People are providing capital, maybe in New York, Pennsylvania, but the railroad's all over the country. And there's too many railroads.
- 3.84** One of the reasons there's too many railroads is government gets into the business. It wants to incentivize railroads. It gives land grants, and it spurs people to start railroads. You get private capital flowing in, you get government flowing in, you get a lot of conflict, and there's basically overproduction of railroad in the late 19th century, which creates its own problems because it's very hard to make a profit when there's so many railroads running, again, caused by the fact that the government pushed railroads and provided them with huge incentives.
- 3.85** And then if you add to the railroad one other element—so you've got coal, you've got the railroads— and now what you're missing is communication. You've got transportation, you've got scale manufacturing. What you're missing is communication. And what fills that in is the telegraph, another innovation.
- 3.86** And the railroads and the telegraph work together because what's the best way to get the rights to be able to draw telegraph cables? The best way to do it is just to follow the railroad. Just put the cable—you see this in the Westerns. If you watch old Westerns, they're building the railroads and they're putting up poles for the telegraph, partially because the railroad wants to be able to communicate with headquarters over the telegraph, and partially because for the telegraph, this is the cheapest way to get point A to point B. Rather than have to negotiate land rights with everybody, they negotiate with the railroad once and they get the whole track.
- 3.87** But suddenly we can communicate. Again, it's hard to believe, but just 150 years ago, there was no way to get from East Coast to West Coast. What was it the Wells Fargo, Pony Express? I mean, weeks, if you wanted to send a letter. Suddenly with a telegraph, you get instantaneous communication. Now, again, today we take completely for granted the fact that we can communicate instantaneously with anybody in the world by video at a marginal cost of zero. I mean, that's unbelievable. It's an unbelievable, beautiful and amazing thing. But that is a very, very recent achievement. And that is a product of the innovation and technology in businesses that we have around us that we love. Love to hate.
- 3.88** So if you add railroads, and telegraph, and coal, what you get is the incentive for businesses to gain scale, to centralize operations, centralize production, and then ship

everything out from a centralized location. And you get a marketplace that drives particular regions to specialize in particular products because they're huge networking effects, huge information effects, huge benefits to local knowledge.

- 3.89** So for example, automobile industry clusters. There are dozens of automobile companies in the early 20th century, and they all cluster around Detroit. And there's a huge incentive. If you're going to be a young entrepreneur in the automobile industry, where do you go? Detroit. Same reason today we have all tech clustering around Silicon Valley. Massive benefits of being in Silicon Valley. Who you know, where the capital is, where the startups are, other engineers, other entrepreneurs, other founders. It's all there. You don't have to go find them, running around finding them. They're all in one place.
- 3.90** Same thing happened with Detroit. Same thing also. We'll see afterwards why automobile manufacturing splintered and went away from Detroit because it did. A lot of American auto manufacturing today is done where? In the Southeast, not in Detroit. Why is that? What do we have in the Southeast that you don't have in Detroit? Or the other way around, what do you have in Detroit that you don't have in the Southeast? Unions.
- 3.91** So in South Carolina, in Alabama, in Georgia, you can build a car without unionized labor. In Detroit, in Michigan, you cannot. Cost is huge. It turns out that a lot of the jobs lost in the Rust Belt, that people bemoan that China took from us, a lot of those jobs didn't go to China. They went to Alabama, and South Carolina, and Georgia, which boomed as Detroit faded.
- 3.92** During this early period, when scale has been created, managers are becoming specialists. Hierarchy is managerial, hierarchy is being basically created, the whole idea of middle managers, upper managers. The whole structure of American corporations, as we view them today, was really born during this period, the late 19th century, as a consequence of all this activity, a consequence of all this entrepreneurship.
- 3.93** And you started getting different structures of businesses. So in the beginning of the 20th century, you got a lot of vertical integration. A business making something would buy the source of the raw material. It was one company, raw material mining, for example. Then you would produce the product, and then you could also buy the distribution, and you would have the whole path.
- 3.94** That's a vertically integrated company, is a company that integrates across all the different functions that lead to the product and get the product to consumer. That's one form of business that evolved in the 20th century. Raw material, mass distribution, all coordinated by specialist managers who would coordinate it internally versus other companies that buy raw material from a bunch of producers of raw material and could choose and pick between suppliers.

- 3.95** Bringing it in-house has advantages, but it has costs. What's the advantage of having multiple sources? Somebody might invent a better way to mine and might be able to do it cheaper, and then you can benefit from that cheapness versus if you have it in-house, if you don't have the knowledge how to make it cheap, you're stuck with the higher cost. But the benefit of having it internally is you always have a dependent source. It's not going to go away.
- 3.96** Other companies diversified horizontally, into products that were related to the product that you're doing— like the cloud computing—but were not directly the product. I'll give you another product from the meatpacking industry, which is interesting: the first manufacturers of soap. Soap was a byproduct of animals, and they made it from the leftovers when they produced meat. Their primary business was meat. But then there was stuff left over that could be made into soap. They made it into soap. Suddenly, the meat packers were in the soap business. That would be horizontal growth.
- 3.97** And a lot of the biggest companies in the United States in the early part of the 20th century also created their own internal R&D facilities, their own labs, their own research labs. And a lot of the horizontal growth was a product of the inventions that came out of those labs; so new products that they thought were related to their core business and that they could grow that way.
- 3.98** At some point, DuPont, I don't know if you've heard of DuPont, a big chemical company, used to be a dominant company in the United States, created what's called the M-form. M, the letter M. The idea is you have corporate managers up here, and then you have these divisions, and each division has a lot of autonomy, runs its own deal. Managers up here maybe only do strategy.
- 3.99** During history, the relationship between divisions doing different products and the people up on top, changes. Sometimes they were only doing strategy, sometimes they also do your accounting and everything else. Sometimes it's at the division level. And a lot of that is determined, as we'll see, by antitrust law. What is in a division and what is at the headquarters, and whether you want divisions and headquarters or whether you want to integrate it all into one entity. Again, another way in which business evolved and changed.
- 3.100** All of this is affected, of course, by external factors: by government, by macroeconomic factors, by the dynamics of the market. For example, at the end of the 19th century, because all markets were new, there was very little outside of you you could subcontract, you could go out there and get. Almost all companies, almost all companies, integrated vertically and horizontally. It was the way in which they grew. It was the way to become profitable. They took advantage of the reduced transportation costs, the reduced energy costs, the reduced communication costs.

- 3.101** At the end of the 20th century, 100 years later, it's the exact opposite. Companies basically shrink, focus, subcontract—a huge amount of what they do. As we said, Apple doesn't produce anything anymore. It subcontracts everything. Maybe the original for that was Dell.
- 3.102** You know how Dell got started? He started in his dorm room. Dell computers. You know Dell computers? Dell got started in his dorm room at the University of Texas in Austin. He would basically buy components of computers and assemble them and sell them to his buddies and friends and people he knew. He didn't make anything. He assembled stuff from different suppliers, and he turned that into a business.
- 3.103** Today, if not the largest computer manufacturer in the world, maybe one of the biggest. And yet they don't produce anything, in a sense. You go on the Dell website and you pick the processor, you pick the monitor, you pick the—I don't know—the size of the hard drive, you pick all these things. They have outsourced that, but they're so good at it today that when you pick them, it's not even going through a Dell website. It's literally going directly to the supplier. The supplier knows you've ordered it. It's already shipped to Dell. As soon as you click send, it's shipped to Dell. All Dell has to do is assemble it all, and there you have your computer. And today, they don't have to deliver it themselves. They don't have to own the railroad that delivers it. They just do FedEx or UPS.
- 3.104** That's a recent innovation, FedEx and UPS. Up until not that long ago, the only way to ship stuff was either by the railroads, and stuff like that, or by post office. FedEx and UPS have changed the way we live our lives. We just take it for granted that we can get Amazon Prime, and the next day it just shows up.
- 3.105** Amazon is not happy with UPS and FedEx anymore. So what are they doing? They're creating their own delivery system: Amazon delivery system, Amazon trucks that deliver stuff to your home. That's that vertical integration. They're taking over the distribution because they don't—I don't know—they can't negotiate a good enough deal with FedEx and UPS. They're taking on the business themselves. You see how this constantly is changing and evolving.
- 3.106** So we go through periods where there's massive integration and then disintegration, integration and disintegration. And it's depending on the economic and political forces at play at any given point in time.
- 3.107** In the early part of the 20th century—in the late part of the 19th century, early part of the 20th century—a huge move to grow big businesses, massive merger booms, massive consolidation. This is the period of the robber baron, the Rockefellers and the Carnegie and Mellon. They create these big businesses because it makes sense at that point.
- 3.108** And then you have a period between 1914 and basically 1973, where there's somewhat stagnation among these corporations. They don't quite grow in the same way. They

diversify. They go into completely unrelated businesses. And a lot of that is driven by the dynamics of the Great Depression and the two World Wars and everything else going on during that period.

- 3.109** And then you go from 1973 to today, you get, as I said, you get these companies broken up, you get people much more focused, and you get much more of this subcontracting, much more of relying on markets and relying on internal production.
- 3.110** So the market constantly, because of innovation, because of changes, because of new companies, because of new possibilities, is constantly changing the transaction costs involved in how you do business. And the consequence of that is the structure of corporations, the structure of businesses is constantly evolving. If you're stagnant in business, you die. I think that's a general principle in life. If you stagnate, you die. Maybe the death is very slow. Some people die very, very slowly. But stagnation equals death.
- 3.111** To grow, to succeed, to profit, to live, you constantly need to be willing and able to change. Constantly be willing to look around, see new opportunities and new threats, constantly be willing to adjust and adapt to what is going on around you. And that's the market. And the market will always change because as long as we innovate, as long as there's innovation, as long as there's progress, there will be changes. Nobody can rest on their success of the past.
- 3.112** All right, so tomorrow, next class, we will talk about the governmental factors that impact the evolution of corporations. As I've said, we'll talk a little bit about the macroeconomics, Great Depression, things like that. We'll talk about antitrust, regulations, and industrial planning. Then we'll take a special look, and make a case study of tech. I think it's interesting and it's current. We'll look at the tech industry basically from the 1970s till today and how it evolved and how it matured. Then we'll end with a look at the future of the corporation, the future of business, and what it looks like today and where we're heading. Thank you.

Lecture 4: Antitrust

At a glance — Antitrust and its rationale.

- 4.1 All right, so in our previous classes, we have talked about the fact that the corporation is not a state creation. It is actually a market creation, a creation of entrepreneurs contracting with providers of capital, contracting with employees, creating a new legal entity called the corporation. We've talked about the fact that this market creation, its goal is basically to serve the interests of its owners—the entrepreneur, to begin with; later, those investors who become owners, shareholders, and who are basically determining the course of the corporation.
- 4.2 It is their responsibility to determine the direction of the corporation. At the end of the day, given dispersed ownership, the assumption is, owners want a financial return. The goal of the corporation is to provide that return. It's to maximize economic profits. It's not to serve some kind of social responsible goal, to serve the public interest, or to serve the state in any way.
- 4.3 And that, ultimately, corporations take many different forms. They can be structured in a variety of different ways. They integrate vertically and horizontally. They could be lean, they could be expansive. And this changes, is constantly changing, depending on market forces, really determined by what is going on in the marketplace, competition, innovation, new ideas, new discoveries about not just products, but about organizational forms.
- 4.4 So, the corporation in its ideal form is determined by a market, by interactions within the market. But we don't live in that ideal form. We don't live in an ideal place. We don't live in a place where the markets are allowed to function free of controls and regulations. A big part of what determines the fate of business, of what determines the structure of business, of what determines who wins and who loses, who survives and who fails, and the behavior in the meantime is government.
- 4.5 Government through antitrust laws, through regulations, through subsidies, industrial planning, a variety of different forms. Ultimately, government is going to shape the way corporations evolve. And certainly in American history, government has had a big role in shaping the evolution of corporations. In this class, we're going to talk about one major way in which government has done this, and that is through antitrust laws.
- 4.6 So, let's think about what antitrust laws are there to do. They're supposed to protect us or restrain what are called monopolies. But what is a monopoly? There is a big dispute about the very nature of monopoly in the economic and the political literature. Traditionally, and many economists still hold to this view, monopolies are creatures of the state. Monopolies are something you are granted by the state to protect you from competition. A monopoly is a position where your competitors are barred from competing with you.

- 4.7 Examples of this are plentiful from mid-19th century and before. The East India Company gets a monopoly, a charter from the king, to trade with the East Indies, and it's the only one allowed to trade with the East Indies. If you compete, if you buy a ship and sail to the East Indies and bring product back to England in competition with the East India Company, you will go to jail. You're in violation of the law. So competition is indeed barred.
- 4.8 Now, it doesn't mean it doesn't exist. Whenever you bar something in a marketplace, what happens? You get black markets, right? You try to bar alcohol and you get a mafia. You try to bar drugs, you get drug cartels. You try to bar anything, it still happens. And indeed, there was competition at the East India Company, but it was all illegal, and therefore expensive, and therefore dangerous and very restricted. And that is true throughout.
- 4.9 When I gave the example in one of the previous classes of one of the first corporations in Massachusetts was a monopoly over building a bridge over the Charles River. Nobody else was allowed to build a bridge over the Charles River. So you could get all the tolls and you could get all the benefits. And competition was barred. That ultimately went away, and competition arose.
- 4.10 But that is the traditional view of what monopoly is. It is a government grant. It is a government favor. It is the government use of force in order to enhance your economics, your ability to do the project. And so monopoly, by this idea, cannot really arise out of the marketplace. The idea here is that this is always competition. There are always competitive forces in a real marketplace. If we evaluate a marketplace, if there are no legal barriers to entry, then there's always somebody who could compete with you. And even if they're not, the fact that they could would change your behavior.
- 4.11 But that, of course, is not the common view of monopolies. The common view of monopolies, and the common view of monopolies that has shaped really the last 140 years of American business, is that they do arise out of markets, that they can arise, and that they have a substantial impact on the welfare of people.
- 4.12 Now, there's no coherent view out there of what that exactly means. If you talk to a variety of different antitrust experts, and we will see this as manifest in the antitrust laws, they will all have a different version of what kind of behavior constitutes monopolies, and they all have a different version of how they come about and what needs to be done about it. There is no coherent view of what it is and what should be stopped, what we need to do in order to prevent these monopolies from arising.
- 4.13 Much of the antitrust law arises not out of any kind of real economic analysis that shows that there is a real problem. Much of the economic, much of the antitrust law comes about out of a sentiment of voters, populist demands, out of compromises by politicians that are trying to deal with a variety of issues, but mainly out of a general fear and distrust by the public of large businesses, sometimes based on the fact that, I

don't know if you remember, about 20 years ago, when Walmart was really expanding and going into small towns and building out this massive network of these superstores, there was a huge backlash.

- 4.14 People were very upset. Even though it lowered their costs and increased variety, a lot of small businesses went out of business because they couldn't compete with Walmart. And this upset the local community, not because of an economic harm to them, but because of the fact that some of their neighbors were losing their jobs, and this feel of a small business was going away. A lot of that kind of sentiment, a lot of that fear, is what drove much of the antitrust.
- 4.15 Another perspective in terms of antitrust came from the progressive movement. The progressive movement is an intellectual movement in the late 19th century that really dominated American intellectual life and ultimately led us to the presidencies of Teddy Roosevelt, Woodrow Wilson, and ultimately to FDR. Sorry. Ultimately to, yeah, Franklin Delano Roosevelt, FDR. So, the progressive movement actually liked big business, and they wanted to use antitrust in a perverse way to protect big business from small businesses.
- 4.16 How did small businesses compete often with big business? Well, they often formed cartels, or they often formed contractual agreements between them to coordinate, so that they could deal with a big business. And that was something the progressives didn't want. They liked big business because they loved the idea of, in essence, central planning. They loved the idea of scientific management. They loved the idea of bigness.
- 4.17 They actually wanted to input that into government. They wanted government, ultimately, to be managing them all. And it's much easier, if you're a central planner, it's much easier to manage 20 big businesses than 2,000 small ones. So the progressives had an incentive to see some big businesses come together. So all kinds of ideas, all kinds of conflicts are coming together here.
- 4.18 But in the late 19th century, you've got a number of forces at play that lead us ultimately to the Sherman Antitrust Act of 1890, the first real antitrust legislation that is passed, and the first attempt by government, in a systematic way or an attempt at a systematic way, to rein in big businesses. The big industry during that period of time and where you see the most conflicts are the railroads.
- 4.19 There are many railroads in the United States, as we talked about in a previous class, many more than probably the market would have determined because the government had subsidized and had given land grants to create more and more and more railroads. And there was a lot of what you would call overinvestment in the field because of that. There was vast competition. Railroads often in the same region, there were multiple railroads, and then between regions, there were railroads.
- 4.20 And then there was the issue of how do you get a train from one system to another system, so they could travel long distances. So there was a vast array of railroads with a

vast array of different kind of contracts. And generally in the railroad business, a railroad business is a very high-capital business. You have to invest a lot of money upfront, build the rails. And then you're trying to make a profit to pay for that after the fact.

- 4.21 But now you're trying to make a profit off of that in a very competitive environment, where profits are being driven down. Remember, competition drives profits down. If I try to hold prices up, my competitors can lower them, forcing me to lower them in order to sustain the business. So, generally, competition lowers profits. So we've got a railroad system that is—these railroads are maybe breaking even. They're certainly not returning capital that was invested in building the railroads themselves.
- 4.22 And many of the—and the railroads deal with this in two ways. One is, they start forming cartels. They start organizing around regions and saying, "Let's keep the prices high. Let's unite to keep the prices high." Now, people don't like cartels because suddenly prices are going up, particularly farmers who are suddenly discovering their ability to export goods outside of their little region into the rest of the United States. They don't like that prices have gone up, so they're complaining.
- 4.23 And other shippers, everybody who's got to ship goods doesn't like these cartel arrangements, so they are complaining to Congress. In addition to the existence, to the fact that these cartels raise prices, the reality is that cartels never last. This is one of the amazing things about cartels and why people's hysteria about cartels is so interesting. Cartels almost never last because somebody always cheats. There's too much of an incentive.
- 4.24 If we raise the price to 10, it used to be 6. Well, when nobody's looking, I'll lower my price to 9 and see what happens. And then everybody catches on, and it's 8, and it goes back to 6. So, cartels don't last. I mean, OPEC, which is the most well-known cartel today in the world, is constantly having to fight the cheating. It's constantly struggling with that. And it cannot actually, for very long, maintain the price of oil. It can for short periods of time. But over the long run, cartels break down.
- 4.25 So the second way in which railroads dealt with this is what economists call price discrimination. It's the idea that we won't charge everybody the same price. We'll figure out ways to charge high-value customers lower prices, and low-value customers higher prices. For example, we're going to charge on where there's real competition. On long freights across the country, we will charge relatively low prices.
- 4.26 And on short, where I control it, like from this farmer to this hub, nobody else competes with me. The farmer, to get competition, would have to go to, quite a while, to another railroad. I'll charge high prices. Sometimes it would cost more to ship across state than across country. So that's one form of price discrimination, is different distances, different lines, where you face more competition, less competition, no uniform pricing.

- 4.27 Another form of price competition, which today, to some extent, we take for granted, but in those days was kind of revolutionary, and people got very upset about it, was charging less for bulk shippers, for people who shipped a lot. Standard Oil, Rockefeller's Standard Oil, got into a lot of trouble because they negotiated deals with the railroads to dramatically lower prices on shipping oil.
- 4.28 Oil in those days was not shipped in pipelines. That's a later development. It was shipped primarily by rail. And Rockefeller negotiated with the train companies to lower those rates. And when that became public, people were very upset. His competitors, of course, were very upset. Now, there are a number of reasons why a train company would want to give Rockefeller a discount, a volume, and guarantee a volume, because he was so big and he could guarantee a long-term contract on long-term delivery.
- 4.29 But also, Rockefeller was very efficient in delivering the oil to the train companies. The train company actually expended less money on the logistics of actually loading up the cars and shipping them. In the time of barrels, Rockefeller actually went out and built his own barrels, and the barrels were better than the competitor barrels, and they didn't leak as much, and they were just better for the railroads.
- 4.30 But ultimately, what Rockefeller innovated was the rail car. Again, we talked about this, which was a—you could fill it up with oil, a container. And that reduced the cost of shipping of oil dramatically. While his competitors were still using barrels, he was shipping whole containers. So, you've got this struggle. People are not happy with it. Competitors are complaining. Everybody's seemingly upset.
- 4.31 Remember, too, that this is the early days of what some people call the second industrial revolution, the big business, big products, railroads, steel, metal, oil, coal. And so this is a time where there's a lot of change. People are not accustomed to this change. They still don't know what this new economic thing is doing and how it's working. And there's a lot of frustration and a lot of challenges.
- 4.32 As a consequence of this, even before you get the antitrust laws, Congress passes something called the Interstate Commerce, or creates the Interstate Commerce Commission, which is in charge now, and will be in charge for the rest of the 19th century and the 20th century, of the railroads. So it's a regulatory body that now is going to determine rates, is going to make sure cartels don't form, is going to heavily regulate the railroad industry.
- 4.33 You could argue, and many have argued, and I agree with this, that, to a large extent, the United States has a fairly primitive railroad system, particularly when it comes to passenger railroad in the U.S. today, as compared to some other countries. And railroads have never really taken off the way they could have, to a large extent because they're being so regulated and so controlled, and their business model has been so constrained by the Interstate Commerce Commission.

- 4.34 In many respects, this is the first big significant regulation, aside from banking, that the U.S. engaged in. And even in banking, banking was always regulated at the state level. This is the first federal big-time regulation that passes. This is 1887. So it's just three years before the Sherman Antitrust Act. And this was all leading towards the Antitrust Act.
- 4.35 So this is going on in the railroads. But in the meantime, big businesses, businesses are consolidating. As we described last time, you've got big businesses can now consolidate manufacturing, and they ship. Before, you had to produce locally because you couldn't get it to customers otherwise. Now with the railroads, you can produce centrally and ship out to different constituencies. And indeed, that is what's happening.
- 4.36 Businesses are consolidating, building these, and a lot of small businesses are worried, a lot of farmers are worried. And a new term comes out, which is called concentration of power. Nobody exactly knows what this means and how to define it, but now, monopoly is associated with concentration of power, and monopoly is attributed to all these businesses that are growing and succeeding. And the smaller guys are complaining about monopolies and concentration of power.
- 4.37 Now, this has also appeared where a lot of other stuff is going on, in terms of economics and in terms of politics. This is a period of rising populism, a little reminiscent of today, but the populist agenda is a little different. The populist agenda is primarily focused on more inflation. During the late 1800s, what you get is declining prices. What you actually get is deflation.
- 4.38 And the reason you get declining prices is because productivity is growing at such a fast rate that the money supply is not matching it. The money supply is constrained by gold in these days. We're in a gold standard. And the money supply cannot grow enough to match productivity. And the consequence, if you have less money, chasing more goods, then the prices of those goods will have to decline, and do decline. So you get slow—this is market deflation versus other forms of deflation we'll get to when we talk about the Great Depression.
- 4.39 This is, by all economic trends, generally a positive thing. But if you're a farmer and the price in which you're selling your goods declines, an economist will tell you, "Don't worry, your dollars are worth more. You can actually buy more with it, so you're actually better off." But it's very hard for people to actually internalize that and completely accept that.
- 4.40 Sometimes, objectively, prices don't move at the same time. Some prices might decline sooner than other prices. For example, you might sell your product at declining prices, and your inputs, what you use in order to produce the product, might be also declining, but at a slower rate, at least initially. So, again, relative prices don't all move together. It creates—and it's new—it creates angst. It creates challenges for people, people upset.

- 4.41 And so, what the populace is demanding at this point is inflation. There's the famous William—was it William something Bryan, who, "the cross of gold. We're all being crucified on a cross of gold." This is the idea of the gold standard is killing us. The gold standard is bad for us. So all we want is, we want silver. We want to add another monetary unit to add money supply. And the idea was to add silver to the monetary supply to create inflation, to increase the amount of money in the economy.
- 4.42 So that's going on. There are certain interests that want the gold standard, certain interests that want to dilute the gold standard with silver, and they are fighting. At the same time, there's a big battle going on around tariffs. This is a period in which, in the past, pre-Civil War, tariffs had been basically pretty low and it'd be used for revenue. But in the post-Civil War era, tariffs are starting to be used as a means to protect industry, as a means of industrial planning.
- 4.43 And this is a period where there's a lot of fighting about tariffs. If you're an exporter, like most of our farmers who are sending grain to Europe and other places, you don't want tariffs. You don't want a trade war. You don't want the price of your goods to go up. Of course, if you're a farmer and you want inflation, the value of the dollar goes down, the value of other currencies goes up, they can buy more stuff from you. So you want those two things, right? You don't want tariffs, you want inflation.
- 4.44 On the other hand, there are plenty of interests, primarily, as it turns out in the northeast United States, which is the industrial hot land at that point of the U.S., who want tariffs to protect certain industries and want to stay in gold. They want a strong dollar. So you've got these conflicting interests. Most of the debate at the time is about these issues. It's about tariffs and monetary policy. Imagine a populist movement that actually talks about monetary policy. I don't think that's consistent with the way it's called today.
- 4.45 One way in which—so one way just to deflect a lot of this discussion was, in a sense, to throw a bone to some of these interests. And it turns out that the Sherman Act was to, a large extent, throwing a bone to some of these interests. It was to appease these various interests as the U.S. raised tariffs. It made farmers upset. But then it told farmers, "Don't worry, we're going to deal with the cartels. We're going to deal with the big business, and we're going to constrain them through this antitrust law."
- 4.46 So it was not the focus of legislation at the time. Most of the focus on legislation at the time, again, was, should we introduce silver? And they do, but they dilute it. They don't really increase the money supply significantly until the early part of the 20th century, when there's a huge new gold find, and then you get a little bit of inflation. But they do increase tariffs.
- 4.47 I mean, Trump talks often about the McKinley economy, where these are the McKinley tariffs that are passed in the late 19th century. In those days, just like today, Republicans were pro-tariffs, Democrats were anti-tariffs, and the administration of

McKinley, a Republican, raised tariffs significantly. So the Sherman Act was passed in 1890 to try to appease all these groups and try to get everything settled as part of this package with tariffs and with an insignificant introduction, reintroduction nevertheless, of silver.

- 4.48 And this is where the act—so the act forbids contracts, combinations, conspiracies in restraint of trade or commerce, as well as, the monopolization or attempt to monopolize commerce. And of course, part of the big debate that follows is, what does it mean to restrain trade? What does it mean to monopolize? What do these terms actually mean? The Sherman Act does not define them, and does not define them clearly at all.
- 4.49 The phrase "restraint of trade" comes from common law. It doesn't really refer to monopolies in common law. It refers to different practices that have to do with, primarily, cartels. But cartels are often a way in which small businesses deal with a big business. They're not a way in which a big business tries to monopolize. So again, there's a lot of tension between the different clauses here and in terms of the actual application.
- 4.50 And the Sherman Act gives no real clarity. Indeed, it really becomes an empty vessel in which, whatever the administration of the time wants to fill it with, it fills it with. Different presidents will appoint different people to the justice department to prosecute antitrust with different views on what it actually means. And therefore, what this does is create massive uncertainty for business. They don't know what it means. They don't know when they're violating antitrust and when they're not violating antitrust. And you might be violating antitrust in the 1910s in one way, but then in the 1930s, that was OK, and now something else is not OK.
- 4.51 And indeed, you saw that just recently with the appointment of Lina Khan in the Biden administration. You saw a complete shift of what antitrust means and the interpretation of antitrust, and things that were completely legal and OK from 1980 to 2000, suddenly from 2000 to 2024, not OK. And it seems, at least for now, that the Trump administration is continuing the line that Lina Khan started. So we're now in a new regime of antitrust. We'll talk about that in a minute.
- 4.52 So, Sherman Act becomes a vessel. The Supreme Court originally rules that manufacturing dominance is not a, in manufacturing, it's not trade or commerce. The Sherman Act doesn't apply to it. Later on, it will reverse that. But originally, manufacturing, you could have total domination of a particular manufacturing field. It's not antitrust because it's not commerce and it's not trade; it's manufacturing. So there's a lot of quibbling over these details.
- 4.53 So, what does it actually preclude? Really, again, as we said, cartels, but it allowed the creation of dominant firms. So you could actually buy up a lot of your competitors and create a firm that own 50 percent or up to 90 percent of an industry, and that was not

viewed as a problem because it wasn't created as a result of a cartel. Now, as a result of these contracting that creates a cartel.

- 4.54 Now, what incentive does that create? It creates an incentive for competitors to merge, rather than sign coordination agreements, cartel agreements. And what you actually got was a massive merger boom. The late 19th century into the early 20th century is a huge merger boom where businesses are consolidating, creating massive consolidated businesses, like U.S. Steel, which becomes the largest steel manufacturer in the United States and dominates the industry.
- 4.55 American Tobacco consolidates the entire tobacco industry and, at some point, controls 90 percent of all tobacco manufacturing, all as a response to the Sherman Act. Because before, tobacco was very competitive. There were all these cartel deals, and those were now deemed illegal, and you couldn't coordinate prices, and you couldn't coordinate with your suppliers; you couldn't coordinate with your subcontractors. So the best way to deal with this is to buy your suppliers, buy your subcontractors, and buy your competitors, and create these massive firms.
- 4.56 You get this massive consolidation. Now, at some point, people realized this isn't what was intended. This isn't right. And in 1904, the Supreme Court expands the definition of the Sherman Act by fiat. Really, it's not in the law, and starts restraining these massive mergers and massive consolidation. But this is an issue that continues, and ultimately has to be dealt with through the legislation and through, we'll get to that, the Clayton Act.
- 4.57 In the early 1900s, as I said, there was a discovery of gold, and now there's a little bit of inflation. This rising prices is now blamed on whom? Elizabeth Wong did not invent this idea that rising prices are blamed on big business, squeezing more profits out of all of us. They are causing inflation by raising prices. This goes back to the early part of the 19th century. Inflation is small, but big business was blamed for it. So again, populist anger at big businesses is only growing. The Sherman Act does not make it go away.
- 4.58 And indeed, again, the progressives who are now gaining power, gaining dominance. The first progressive president is Teddy Roosevelt, and then you get Woodrow Wilson. They are now the dominant intellectuals and their influence from politics and influencing policy. They want control over business, and antitrust is one way for them to gain that control over business. They want to run the economy.
- 4.59 And again, there are these conflicting ideas. Teddy Roosevelt famously said, "I can tell." I mean, he sounds a little bit like Trump. "I can tell. I'm me. I can tell which monopolies are good and which monopolies are bad. Me, I will decide who to go after and who not to go after." So it was very much centered around "he would do it." Now, he didn't have the tools in those days to actually go after, and he tried to create a commission that would determine which trusts or monopolies were OK, and which trusts or monopolies were not. Luckily, he never got to pass legislation.

- 4.60 But it's very much a period in which the intellectuals and some of our politicians really are contemplating this central-planned, central-controlled economy, that that is some kind of ideal. The progressives remember, come, I mean, their primary intellectual influence is Germany. It's German Romanticism. They're influenced very much by, in those days, the German welfare state.
- 4.61 They're also the first people to actually propose a welfare state in the modeling. Germany is also the first country to institute public education on scale. The progressives pick that up, and we start getting, in the late 19th century, in the U.S., public education on scale. So the progressives are having a huge impact on shaping the American politics and the American economy at this point.
- 4.62 Now, one of the major figures during this period who helped shape antitrust regulation and controls is Louis Brandeis. Louis Brandeis, who then advises Wilson, President Wilson, and then becomes a Supreme Court judge, writes a lot about antitrust and about monopolies. And his perspective is that bigness is bad. He hates bigness. Again, you can see the different conflicting forces.
- 4.63 He believes that America, in a sense, morally and spiritually, is a land of small business. That there is virtue in the idea of a lot of people owning small businesses and working themselves and controlling their destiny through small business. He believed that it is essential for the morals of the nation to have lots of small businesses. And he thought that bigness both destroys morality, but also bigness would ultimately destroy democracy. And so, he was a force at the same time pushing to break up companies, to make them smaller, to force them to divest and to shrink.
- 4.64 So again, you've got multiple different forces going on, different views, different ideas on what should be done. Out of all this, during Wilson's administration, they figured that the Sherman Antitrust Act is just not robust enough. It's not clear enough to be able to achieve their goals, particularly Brandeis' goal of breaking up companies, but also this idea of having more control over the economy.
- 4.65 So, in 1914, Congress passes what's called the Clayton Act, which is the second big antitrust law that shapes our antitrust to this day. What they really want in the Clayton Act, what Brandeis wants, what Wilson wants, what a lot of people want, is a list of all the things you're not allowed to do. They want it to be clear. They want it to be articulate. They want it to be clear to businesses what they can and cannot do, and what the penalties are going to be.
- 4.66 But the reality is, they can't do that. By its very nature, antitrust is a non-objective law. There's no one criteria by which one can measure. What does a monopoly mean? What does restraint of trade mean? None of these concepts actually mean anything. And indeed, different people sitting down to try to do this legislation come up with different things that they don't want business to do that contradict each other. And they literally cannot come up with a set.

- 4.67 I mean, Ayn Rand famously said, "Antitrust is a kind of law that basically makes all business activity illegal." If you charge a high price on your product, higher than the market, supposedly, then how do you get away with that? You must be a monopoly, so we can go after you. The only way you can do that is you have some barriers to entry, some constraints.
- 4.68 Think of Apple today, charging a higher price on their iPhone than all its competitors. It's just a question of time and a question of the right attitude by the right person at one of these regulatory agencies before the government goes after Apple for charging a higher price on the iPhone than an equivalent Samsung. It must have monopoly power. If you charge lower than everybody else, if you charge low prices, then you're trying to drive your competitors out of business and trying to consolidate power around yourself.
- 4.69 This is clearly an anti-competitive practice. Brandeis certainly would have said, "This is a way to destroy your small competitors. You can afford to lower prices because you're big. Then later on, once you establish, once you get rid of all your little competitors, you'll raise prices and return the profit to you." So that's anti-competitive. So low prices are no good. High prices are no good, low prices are no good.
- 4.70 And then, of course, in the ultimate irony, but truth to the whole thing, if you charge exactly the same price as all your competitors, well, then where's the competition? You must be colluding. And that's collusion. And therefore, that's a violation of the idea of antitrust. But this is the point. Antitrust is completely non-objective. They can't actually articulate, "This is right. This is wrong. This is what we approve. This is what we don't approve," because everything is in contradiction. Do we want low prices? No. Do we want hyper? No. Do we want? No. What do we want? Unclear.
- 4.71 Brandeis wants lots of little companies. Others want a few big companies who can compete with one another. Who's right? Is the standard, and this is a debate happening right now, is the standard economic efficiency by some economic model? Do we have an economic model of economic efficiency? And we want to structure business in a way as to make it economically efficient? That could be one interpretation, and that's the interpretation that dominates antitrust law, probably from the late Great Depression through World War II, onto, certainly, in the 1980s and 1990s.
- 4.72 But even there, which economic model are you using? The economic model that the people argued for economic efficiency from 1939 to 1980 were using was a very Keynesian model. But the economic model that people are arguing for economic efficiency from 1980 to Lina Khan was a more Chicago, free market-oriented model. What economic model determines whether an economy is efficient or inefficient?
- 4.73 And then, of course, is that really what we want to achieve? Brandeis and Lina Kahn, who is a student of Brandeis, would say, "No. We don't care about economic

efficiency." Or economic efficiency is not the most important thing. What's important is morality. What's important is fragmented industry. What's important is preserve small businesses. What's important is not to allow bigness to dominate. And economic efficiency should be a secondary consideration.

- 4.74 So again, what's the criteria? How do we determine? How do we determine how to interpret these laws? Nobody can tell you, and as a consequence, it's constantly changing. And imagine being a business person, where you don't know if your deal—you believe that merging with another company makes sense. You don't know if they're going to approve it, not approve it, if you're going to be considered a monopoly or not a monopoly.
- 4.75 This Nippon Steel, U.S. Steel purchase that we've talked about, it's 18 months it's taken for the deal to finally be approved with the golden share, with the control of the government now over U.S. Steel. So Nippon Steel, that's not what they were bargaining for. That's not what's in their long-term plan. This has screwed them up quite a bit. But imagine that over and over and over again, really every business decision on scale is going to be questioned by some bureaucrat, could be questioned by some bureaucrat. Maybe it won't be. Who knows? So that uncertainty is enormous.
- 4.76 So, at the end of the day, when they come up with the Clayton Act, they can't articulate the rules. And again, it lands up being vague descriptions of anti-competitive practices, things we don't like, without real, clear, articulate definitions of what those anti-competitive practices are. So, they say price discrimination is bad. Remember when I said charging different people different prices for the same good? Not good. That's anti-competitive.
- 4.77 Tying, exclusive dealings. Me telling you, "OK, I'll sell you—you can distribute my products, but you can only distribute my products." That's anticompetitive. That's exclusive dealing. I'm preventing you from dealing with other parties that is hampering their ability to use you as a distributor. So any kind of what you could call non-standard contracts, like contracts that limit you in some way in what you do and how you do with my product is now constrained.
- 4.78 It also prohibits interlocking board memberships. It's really trying to dramatically reduce the influence of people like J.P. Morgan and bankers like J.P. Morgan on business. So, basically, in anticipation of the Clayton Act passing, J.P. Morgan and, with J.P. Morgan the banker, and the people at his bank resigned from dozens of boards. And the economic evidence suggests that that is all negative because these are good board members. There's a real value that they bring to be on the board. And the relationship with the investment bank has real value. And yet, that is now cut.
- 4.79 The other thing that the Clayton Act does is it creates something called the Federal Trade Commission. This is what Lina Khan was in charge of, up until Trump won the

election. And it is in charge, again, of supervising mergers, of evaluating mergers and deciding which ones are anticompetitive, which ones should be blocked and stopped.

- 4.80 And they're the ones who sue. They're the ones, for example, suing together. Sometimes the Justice Department does it. Sometimes the FTC does it. It's not clear. They have a division of labor between them. But a lot of their big tech lawsuits are driven by the FTC, have been driven by the FTC.
- 4.81 Now, what impact does that have? Now, think about it. If I can't contract with you to be my sole distributor, that you can't distribute other people's stuff, just only mine, if I can't do that contract, but I still want that to happen, how do I get it done? Well, I buy you, and now you're my sole distributor. But I own you, so it's not anticompetitive. What the Clayton Act really didn't like is the contracts, but if I buy you, that's OK.
- 4.82 What you got is significant vertical integration. That is, companies buying people that otherwise they would contract with on the same product. That's vertical. If my suppliers, I want to buy, I don't know, iron from you, but I don't want this type of iron to be sold to other people, I can't do that anymore. I'm just going to buy the iron source. I'm going to buy the iron foundry if I'm a user of iron, whatever the company might be.
- 4.83 Again, we mentioned this, if I can't form cartels, if I can't try to form a cartel, if I can't come to some agreement with my competitors about how we're going to divvy up a market, then I might as well just buy the competitors and form a new company. And again, Clayton Act can't stop that from happening and allows that to happen.
- 4.84 So, for Brandeis, too low of a price is the bad, not too high of a price. So actually creating big companies that charge high prices, that's fine, because then small companies can compete with them, with low prices, and supposedly destroy the bigger company through lower prices. So again, a mishmash. Nothing really coherent comes out of this. But note that it changes the way business thinks about, changes the way the corporation thinks about, what it should own, what kind of relationships it would have, and with whom. So the structure of business is significantly changing.
- 4.85 We'll talk more about the Great Depression in a little bit. But during the Great Depression, there is more antitrust activity, more going after businesses for antitrust violations in a decade than all the antitrust cases leading up to that. The Great Depression, generally, is a time of incredible government activism, government activism to try to shape business, to try to shape the business environment out there. So the focus is to stop this vertical integration.
- 4.86 The Clayton Act provided an incentive to do that, but now we don't like it because it's creating these big businesses. We want to stop it. And again, it's constantly contradictory motivations, constantly contradictory signals to the business community. In the post-World II environment, economic efficiency is now the standard for antitrust. So some of the moralistic stuff that Brandeis is put to the side.

- 4.87 But again, economic efficiency is determined by an economic model that's quite statist, a Keynesian economic model. It's not free markets. Some of us would think that free markets is economic efficiency. Not having antitrust at all is the most economic efficiently, efficient solution. But that's not what they're thinking at the end of World War II.
- 4.88 One example of a case, there's a company. It sounds funny today, but a significant company in the United States in the post-war era was a company called United Shoe Machinery Company. United States manufactured a lot of shoes. There's a robust, innovative shoe industry in the U.S., a lot of small manufacturers of shoes.
- 4.89 And United Shoe Machinery actually produced, during the late 19th century and early 20th century, they consolidated the machinery used by the shoe manufacturers to produce the shoes. So they provided the machines. And they used a method by which they didn't sell you the machine. If you were a shoe company, you didn't buy the machine, you leased it. And then they would provide services.
- 4.90 Now, this is a familiar business model. If you know, IBM never really sold mainframes. They leased mainframes. That gave them a steady revenue stream, and they provided services, and they provided other auxiliaries to that. Microsoft today, you don't buy Microsoft Office, you lease it. You basically have an annual contract where you get and they give you updates. You get updates and you have a helpline, and you pay a certain amount, and I think you can pay more to get more services, but you basically, it's a subscription model.
- 4.91 United Machinery had a subscription model. And it dominated the industry. There were lots of small guys, but it dominated with the machinery industry. It would make, again, beyond the actually providing services, this is true, I think, of IBM at the time, what they also did was help you set up. If you wanted to start a shoe company, they would help you set up the shop. They had the expertise, they had the knowledge, and they provided that knowledge.
- 4.92 Now, in order to prevent you from free riding off of that—free riding means using that and then not contracting with them, not using their machines—they engaged in all kinds of contracts. For example, you could get them to help you set up a shop, and then you could buy a knockoff machine from some other manufacturer that was similar to theirs, or you could lease it for a year and then replace it with somebody else's machines. So they would have long-term contracts where they would tie you contractually to them.
- 4.93 There's a variety of different problems with providing these services and this model. So there were all kinds of contracts that protected the United Shoe Machinery Company. Well, in 1953, the court decided that these contracts were bad, that these contracts were anti-competitive, that they excluded competition, that the whole model of leasing was deemed anti-competitive. They demanded that they offer the ability to purchase

the product in addition to leasing it. So you couldn't just lease. It would be up to the shoe manufacturer to determine whether they wanted to purchase or lease.

- 4.94 They forbade any restrictive provisioning, any of this tying up. They forbade all of that. And you could really pinpoint the start of the decline of the shoe manufacturing in the United States to this ruling. It basically took away a lot of the benefits that were provided by one company, or one company dominating, at least, providing the services, providing the machinery. Now it couldn't tie up its competitors. They could free-ride, or not competitors, its customers, they could free-ride. Other competitors entered the market.
- 4.95 And soon enough, much of the shoe manufacturing business in the United States went overseas, way before the globalization trend moved industry out of the country. So whatever it was about the particular original contracting that United Shoe Machinery had done, the market said, "Yes, this works. This is good." And when it was taken away artificially, the market basically collapsed, and it was inefficient anymore to make shoes under these constraints that were placed on American industry.
- 4.96 So that's just an example. And there are lots, there are hundreds of these examples, thousands of these examples. IBM, later on, IBM is also, in its antitrust suit, a big deal, a big part of the antitrust suit on IBM in the 1960s, '70s, all the way into the '80s, because this is an antitrust suit that took almost 20 years, is, they go after them for leasing the machines. And IBM, in order to preempt it during the lawsuit, actually starts selling machines.
- 4.97 And you can date part of the decline of IBM as a manufacturer of computers to the change in business model that didn't really suit what IBM was about. IBM was a service provider. It wasn't just selling a machine and go away. And with a lease, you actually bought a whole service package, which IBM provided. So this is, again, a business model that is challenged later on in the IBM case.
- 4.98 Now, things change in 1980, and we'll see other changes that come in the '70s and '80s. It's a pretty dramatic era for American economy and American business in particular. And they change intellectually. There's a real shift in terms of the way people are now thinking about antitrust. This is the era influenced by people like Robert Bork. If you remember, he was nominated to be a Supreme Court judge and voted down. This is during the Reagan era, by Posner and two lawyers and George Stigler, who was an economist. So this era is dominated by their thinking.
- 4.99 And this group is a group that definitely leans towards free markets and is very skeptical about antitrust, generally, and the first real group to understand the economic value of all these unusual contracts. They come out of Chicago. Chicago is also where Ronald Coase, who we discussed in one of the previous classes, Ronald Coase had been teaching. He's the guy who came up with this idea that corporations exist because, within the corporation, the transaction costs are lower than outside the

corporation. And when transaction costs are lower outside the corporation, then the corporation divests functions to the marketplace.

- 4.100 They're very influenced by Coase, but they're also influenced by the growing influence of the Chicago economic school. They're not quite laissez-faire. They're not quite ready to say there should be no antitrust, but they want to limit its use. They want to constrain what it's applied to. And indeed, antitrust enforcement, you could call it, is a lot less from around 1980 through, I'd say, the first or the middle of the first Trump administration. You get a lot less antitrust than you did before. A lot fewer prosecutions. It's a real shift.
- 4.101 They're mainly concerned here about— they are concerned more about rising prices. Remember, Brandeis didn't like low prices. They don't like high prices. High prices suggest lack of competition. Competition from an economic efficiency perspective is good. So they're much more likely to go after big businesses that have high prices and break them up.
- 4.102 This is incompatible with the idea of going after Google, for example. What high prices does Google charge? We all use it for free, basically. It's incompatible with going after the big five tech companies who all are constantly driving towards lower prices, in spite of having what appears to be monopolies. They have domination over their particular areas, and their prices are declining.
- 4.103 So, part of what happens is, in the new era where the big five tech companies are the villains, and they're the ones we go after, we need a different antitrust theory in order to go after. And indeed, again, Lina Khan, who I keep mentioning because I think she's the dominant intellectual, she's not much of an intellectual, but anyway, intellectual force behind the new shift to the new model of antitrust, which the Trump administration has adopted, she is influential and she brings back Brandeis.
- 4.104 And what they're called today is "new constructionists", but you can also call them neo-Brandeisians, and they're concerned with low prices. This is why they go after Google. Google is offering a product for free. Wait a minute. By offering a product for free, nobody can compete with them. That's not good. We don't want Google to be able to offer a product for free. So the whole paradigm has now shifted.
- 4.105 And these lawsuits—and really, I think every one of the big five tech companies, what are they? Let's see. Google, Apple, Microsoft, Amazon. Who am I missing? Meta. Meta. All five are under the, they're all in some process or another, on some issue or another, being sued for antitrust reasons right now. And again, the Trump administration is doubling up on what Lina Khan did, what the Biden administration does. And many of the lawsuits actually started under Trump in his first term.
- 4.106 So for this, we need something different. So for Lina Khan, antitrust law and competitive policy should promote, not welfare, economic efficiency, you could call it, not welfare, but competitive markets. That is, we care about competition for its own

sake, not because competition enhances welfare, not because competition makes the world more efficient, makes the economy better, makes consumers better off. In a sense, we don't really care about consumers.

- 4.107 And this is why, in the Google case, if you read the arguments in the Google case, the government is not making any argument about this hurts consumers. And Google is defending itself, saying, "Look, we help consumers." And the judge is basically saying, "That's not the standard." Consumers are not the standards. Efficiency is not the standard. The standard now in this new phase is competition for the sake of competition. Again, smallness. We want lots of small businesses. That's virtuous, that's good.
- 4.108 So, according to Kahn, there exist non-economic values. She writes, "Including our interests as workers, producers, entrepreneurs, and citizens, these values may trump whether consumers are materially better off. The way to ensure the survival of these values," she maintains, "is to keep power artificially decentralized." We want decentralization. We don't like bigness. And that is the new era we're walking into.
- 4.109 And of all the big tech companies, the one she made her career off of hating, if you will, because she wrote a major essay on this that really elevated her profile, is Amazon. She despises Amazon. She writes, "A market with less choice and diversity for readers amounts to a form of customer injury." Now, that's bizarre. Amazon provides us with more choices, but it's all from Amazon. What she's talking about, choices and where we get our stuff from, and for her, Amazon reduces that.
- 4.110 She despises platforms. If you think about the shoe machinery, that's a platform. And Amazon is a platform in which other sellers come and sell, but they now have to ship through Amazon. Of course, that reduces their costs, and it makes us all better off because we're paying a lot less for shipping because they get to use the bulk deals that FedEx and UPS, that Amazon has settled with FedEx and UPS, doesn't interest her. She doesn't care.
- 4.111 She definitely has been focused on Amazon and their dominance and what they do supposedly to small sellers, even though you could argue, easily, I think, that they've enhanced small sellers, not reduced them. So this is where we are today. We will see how all of this has shaped business, has helped shape business. We saw a little bit of this already, but we're seeing the future classes, how this has shaped the structure of American corporations over these decades, how antitrust enforcement has shaped it.
- 4.112 Tomorrow, we'll talk about the tech industry in particular, and we'll talk about the evolution of tech and how antitrust has played a role there. Because famously, the antitrust case against IBM and the antitrust case against Microsoft. But also just the evolution of that business and how corporations have adjusted to it.
- 4.113 I mean, real competition is freedom. Real competition exists when you allow markets to be free. It's active rivalry, it's striving, it's experimentation, it's entrepreneurship, it's

the discovery of new ideas. It's not about where prices are. It's about new entrepreneurs constantly coming into the markets and changing things up. If you think about who the competitor to Standard Oil was in the 19th century, it wasn't other oil companies in the end. Standard Oil produces kerosene for lighting.

- 4.114 The ultimate competitor to Standard Oil was Thomas Edison, was electricity. It competed Standard Oil out of the business of lighting. Now, it turns out, and Rockefeller knew this, that oil had many other uses, and he constantly invested in refining it better and better and better so that you could extract other products other than kerosene from it. But the reality is that electricity replaced kerosene as the form of lighting.
- 4.115 It's innovations like that. It's a personal computer outcompeting IBM. The personal computer is not a mainframe, but it still outcompeted the mainframe. Basically, there are no mainframes anymore. Very few mainframes used today because of personal computers, because of servers, because of the cloud. So, competition is this constant innovation. New tech, new products, new sources of supply, new types of organization. That's what produces competition. That's what creates real competition.
- 4.116 And that's what creates real economic efficiency, real economic well-being. And that's all a consequence of freedom. That's all a consequence of free market. And the consequence of that is, for consumers, lower prices, better products, exponentially better and better and better products over time. And you can see that in the marketplace.
- 4.117 So we are now ready to look at a couple of things in the next couple of classes. We'll look at what happened to American corporations during the period between World War I and through World War II. We'll look at what happened after World War II, because that's a distinct period, something particularly interesting happened between World War II and, I'd say, sometime in the mid-1970s.
- 4.118 We'll also look at regulations and how regulations impacted all of this. This is a period of heavy, heavy regulation and control, particularly coming out of the Great Depression. And then, to some extent, a lot of that gets unwound starting in the '70s and into the 1980s, and there's dramatic change. So we'll look at that whole period coming close to the period of today in the next two classes. Thank you.

Lecture 5: Early 20th Century Corporations

At a glance — Early 20th-century corporations and the impact of antitrust.

- 5.1** So last class, we talked about the impact of antitrust on the corporation and how it's structured and the incentives and what's been going on, really, for the last 130, 140 years with regard to antitrust. So this class, we're going to talk about other impacts, other effects on the corporation, its structure, and the way it evolved through the 20th century. And here we're going to focus primarily on three big events that really shaped the century and shaped American business and shaped our politics to a large extent. And that is World War I, the Great Depression, and World War II.
- 5.2** All had profound impacts on American business, all changed and set us up for the corporate world we encounter in the '50s and '60s and later on, which we will dedicate next class to. So really what we're covering now is the period from the beginning of World War I until the end of World War II, maybe a little bit post-World War II, and what is going there, and see how corporations respond to these events.
- 5.3** Many people have this notion that wars are somehow good for the economy. Wars, in a sense, bring full employment. There's a lot of money the government spends on building new factories to build weapon systems, and as a consequence, people are employed and are productive. GDP, as a measure, economic growth typically goes up. On top of that, the idea is that there is a lot of innovation that comes out of wars. A lot of money is being spent now on building better planes, figuring out radar, figuring out different technologies that are helpful in war and later get commercialized in the private sector.
- 5.4** It is, though, a pretty bizarre real view of war to think in those terms. War is destructive. It's about blowing stuff up. It's about killing people and blowing assets, destroying assets. Yes, it's true, there's a lot of production that happens, but it's not production that enhances human life. You're producing tanks and war machines. War machines are aimed at destruction, not production, not at consumption, not at bettering human life. Yes, it's necessary. You got to win the war, and if the war is a war of self-defense, it's a necessary thing to do. But it's not enhancing our standard of living. It's not enhancing quality of life. And through every war, standard living and quality of life ultimately go down, not up, in spite of the GDP number—which is heavily influenced by government spending. And since government spending and government borrowing go up a lot during warfare, GDP goes up as well.
- 5.5** But in this case, GDP, while usually highly correlated with the well-being of individuals in an economy; in a war, it's clearly not. You've got a large percentage of population out fighting and dying. Their quality of life has gone way down. In some cases, there's no life left. Think about all the injured troops that come home maimed and whose lives are, to a large extent, destroyed by war.

- 5.6 So there is a certain romanticization, even in economics, of war, which is overwhelmingly false. Even the question of new technologies, one has to really ask to what extent it is real, we don't have a counterfactual. We don't have the alternative universe in which you don't go to war—and in which all that capital that went into destructive activities is being invested in research and development and creating consumer products and actually growing the economy and actually making our lives better.
- 5.7 It's quite, I think, easy to contemplate or to think that that is a much better outcome than the outcome. And yes, some technologies get better during war. But if we had peace, quality of life, standard of living would be going up much faster, and those technologies would probably be accelerated anyway through private investment.
- 5.8 So that, I think, is the context in which we should approach viewing war. And again, a lot of what you read about wars is wrong because of that assumption that somehow wars are beneficial economically. And some among the progressives, even thought the wars were beneficial spiritually, they brought the country together. You got a common mission. We all bonded around a common goal. We even have this idea that the greatest generation is the generation that fought in World War II because they came together and confronted an enemy and fought it, and all that is good. But is that making them the greatest generation, given that they also arguably did many other things other than that that benefited or retarded the quality of life and standard of living in the United States?
- 5.9 But we have a certain high respect for war, which it does not deserve. Really, if you think about the period before 1914—before 1914, the world was globalizing. International trade, in spite of the high tariffs in the United States, international trade was booming. One of the things that mitigated high tariffs in the US during this period was the fact that transportation costs were plummeting.
- 5.10 So even though you raised tariffs, transportation costs were plummeting. On net, prices of inputs didn't go up that much because of the lower transportation costs. The world was then dominated by the British Empire, which secured the shipping lanes, and you could trade anywhere with anyone. Britain itself was a free trade island. They had almost no tariffs, and it facilitated trade on a global scale. And indeed, quality of life, standard of living, any measure of human well-being throughout the 19th century had gone up dramatically and was doing phenomenally well.
- 5.11 To a large extent, the First World War brought that to an end. And really the period from 1914, early until the early 1970s—a period of decline when it comes to globalization, a decline when it comes to trade, and arguably a period of relative economic stagnation, if you take it as a totality, and not the same growth and prosperity that we saw in the era before World War I.

- 5.12** Now, World War I was a war that the United States, just like World War II, entered relatively late. By the time it entered, the war was almost over. The fact that the United States entered made it almost over. So the United States had not participated in the war for very long. But to a large extent, the United States was providing weapons and providing assistance to its allies, the British and the French, from the beginning of the war. And in a sense, the war economy, the organization of the economy towards warfare, was started even before the United States actually entered, because it seems like Woodrow Wilson anticipated the entry into World War I. In a sense, it is something that ultimately he wanted to pursue.
- 5.13** So in the pre-entry of the United States into the war, most of the procurement of weapon systems and aid to the Allies was done through a consortium developed by J. P. Morgan. J. P. Morgan basically became the seller to Britain and to France of weapon systems. He would buy it from manufacturers in the U.S. And sell it, and he had it down. I mean, he was incredibly efficient about this and productive. He, of course, made money off of it, but it was run as a business, and it was not disruptive to what was going on in the United States, and it provided the Allies with help on a very efficient basis.
- 5.14** Of course, as soon as the U.S. actually entered into the war, the Wilson administration scrapped the J. P. Morgan model and adopted the direct procurement of weapon systems by the U.S. government. Most of that they attempted to do in a centralized manner. Top down, they established War Industries Board that basically dictated to everybody what should be produced, how much should be produced, where it should be produced, when it should arrive.
- 5.15** Created massive inefficiencies. The government itself tried to produce things like gunpowder. Instead of letting DuPont, who was the main producer of gunpowder, do it themselves. The government tried to do it. By the time the government built the factory, got everything running, got it to the point where they could actually produce the gunpowder, the war, of course, was already over. So the government spent a huge amount of resources to both buy stuff from private industry, but also to produce it itself. This is an era in which—this is a period in which the government probably did more economic activity than maybe in any other period in American history. It had done it all from an attempt to do it all centralized.
- 5.16** But the reality is that this centralized approach failed very early. And what ended up happening is that they would assign contracts to regional associations, industry associations. And when there were no industry associations, the government helped create industry associations. And a lot of the actual production, a lot of the actual coordination, a lot of the actual work was done at the localized level, in a decentralized way, even though, again, the War Production Board took all the credit for everything that happened. And then many people who go back at this and look and say, and I think to a large extent, many people during the Great Depression looked at it and said, "Look,

central planning works. The government did a good job in procuring the machinery, the weapons, the things that it needed during World War I."

5.17 And the lesson learned out of World War I, I think the wrong lesson learned out of World War I, looking at the wrong data, not really considering everything that was going on, was that central planning is probably the future. Many of the people around Roosevelt, particularly in his first administration during the Great Depression, really believed this and strove, in a sense, to mimic a wartime economy during the Great Depression. And thus marginalize and minimize the role of the corporation, the role of private businesses, the role of markets. And the story of the Great Depression in World War II is very much that—particularly the Great Depression, of marginalization of markets and the elevation of private businesses only to the extent that they could be controlled and regulated and thus not really private.

5.18 So World War I—many of the lessons that came out of World War I were central planning works. One of the other aspects of World War I was taxes went up dramatically. So the first income tax in the United States required an amendment to the Constitution because it had been ruled unconstitutional by the Supreme Court on several occasions. There was a short period where there was an income tax during the Civil War, but that went away after Civil War. A persistent income tax was ruled unconstitutional by the Supreme Court.

5.19 There was a Constitutional amendment in 1914. We get the first income tax in the United States. And when it was first instituted, the idea was only the very rich will pay it, only them, and it'll be very low. Like I think the first income tax rate, top marginal rate was 8%, and very few people paid it. As soon as war broke out, marginal rates skyrocketed, and the number of people paying income tax increased dramatically. In a sense, again, World War I is a war that gives us the modern income tax. It gives us the kind of income tax that we have today, both in scope and in terms of the rates—or rates were even higher during World War I.

5.20 But it took just a couple of years, and the promises of that first income tax that was passed in 1914 were completely pushed aside in the name of war, but it never went away. Not really. Not in terms of scale and not even in terms of the size of the marginal rates. Again, they go up and down over history, but the possibility of very high rates is always there.

5.21 In addition, during the war, there's always this fear that our businesses are going to make a lot of money off of war, and that is considered bad. That is considered unfair. So the government imposed both in World War I and World War II, what's called an excess profits tax. So the government decided what a reasonable profit would be, and anything above that was taxed at very, very high rates so that businesses could not do war profiteering.

- 5.22 I don't know if you've watched a movie on Howard Hughes with DiCaprio. It's a good movie. It's worth watching. But Howard Hughes, after World War II, was brought in front of Congress and accused of war profiteering. And a big deal is made out of the fact that he made money during the war. He actually stands up to Congress and gives them an earful. One of the best, I think, counter-Congress— of a businessman standing up to Congress that you've seen in American history was a relatively young Howard Hughes in the late 1940s.
- 5.23 World War I basically leads to these conclusions among certain intellectuals. Again, central planning is a good thing. But in the post-World War I era, they are not the dominant political force. They are not in power. They are not very influential. And indeed, one could argue the 1920's are a real golden age for American capitalism, a real golden age for American business, a golden age in terms of innovation and progress, and a golden age for the corporation.
- 5.24 The 1920s are relatively hands-off period in terms of antitrust enforcement, in terms of regulations, in terms of intense government involvement in the economy. And primarily responsible for that are Coolidge and Harding, the two presidents that are during this period. There is inflation after World War I that then is dealt with, and there's a severe recession in 1920, '21. The government does very little during that recession. It responds in a very different way than they do later on for the Great Depression and further into the future in the Great Recession. They let things go. And there's a quick recovery out of that Great Recession. And what continues after that is a significant economic growth during the 1920s.
- 5.25 The Federal Reserve, another creation of the mid-1910s, is relatively young at this point, and we'll see how that plays a role in the Great Depression. But during the 1920s, at least during the first part of the 1920s, doesn't do much, keeps interest rates relatively low, doesn't try to play games with the economy, keeps things fairly stable from a macroeconomic perspective. And behind all that, that facilitates, again, real economic activity, real progress. Growth in terms of standard of living, quality of life, productivity skyrockets. Overall, factory productivity grows at phenomenal rates. This is productivity from all different factors of production.
- 5.26 The one real negative—I think it's a negative—that happens in the 1920s is immigration is dramatically constrained, which probably actually puts a hold to some of the increases in economic growth because the population is not growing as fast as it could. But other than that, it is an era of, relatively speaking, hands-off by the government, and as a consequence, an era of prosperity.
- 5.27 This is an era in which you get a lot of innovation in the automobile industry, a lot of growth in the automobile industry. Ford, of course, has already introduced the Model T. This is an era in which General Motors comes into existence as an amalgamation of a number of different auto companies and brings to the forefront what is described as— General Motors becomes the first corporation that is considered scientifically managed.

- 5.28** Alfred Sloan, who becomes a CEO of General Motors, is thoughtful about every aspect of production and about measuring everything and about being scientific in terms of how the business is constructed. He takes each of the divisions and he creates a vertical around each division. He leaves those divisions relatively autonomous in terms of—so they are close to their customers, close to their suppliers. They can each innovate themselves. Although there is a layer of oversight provided by the General Motors, the corporation, that provides the accounting and the supervision and some strategy in coordination with these subdivisions. Sloan really sets off General Motors to be a model, really a model of an American corporation and a model in the auto industry. It becomes the largest automaker in the world, or at least in the United States. It overtakes Ford.
- 5.29** And he really creates this system, this multidivisional system that is often called an M form. We talked about this in a previous class. Each division is independent or autonomously run, or has a lot of autonomy. But there is centralized monitoring and centralized information gathering for all these divisions. This becomes a model for American corporations. And this idea of scientific management becomes a model for how corporations should always be in the future.
- 5.30** Unfortunately, this is taken as, "this is proof of the ability to centrally plan" and a thumbs up to very big corporations with lots of different divisions. We'll get to the problems that that ultimately creates. But at this point, at least in General Motors' history, they're very focused. They might have many divisions, but they're all automobiles. They're all focused on the same market. They're all making the same kind of product. Each automobile is targeted to a different segment of the population, and they do phenomenally well.
- 5.31** And at the same time, other automakers are coming into their own and building their own cars. At this point, there's still many automakers in the United States. Almost all the automakers are in the Michigan, Detroit area or in other parts of the Midwest in that area. It's a very concentrated industry in terms of geography. There are a lot of benefits to that in terms of knowledge, in terms of talent, in terms of learning. And the auto industry thrives during the 1920s.
- 5.32** Another industry that is coming into its own during this period is radio. AM radio, the ability to transmit voice over the air. This is a big deal. It somewhat comes out of technology that was developed for communication during World War I, and it's picked up by hobbyists, and ultimately integrated into radio stations. Partially because of its use during war, the government feels like this is something it needs to be involved in, really, from the beginning. And this is one area in radio where the government regulations, government controls are there from day one.
- 5.33** One of the big mistakes government makes, one of the mistakes that has consequences really all the way to today is the fact that the wave bands in which—the frequencies in which radio broadcasts—are not sold. They are leased to various companies. They get a

license to broadcast. And now government is in the business of deciding who can broadcast and who can't broadcast.

- 5.34** It also is in the business of deciding what can be broadcast and what cannot be broadcast. Serious First Amendment issues here which were never really fleshed out. But the whole idea of radio has to be wholesome. Radio has to have a certain—certain views can be expressed on the radio, and certain views cannot. Holds until well into the 1980s and beyond.
- 5.35** Government also gets into the business of preventing competition from rising for these AM radios. It gets to decide how many stations each market should have, whether they can consolidate, whether you can form a national radio station, who gets to make the content and then the equipment for the radio stations. Sadly, this industry is really, from the beginning, heavily, heavily regulated and heavily, heavily controlled. And yet it's such an exciting new product. And people respond to it so eagerly that it grows, and it is a huge success in the United States during the '20s and then on from there, in spite, if you will, of these regulations.
- 5.36** Other area that you get a lot of growth here is, again, something we take for granted. Supermarkets, distribution of food. There were no supermarkets in pre-1920s, really. There was no mass distribution of food. Here you get the first supermarket to spread across, ultimately becomes A&P. Some of you might remember A&P as a chain. And again, they're using the new technology of refrigerated rail cars, and the railroads are very effective, and distribution networks, and the ability to monitor all this, and the ability to follow all this. So this is another area in which you get massive explosion of growth in this period.
- 5.37** And on top of that, you get chain stores. You already had Sears, which was already doing catalog sales. The Amazon of its day—you get a catalog in the mail, you order by phone, and it would arrive. And that was like unbelievably revolutionary at the time. People didn't have to go into town or—if they wanted something really, for them, unusual—into the city to go and buy something. Now it could just be delivered to them. And chain stores became very efficient during this period at distributing goods and going into cities and small towns and with all the wares and goods that people needed and increasing convenience.
- 5.38** Again, stuff we take for granted, but we forget how difficult life must have been, or we don't know how difficult life must have been before this existed, before these conveniences that we all take for granted existed. This was all growing dramatically during this period. It's a period of innovation.
- 5.39** A lot of the vertical integration that we talked about last time that was happening was companies were—partially because of Clayton Act, partially because of market forces—they were letting go, and they were relying much more on market transactions and contracting during this period. So the corporations were not growing dramatically in

terms of vertical integration. They were growing just in terms of innovation and in terms of new markets and in terms of growing geographically because of the new means, both of communication and the new means of transportation. Again, a period of very little government involvement.

- 5.40** But all that, of course, changes with the beginning of the Great Depression. Now, there's a lot one can say about the causes of the Great Depression and what exactly happened and why it exactly happened. But I think there's fair consensus today among economists on a few points.
- 5.41** One is that the Federal Reserve, a young Federal Reserve, an inexperienced Federal Reserve, and maybe driven by economic models and politics, they were wrong. And of course, the challenge of being a central planner when it comes to money. Maybe the thing that one can least centrally plan is money. And yet the Federal Reserve was given this responsibility of being a central planner of how much money, what interest rates should be—a responsibility that I don't think anybody could execute on.
- 5.42** But it's clear that the Federal Reserve executed about as badly as one can imagine in the period leading up to the stock market crash in 1929, and even more so in the period post-1929. I think there's fair agreement among economists today, among a majority of economists today, that the Great Depression was a monetary phenomenon. It was caused by the Federal Reserve. It was caused by mismanaging the supply of money. In the lead up to the stock market crash in 1929, raising interest rates too fast. Then when the crash happened, what a modern central bank would do is actually increase the money supply to compensate for the loss of, in a sense, the loss in the money supply that has happened because of the collapse of the stock market.
- 5.43** The Federal Reserve did the exact opposite. It constrained the money supply even further, going into 1930 and all the way through 1933. There was this massive contraction of money which led to a domino effect, led to people not being able to pay back their loans. It led to banks collapsing. It led to banks collapsing, contracting the money supply even further. Because if you think about banks through the fractional reserve system are actually increasing the money supply with the lending that they're doing. But because there's now deflation at big scale, everybody's paying back their debt, and some can't pay back their debt because companies are failing and consumers are not buying as much because we're in a recession.
- 5.44** And the whole thing just implodes in the early 1930s. You get this spiral of bank failures, more deflation, more bank failures, more deflation, more bank failures, more deflation. And it is truly catastrophic. Catastrophic in ways that I think are very hard for us to really get a handle on because none of us have experienced anything like it from an economic perspective.
- 5.45** By March 3rd, 1933, The Dow Jones Industrial, the dominant index of the time, had lost 86% of its value, 86% of its value from 1929 when it hit its peak. So in three and a half

years, 86 % of the wealth that was in the stock market had disappeared. Real GDP, GDP adjusted for inflation, was down 29%. The economy had contracted by 29%. Basically, GDP in 1933 was at the same level it had been in 1901, in some way, wiping out the growth of 30 years of industry, of 30 years of production.

- 5.46 The number of banks was down from 25,000 to 14,000, not because of consolidation—that would have been a good thing—but because of bankruptcies. It's a whole other topic of "there were too many banks." But I think there's a course on here by Charles Calomiris that deals with banking that's very good. I recommend it if you're interested. And unemployment in 1933 was 25%. One in four people in the U.S. economy were unemployed.
- 5.47 So the economy had been devastated. All as a result, of really, mostly monetary mismanagement. Monetary mismanagement that was not helped by things like tariffs increasing by Hoover, taxes going up during the Hoover administration, which only accelerated, made worse what was already happening in the economy. But the primary blame for the Great Depression falls with the Federal Reserve and the attempt to centrally plan money and the failed models that they had for what that actually means.
- 5.48 The consequences of this were primarily felt by small businesses. Most of the bankruptcies and the failures were small businesses. They couldn't get loans from the banks. They couldn't access capital markets in other ways. Capital markets were basically closed. There was no access to fresh new capital. And small businesses were going out of business on a large scale. It was big companies that managed to survive during this period. And they managed to survive by retaining earnings, by keeping cash, by growing their cash reserves. Indeed, during this period, large businesses grew their cash reserves dramatically. They kept capital inside. They didn't pay out dividends. They didn't invest. They kept cash.
- 5.49 And really, the economy during the Great Depression reoriented towards large corporations. And coming out of the Great Depression, the economy was dominated by large businesses, by large corporations that were favored, to some extent, by the Roosevelt administration. And this is the other aspect of it. FDR's response to the Great Depression was to try to control as much of the economy as he could. To initiate processes by which he could take over the running of businesses. His initial attempt, which is called the National Industrial Recovery Act, was basically a bill that gave him massive power to basically control and regulate American corporations, American businesses.
- 5.50 It ultimately was deemed by the Supreme Court as unconstitutional because it had given him so much power. But what he ended up doing instead of that was focusing on smaller bills, smaller regulations of particular industries, still heavily controlled, heavily regulated, heavily top-down from the government down.

- 5.51** What broke down during the Great Depression—broke down because government broke it—were markets. Markets broke down. The ability of businesses to contract with smaller businesses all broke down. And to the extent that businesses, that large businesses had cash, they spent it on buying some of the small businesses and growing—still trying to preserve as much cash as possible for contingencies in the future.
- 5.52** If you look at before the Depression, most innovation in the economy, most new IP, intellectual property, was actually created by individuals and small firms, by inventors. And what would happen is the larger businesses would buy them out, would purchase that. So if you look at the IP portfolios, the intellectual or the patent portfolios of large businesses, almost all of them came from outside, came from outside the business that they had purchased. A lot of that disappears during the Great Depression.
- 5.53** And a lot of what happens is that large companies set up their own research and development arms. They start heavily investing in research and development internally. Almost all the IP that is generated starting in the 1930s comes from inside rather than outside. Now, you can see that, by the way, reversed today. We're back in some sense, or have been in the last 20 years, back much more to the model of the 1920s. Think about all the startups. They create original IP. And what does Google do? Google has a lot of research and development, but most of the new stuff Google does, has, comes from purchasing startups and bringing the IP into Google.
- 5.54** So we're back—and we'll talk about how we got back—we're back to a much more market-driven innovation, market-driven progress, rather than the market is being decimated by the Depression and everything has to be internalized into the corporation. The Corporation itself is heavily, heavily regulated and controlled by the central government, by the Fed.
- 5.55** During the Depression, the large R&D-based big company becomes dominant. At the same time, as we said last time, antitrust is going full blast. Again, they're using antitrust to recarve industries and to control and to manipulate what they're seeing in front of them. Again, this is very top-down. This administration that wants central planning, wants to run the economy, and is using antitrust as a tool to do that.
- 5.56** And the goal now is not a Brandeisian "Small is good; little guys are great. It's good morally," or whatever. Now it's all about economic efficiency from the perspective of a central planner. This company, it's too big by our estimates, so we're going to break it up. Here it's too small, so we're going to allow combinations. It's all this mentality of, "We know what's best. We're not going to allow markets to work. We're going to guide markets as much as we can." And indeed, by the end of the 1930s, a big, big portion of American business is now regulated and controlled in some way or the other, and it always has an antitrust—the threat of antitrust sitting above, looking over their shoulder.

- 5.57 As we talked about the airwaves, heavily regulated and controlled by the central government. Transportation. We've talked about the fact that railroads were the first real business to be regulated by the government. That does not go away during this period. If anything, what happens during the Great Depression is now trucking, which is a competitor to railroads, becomes regulated. It's a relatively fragmented industry. The government wants to keep it that way. It controls mergers and acquisitions. It controls, to some extent, pricing. The railroad pricing is all set by government. Transportation is heavily, heavily regulated during this era. And of course, that will ultimately determine heavy regulations on airlines.
- 5.58 So airlines become very regulated once they become more consumer friendly, which is starting in the 1930s, but really takes off in the 1950s. The government is setting which routes you can do. The government is setting maximum prices you can charge. The government is determining—for example, the whole hub and spoke system that we have today with the airlines was not allowed by the government in this period. So airlines are heavily, heavily regulated going into the 1950s and the 1960s, and as we'll see, only get deregulated in the big deregulation move in the 1970s. So prices stay relatively high during this entire period.
- 5.59 Finance continues to be heavily regulated. It's, to some extent, always been regulated. But in the 1930s, they just go all out. We get 1933, 1934. We get major regulation of American banks, investment banks. Commercial banks, and investment banks are separated. The power and the influence of investment bankers declines significantly. This is partially why capital is so scarce. It's hard to raise capital during this period, partially because of the depression going on, partially because all these banks are being weakened dramatically by regulation.
- 5.60 You also get something called the Public Utilities Holding Company Act, which tries to eliminate holding companies, is trying to eliminate the practice in business of creating a holding company with lots of subsidiaries. So now you have to integrate it into one company. That bill also restricts bankers from sitting on boards of directors of railroads in particular. And there's actually been some economic studies about this because the bankers—you can test this. All at once, all the bankers leave the boards of directors of railroads. So you can test: how did railroads look before and how do railroads look after? And it's clear that railroads are worse off after. They have less access to capital. They're less professionally run.
- 5.61 Bankers are incredibly good board members. We keep coming back to that, but it's true. The J. P. Morgans of the world really add value. Not only do they add value in terms of their knowledge of business, because they sit on a number of boards, but they also add value because of their connection to the bank and the ability to provide capital when needed, and the knowledge of the business so that they can evaluate under what terms to provide that capital. Suddenly, they're outsiders. It's much more difficult for

an outsider now to figure out what's exactly going on, how much capital under what terms to provide. It's easier to back off.

- 5.62** So you get dramatic regulation of banking, really of the entire financial sector going all the way from 1933 to 1940. Just a series of bills that are still on the books, mostly that regulate finance, almost every aspect of finance. And again, some of that starts getting deregulated in the 1970s, but much of it is still on the books to this day. It's there. For example, you get deposit insurance, which we still have today.
- 5.63** They create the FCC, which is the Federal Communication Commission. And it is responsible for regulating not just radio and broadcasting, but ultimately, it's also responsible for regulating telephones. And AT&T becomes basically a protected monopoly. So the deal with AT&T is, they basically allow AT&T to maintain the dominance in the industry. Actually, the government prohibits competition from coming in and challenging it. And again, that doesn't change until AT&T really is broken up in 1982, which is arguably not an issue of antitrust, but more an issue of deregulation, because the structure of AT&T, the dominance of AT&T, the size of AT&T is all protected and guaranteed by government, by government regulations. And one of the deals that is cut, for example, is that long distance rates are going to be relatively high and short distance rates are relatively low because the constituency for the short distance is much more significant than the long distance. So it's a political thing. And as long as AT&T plays by the rules, the government protects it from competition, basically.
- 5.64** And in all these areas, ultimately, what happens is that it suppresses innovation. If you think about radio, FM is discovered way before, like two decades before, FM becomes actually popular, which is the 1980s. You only could really get FM in the 1980s. And yet FM is known to be in existence. They won't give it the frequencies. They won't create the competition to AM, even though FM is, in terms of sound quality, dramatically superior to AM radio.
- 5.65** The same thing happens with television. We could have gotten television earlier. We could have got color television earlier. But the government controlled every one of the steps in the innovation process. Even when it comes to broadcasting and television, they had their favored big business that they established and helped promote and helped sustain their market position, which was RCA. RCA was the favored both in radio and ultimately TV, in broadcasting and telecommunicating in that form of telecommunication.
- 5.66** So the government very much took, during the '30s and '40s and spilling into the post-war era, a position about certain companies having and sustaining dominant positions in the area and really made a real effort to suppress competition and suppress innovation in those areas. We could be decades ahead of where we are today. If we had allowed, for example—we could have allowed microwave long distance phone calls through microwave significantly earlier than they did, which was in the early 1980s. Who knows if we could have gotten an Internet and a lot of other things earlier than we

actually did. Again, during this period, leading up to the 1980s, government really suppressed innovation through regulation, through control, through picking winners and protecting those winners. AT&T being a big one, RCA being a slightly smaller one. But those businesses were protected by government.

- 5.67** All of this, of course, the centralization and favoring large businesses, was all dramatically, in a sense, accelerated during World War II. World Wars, when you shift the entire industrial base of the country towards manufacturing weapon systems, when you ship a big chunk of people who are working, who are being productive overseas to fight in wars, it tends to concentrate power much more in the hands of government than before, and concentrate power in the hands of a small group of large corporations that can actually take on the production of these weapon systems. Detroit recalibrated much of its automaking to build weapon systems. Even airplanes were built by some of the auto manufacturers. The entire economy shifted.
- 5.68** And while, again, there was a big attempt by the government to centralize it all, to have it all guided by some central planner, at the end of the day, it all got fragmented. And a lot of the innovation, a lot of the new weapon systems that came out of it were not planned from the top, but came from the ingenuity of these companies that were used to dealing with markets, used to dealing with innovation for markets, and actually creating new products at the lower level. The B29 is an example of that, but there are many others.
- 5.69** During this period, a series of large corporations did very well because they got a lot of government contracts and they grew as a consequence. They could hire people, they could keep factories open. They could run with it. A lot of small businesses, just like in World War I and just like in the Great Depression, couldn't compete and disappeared from the marketplace. Again, this was an era of consolidation of large businesses, government contracts, and of this idea of "We can centrally plan," even though in the field, in reality, that's not what actually happened. What was actually happening was a lot of these big businesses being innovative and independent and producing the stuff that was needed for the war without the real micromanaging that the central planners tried to impose on them. A lot of that was ignored, and they just went ahead and produced and created and built.
- 5.70** Really, coming out of World War II, we have a business sector that is dominated by large corporations, and we have a government that is used to central planning and is being committed to central planning. And we're also coming out, from a macroeconomic perspective, with enormous debt. And the prevailing economic view was: as soon as we come out of the war, government spending is going to plummet. We're going to get millions of young men coming back, and they're not going to have jobs. And we're going to go into a deep recession, maybe even just a continuation of the Depression.

- 5.71** And indeed, you could argue that the war didn't end the Depression. The war was a continuation in many ways of the Depression. And the expectation was that at the end of World War II, the Depression would actually continue. Because all the macroeconomics seemed— if you're Keynesian, suddenly you take away this massive government stimulus of spending huge amounts of money. What would these corporations do? Suddenly, all these businesses are going to have to convert from building tanks back to building automobiles, from building airplanes back to doing something else. How is the economy going to survive?
- 5.72** And one of the amazing things that happened post-World War II—which surprises a lot of economists, although it doesn't seem like they learn from it—is the economy thrives. Government spending comes down dramatically all at once. The government runs surpluses and starts paying back the debt. The men coming back actually have jobs. Corporations quickly convert back to civilian applications and thrive. And the economy actually booms coming out of World War II—proof that the economy doesn't thrive because government spends money. Quite the contrary. A lot of the regulations and the controls that were imposed on business during World War II go away. A lot of the regulations stay, but a lot of them go away.
- 5.73** Monetary policy is stabilized because of Bretton Woods. Bretton Woods is a new monetary system globally. They stabilize monetary policy, takes a little bit away of discretion, away from the Federal Reserve. Federal Reserve can't do some of the stupid things that they did leading up to the Great Depression and during the Great Depression. So you get relatively stable dollar. You get declining government spending, dramatically declining government spending. You get less regulation. And surprise, surprise, the economy takes off. If you think about consumers that for years, during the Great Depression, and then during the war were not consuming, there was a huge pent-up demand to go out there and build, create, make, and consume.
- 5.74** And all of that came together in a booming economy in the '50s and '60s. Of course, a big part of that, as we'll see, was the fact that the United States really was the only country in the world with real productive capacity that had not been destroyed. So it became a dominant economy in the world, and countries all over the world depended on the United States to export to them the stuff that they needed, which again helped companies become incredibly profitable.
- 5.75** Okay, so next class, we're going to talk about what happens to corporate America during this period of boom, this period that so many people look back with nostalgia at, a period of relatively fast economic growth, of high profitability. What happens to the corporations during this period of time? And then what happens? What leads to—what we've talked about a little bit—the deregulation of the 1970s. And the real change is a shift from corporation-centered, central planning-centered to a much more economic-driven economy of the 1980s, and extent to this day.
- 5.76** I'll see you next class.

Lecture 6: Post-War Corporations

At a glance — Post-war corporations after the World Wars and Depression.

- 6.1 So last lecture, we talked about the effects of World Wars and the Great Depression on the corporation, and really on the, at least to some extent, on the macroeconomics of the United States.
- 6.2 Coming out of World War II, the economy of the United States does really well, partially because of less controls, less regulations, better monetary policy, and the inflow of young men eager to get to work. And indeed, the country grows, business grows, and those big corporations that came out of World War II thrive in this environment. To a large extent, they now have opportunities that they did not have before to enter the global market.
- 6.3 There is a massive shortage in the world of productive capacity. Germany has been blown up. England was a war economy for six years, and much of it was blown up. France— it's going to take France time to recover. Spain has just undergone a civil war. And of course, Eastern Europe is now Communist and completely cut off from the world. Japan, which was industrializing significantly before World War II, again, has been basically destroyed. Its industrial capacity is almost nonexistent. China, during this period, if you remember, is falling under communism. It is off the table in terms of production. The Korean War is about to break out, so there's no production going on in the Korean Peninsula. Today, we assume Korea is this productivity hub.
- 6.4 It's the United States—the United States is there to help provide the material and the production to rebuild Europe, to rebuild Japan, and to provide goods to much of the world that has not fallen under communism. And even the communists would have a really hard time if they couldn't once in a while get help from the United States during this period in terms of food aid and other aid that the U.S. supplied them.
- 6.5 The 1950s, and in particular, the 1960s, turned into really the heyday of big corporations. Stock market is booming, so capital is relatively cheap. It's relatively easy for them to raise capital. Interest rates are low, again, because the Fed is controlling monetary policy. Until probably the late 1960s, interest rates are low. Capital is cheap. They can grow; they can expand.
- 6.6 It is true that finance is still heavily regulated. So in many respects, these companies do not have the monitoring and the oversight of financiers like J. P. Morgan and so on. They're not allowed on boards, and their power has been diminished significantly because of the regulations coming out of the 1930s. And it's also true at this phase that the investor market is primarily very dispersed. There are very few of what today we recognize as institutional investors. There are very few big pension plans investing in the stock market. Those only become large going into the in the 1980s. There are very few insurance companies in the market. There are very few blockholders, big investors,

again, who can serve as monitors, supervisors over the business practices of the corporations.

- 6.7 To a large extent, CEOs during this period have a free rein. They can pretty much do what they want. The boards of directors are very much, as we discussed last time, basically appointed by the CEO, and there's very little pressure from shareholders for them to make the CEOs behave because stock market's going up. Everybody seems happy. Things are going well.
- 6.8 The consequence of this is that CEOs are appointed in many large businesses in the United States who are not shareholders, who do not own significant amounts of stock. Often, the directors are compensating these CEOs, not based on stock market performance, but rather based on things like sales and revenue and asset size of the company, which provides them a strong incentive to grow the company.
- 6.9 There are other incentives to grow. These companies are generating huge amounts of cash. They're very profitable; they're generating a lot of money. The question is: what do you do with the money? You could return it to shareholders. But there is a strong incentive to use that money to, in a sense, grow the company, enhance the reputation, enhance the prestige of the CEO. Also, if you can grow the company, your income will grow. A lot of people within the company are basically being compensated based on the size and the growth of the entity.
- 6.10 Now, where do you spend this cash? There are two factors here that are going on. One is antitrust regulation has evolved at this point to say you cannot vertically integrate. Vertical integration is not allowed. So you can't buy your suppliers, you can't buy your customers or the people one step before your customers. You can't integrate vertically, so you can't go that way. But you also are not allowed to buy your competitors. So you can't integrate horizontally. So what do you do? How do you grow?
- 6.11 Well, what happens is American businesses start buying businesses that are completely unrelated to what they actually do. They start diversifying. They start buying all kinds of businesses. Now, there are a number of reasons for that. One, it's a path to growth. If you want to grow, this is the way to grow. You can't grow vertically, you can't grow horizontally, antitrust won't allow that. You've got the cash and you've got capital markets. Interest rates are low. As I said, stock prices are going up, so you've got cheap capital. You can use that to grow. So why not grow? You're being compensated based on growth.
- 6.12 In addition, diversification is—it makes sense for a manager in a perverted kind of way. The probability that you're going to go bust is much smaller if you're running a diversified business. There might be a year or a few years where one of the businesses does really, really poorly, but it's likely that during that year, another business does okay. The more you diversify, the more businesses you have, the less affected you are by any particular movement in the market. You're diversified. It's why as investors, we

always tell you, if you're investing for your time, diversify because it's low risk. And it actually, on a risk-return basis, it's the optimal strategy.

- 6.13** And managers can do that by buying lots of different companies in lots of different industries. They reduce the risk to themselves of losing their job, of not performing. Now, this is bad economically, if you will, because the reality is that the owners of the company can much more easily diversify than the company itself. If I want exposure to airlines and hotels and steel and pharmaceuticals, I can buy a share in each one of those, and it's cheap. But if I buy a share in a company that then buys a pharmaceutical company and an airline and a toy factory and a steel factory and all these things, that's much more expensive. From an economic perspective and from a shareholder perspective, it doesn't ultimately make sense for the CEO to diversify. It makes sense for the investor to diversify.
- 6.14** But during this period, investors were willing to give CEOs the benefit of the doubt and were willing to allow them to experiment with this. And again, one has to remember that a lot of what we're talking about—markets, capitalism, corporations—it's new. It's been around for 100 and something years, and we discover a lot through experimentation. And the 1960s, were one big, massive experiment in conglomeration, in creating conglomerates, conglomerates of unrelated businesses. It's an experiment that failed, and maybe we'll learn from that something. But it's an experiment, again, incentivized by antitrust and incentivized by a relative freedom that CEOs had that they had not had before and have not had since.
- 6.15** The other aspect of this is, a lot of these companies, the owner, the entrepreneur, who had founded the company was now retired or dead or gone. The management of these companies was passing to a group of professional managers. Many of them were very ambitious and wanted to prove themselves. And one of the ways to prove yourself was to show that you can handle 5, 10, 15, 20 different businesses all at once.
- 6.16** A number of examples of this: ITT, a company called International Telephone and Telegraph, which was founded in 1920, specialized in basically what AT&T did, building telephone equipment and telephone networks, running telephone networks, but outside the United States. Even though they were incorporated in New York, almost all their business was outside the United States. They owned a telephone network in Puerto Rico. Then they owned a lot of the telephone networks in Latin America. They built equipment for companies all over the world in telephone equipment. And they had done well. They had grown and done well.
- 6.17** In the 1950s, the founders retired, and a professional manager was brought in to take over. Again, a manager who did not necessarily have shares. Because if you have shares and you're thinking long term, it's not clear that that diversification makes sense if you're an owner operator versus if you're just, again, being compensated. Your primary compensation comes from, "How big of a company am I running? The bigger the

company, the bigger my compensation is going to be, and my stock price is less important to me."

- 6.18** So came on board in 1960. Between 1960 and 1977, ITT bought 350 different companies. Now, remember, this was a telephone company. This was very similar to AT&T, just internationally. It bought Avis, Sheraton, Hartford Insurance, Continental Baking—people who made wonder bread—and again, 300 other companies, 300 and something other companies in unrelated businesses to this. A lot of this was organized around groups, and then they had their own divisions, and this M-form was very much put in place. But the reality is that it takes a lot of talent and a lot of skill to manage one company in one industry. And certainly an enormous amount of talent and skill to manage a very large company in one industry, let's say, managing GM, just doing automobiles.
- 6.19** GM, by the way, also diversified heavily during this period into finance— GMAC—but also into a lot of unrelated travel brands, hotels, car rentals. Everybody, for some reason, wanted a car rental company, every one of these. Ultimately, all those were spun off. But it takes a lot of skill to run a GM with five, let's say, different types of automobiles and coordinate it all and run it all. But now you're talking about 350 in a variety of different fields. I mean, you could be an expert on cars, but does that make you an expert on hotels? Does that make you an expert on running a rental car [company]?
- 6.20** And in theory, you could say, "Well, each one of those division managers are going to be left alone. They're going to be autonomous. We're just going to provide the oversight." But the reality is that that is very, very rare that somebody actually allows for that. You're the CEO. You're going to want to have a say. You're going to want to intervene. You're going to want to help strategize. But what are you adding? It's not clear that you're adding anything, maybe even detracting.
- 6.21** So indeed, going into the late 1970s, almost all American big businesses in the Fortune 500 were conglomerates, almost all of them. Even companies that today are fairly lean and focused, were not back then. They were a conglomerate of a variety of different businesses. Just to give you a sense of the growth, the ITT went from 760 million in sales to 17 billion in a matter of 12 years. And most of that, almost all of that is to acquisition. But you can imagine if you're running a \$17 billion company or you're running a \$600 million company, a \$700 million company, your compensation is going to be different. A lot of these companies, once they got big, one of the perks of getting big is a private jet for the CEO, maybe a nice corporate apartment on Fifth Avenue in New York City, and the perks that come along with that. So this is perk heavy.
- 6.22** R.J. Reynolds—a tobacco company which became a conglomerate; it was R. J. Reynolds Nabisco in the 1980s—was famous for the fleets of corporate jets that they had, not just one. Most of the senior managers traveled by a corporate jet around the country, and for the beautiful penthouses and nice accommodation they had in various cities

around the country, particularly in New York, for the CEO. And just generally, the unbelievable perks, high salary, and benefits that they got.

- 6.23** During this period, there aren't these blockholders, shareholders, and bankers who might say, "Oh, wait a minute. This is no good. You got to stop this." And because a lot of this was generated cash from within, there wasn't as much need to go to the public market. And even when they went to the public market, public markets were very forgiving of this behavior. Again, until about 1980, this was accepted. Even at ITT, in the 1970s, there was already some suspicion about this, and they tried to spin off a little bit, and they tried to restructure a little bit, but nothing major happened until the 1980s.
- 6.24** Another company, for example, is LTV. Companies you've never heard of, probably, but in those days, they were at the top of the list, and some of the largest companies in the world. LTV was an electronics company. During the 1960s, they bought aerospace, Wire and Cable, Wilson Sporting—the people who make tennis rackets—pharma companies, chemical companies, steel companies, and airline and resorts. They own Braniff. Remember Braniff Airlines? It makes no sense, and it made no sense. This is an era of the 1960s, where it seemed like this was working because everything was going well.
- 6.25** And then you got some macroeconomic challenges. One of those macroeconomic challenges was the Vietnam War and the welfare state, the introduction of Medicare and Medicaid, the war on poverty, which all led to large government spending, big deficits, and the beginning of inflation. Inflation did now put pressure on this relatively stable monetary system. They were being created under Bretton Woods, where dollars on an international basis could be exchanged for gold at a fixed rate.
- 6.26** And now the U.S. government was printing a lot of dollars. And foreign countries were saying, "Wait a minute. You're going to—this is inflationary." The U.S. government said, "No problem. Don't worry about it." And the French basically said, "You got to be kidding." They basically took all their dollars and started converting into gold at the Federal Reserve. Gold was flowing out of the United States. Ship loads of gold were going out of the U.S., primarily to France, and it was unsustainable. You would have emptied all the reserves of gold in the U.S., and it still would have been a lot of dollars that potentially could have been exchanged for gold. And the U.S. would have defaulted on the exchange.
- 6.27** So in 1971, Richard Nixon went off the gold standard completely, abrogated the responsibility of converting dollars into gold. Dollar became completely floating. Exchange rates became floating versus the dollar, which created a massive shock in the market. It also allowed—freed up inflation. Nixon tried to hold back inflation with price controls and wage control. It was a pretty bad period of time from a macroeconomic perspective.

- 6.28** And of course, it now was much more difficult to run these conglomerates. These conglomerates were now being shaken up. Stock prices were going down. The 1970s saw a huge volatility in the stock market. And generally, the stock prices from, I think, 68 to 82 were basically flat during that period, but with a lot of volatility in between, mostly on the downside. So suddenly, the challenge of running these conglomerates increased significantly.
- 6.29** Two other things were also going on during this period that are related somewhat—but are important. First, you got massive foreign competition for the first time. Up until the 1970s, the U.S. was cruising; it was a dominant producer in the world. It's hard to believe today, but the U.S. produced most of the televisions. Certainly, all the televisions sold in the United States were American televisions. There were about a dozen television manufacturers in the U.S. in the 1970s. RCA was a massive producer of television, Zenith and many others. Almost all consumer electronics in those days were produced in the U.S.. Yet suddenly, product was coming in from Japan, better and cheaper, and driving companies literally out of business.
- 6.30** By the early 1980s, the United States was not producing televisions, was not producing radios. All of that was being imported from Japan. How many televisions does the United States produce today? Zero. Most of the consumer electronics that you own today are either from Japan or China or South Korea, with one exception, of course, which is Apple, which is a consumer electronics company. But it's not just Apple; you've got Google and you've got the big five. You got tech. With the exception of the tech, all the other consumer electronics are coming from overseas.
- 6.31** So suddenly, the U.S. faces competition and a declining demand for U.S. exports because countries are producing it themselves. The Germans don't have to import American cars; they have their own cars. The Japanese are producing their own cars. And indeed, the automobile industry in the United States comes under incredible pressure during the 1970s. Chrysler basically goes bankrupt in the late 1970s, early 1980s, and is bailed out by both the Carter administration and the Reagan administration with a big loan that they pay back very quickly. They get maybe lucky, you could call it, but they come out with a minivan, and the minivan becomes this massive success. This is before SUVs. There were no SUVs on American roads. The minivan is the first big family car other than the station wagon. It's a huge hit, and it basically saves Chrysler.
- 6.32** So this is coming under major, major challenges from overseas, primarily the Japanese, but also the Germans. The Germans are competing on a lot of the heavy industry and a lot of the fine tools and the things that they're still good at or to some extent good at to this day, and of course, automobiles. It's interesting because—I'll say a few words on the Japanese corporate model.
- 6.33** Japan has evolved differently than the United States from a corporate perspective. Japan's corporations were not basically formed by antitrust. There really was no

antitrust in Japan, certainly in the early days. In many respects, there was little regulation in Japan in Japanese business coming out of World War II and into the 1970s. And the way Japanese companies formed is much closer and much more reminiscent to the way American companies were forming before the antitrust stepped in—a heavy role for bankers, a corporate structure that was more of a holding company and subsidiary companies, groups, rather business group, all integrated with a bank.

- 6.34** Now, ultimately, the Japanese model failed. It failed because those banks were heavily, heavily regulated by the government, and the government started telling the banks who they could lend money to and who not. And basically, when the Japanese bubble burst in 1991, the Japanese government didn't let companies go bankrupt. They forced the banking system to bail everybody out. And that bailout has basically caused the Japanese economy to be in a stagnation over the last—what is it—since 1991, so 30 years.
- 6.35** Basically, Japan has been in stagnation, primarily because of the government forcing a bailout on everybody. Instead of allowing the bubble to burst—it was a real estate, primarily financial bubble—and then let bankruptcies clean out the unproductive and refocus that capital to the more productive, it kept unproductive alive. And that turned out to be a disaster. Bailouts, generally, are destroyers of wealth and destroyers of long-term economic growth. True in America, true in Japan. If you want stagnation, bail everybody out. That'll give you stagnation.
- 6.36** Because you need what Schumpeter called creative destruction. You need this process of companies, new companies, coming with new technologies and other companies going bankrupt. You need bankruptcy. Bankruptcy is really, really important for economic growth and for innovation. Indeed, one of the things that is unique about the United States that has always been good about the United States is that we have very developed bankruptcy laws. You can go bankrupt, you can restructure the company, how bankruptcy works is very transparent— everybody knows. All the different financial stakeholders know how they're going to be treated in bankruptcy, approximately. And it's a very smooth process in the United States.
- 6.37** These bankruptcy laws, until very recently, don't exist in Japan, didn't exist in China, in Korea. And that hampers their economic growth long term. It hampers their ability to deal with crisis, to deal with the fact that some business has to go out of business. And sometimes they have to go out of business and disappear. Sometimes they have to go out of business and restructure. And the American bankruptcy system allows for both those paths in a very efficient way, which other countries do not have.
- 6.38** So a number of things are happening in the '70s. One is competition from outside, a competition that really hurts. The auto industry in the United States is really hurt by Japanese competition. Without the government bailout in the '80s and a bail out again

in the Great Financial Crisis, it's doubtful how much of an auto industry we would have in the United States today, if not for that.

- 6.39** And of course, the other part of the bailout of the automobile industry was import controls on Japan that were imposed by the Reagan administration. Reagan actually got the Japanese to agree to a quota system where they wouldn't import to the United States more than a certain number of cars. And that held. And what that caused was the Japanese started building automobile companies in the U.S., and where did they build those companies? They didn't build them in Detroit. They didn't build them in the Midwest. They built them all in the South.
- 6.40** Almost all the Japanese auto companies and German companies—BMW has a huge plant in South Carolina—they all built them in the South. Why in the South? Relatively low regulation and very low union participation. So non-unionized shops with relatively low state regulation. So again, I think I mentioned this in another class: much of the de-industrialization of America people are talking about these days is not the de-industrialization of America; it's the de-industrialization of the Midwest in favor of industrialization of the South.
- 6.41** We don't have time in this course to talk about it, but one of the causes of that and one of the big issues that have plagued corporations since the beginning is unionization. And even here, there's a difference between unionization in America and unionization in Japan. In America, you have an auto union. It covers all auto companies, one union. In Japan, Toyota has a union, Honda has a union, every company has its own workers union. They're much better integrated in terms of incentives into the particular company. The negotiations are easier. The incentive structure—they built incentive structures to align the incentives of workers' with management.
- 6.42** In the United States, they've created a confrontational system, a system of conflict. We've entrenched that system into laws, into various laws over the years, particularly, again, the Great Depression, which has strengthened the power of unions over American business. And that has, again, hampered businesses and shaped how they're structured and, of course, shaped whether they're able to respond to things like competition from abroad, whether they respond to these incentives. So one is competition.
- 6.43** The second thing that happens in the 1970s is deregulation. This is pretty amazing, actually, to read about the deregulation that happens in the 1970s because it's not superficial. These are laws that get rid of regulations, so they can't be easily imposed. Right now, when Trump comes to the office or a Republican gets in versus a Democrat, the regulatory agency, they deregulate. And then when the opposite party comes in, they undo what was done before because it's not a law. You don't have to go through Congress to actually pass it.

- 6.44 What happened in the 1970s and into the 1980s was that laws were passed to actually get rid of old regulations. A lot of times it's surprising where the deregulation came from, because first of all, most of this happened under Jimmy Carter, under a Democratic President, Democratic Senate, and a Democratic House. They controlled all levels of government, and yet they were the ones who passed the deregulation.
- 6.45 One of the things that surprised me was that one of the biggest pushes for deregulation during this period was Ted Kennedy of all people, pretty leftist Senator. I think it really boiled down to the fact that there was a real constituency complaining about the state of business in America. Inflation was causing a lot of angst among people, and they wanted change, and they needed change, and they thought that these regulations—they thought correctly—were keeping prices higher than they should be. There was a lot of pressure on politicians to do something. And to do something in this case was move towards more freedom.
- 6.46 So you saw deregulation of airlines and prices plummeted. The number of flights dramatically rose. We pay a lot less in real dollars today to fly than we did back then in the '60s and '70s. It's much safer, it's much more convenient, and you have many more routes. And it's not just the U.S. deregulated airlines. Europe airlines are even more deregulated than the United States in some way. And it's a global phenomenon of having airlines set their own prices, set their own route systems, run the business for the most part the way they do. Regulation is mostly limited to safety, and there is some regulation of airports in terms of how many gates each airline has. So there's definitely regulation of that rather than allowing a pure market system. But other than that, airlines are left alone. Not true up until the 1970s. Heavily regulated before than.
- 6.47 Trucking was deregulated: set their own routes, set their own prices, allowed for consolidation. So suddenly, you get consolidated, bigger truckers—much cheaper. The number of trucks and the amount of goods being transported in the trucks grows dramatically. I think I mentioned the other day about the innovation of containers—the innovation of creating a standardized platform to ship things. This is a massive innovation of the 1950s, innovated by a trucking guy who came up with this and ultimately had to buy a shipper to prove that it all works. He had to integrate to show that it worked, but then it became the standard all over.
- 6.48 And of course, once you deregulated trucking, that helped railroads. Railroads were deregulated somewhat, not as much as they should have been. And suddenly, transportation became much, much easier, much, much cheaper, much more intense, and this allowed also for greater trade. This is part of what allowed for globalization. Much greater trade was made possible. Shipping costs plummeted, went down dramatically.
- 6.49 In finance, brokerage companies would deregulate. It used to be that you paid a fixed rate on no matter how many shares you sold or bought. It was very expensive to trade in the marketplace. All of that was done away with. Trading in the marketplace—cost

plummeted to the point where today, how much does it cost to trade? If you're on Robinhood, it's zero. You put in trades for zero. Nobody could have contemplated that in the 1970s. We couldn't even imagine a system like that.

- 6.50** Suddenly it became a lot easier to accumulate shares. It became a lot easier to trade. It became a lot easier to be an investor. The number of people investing in the market grew dramatically. The whole industry changed. Hedge funds came about. Mutual funds became much bigger—they existed beforehand, but they became much bigger. And finance, generally, as we'll see, got a real boost in the 1970s and 1980s. The whole industry changed dramatically, began asserting itself in American business in a way that it really hadn't done since the 1920s and maybe even since the 1910s. Suddenly, American finance was back and dominant and imposing itself on many of these corporate structures, as we'll see.
- 6.51** And of course, communication was deregulated with a breakup of AT&T. Whether that was the right way to deregulate, literally breaking AT&T up, is an open question. But the reality was that the effect that that had is that it allowed for competition. So you got MCI, you got Sprint—MCI later became WorldCom, went bankrupt, came back in other forms—but basically, they both created these long distance phone lines with microwave. But they also dug up the United States and put fiber optics in the ground. Not because they knew the Internet was coming. They put fiber optics in the ground because they wanted to compete with AT&T on long distance, and they weren't going to replicate the copper wires. That wasn't going to be efficient. Instead, they put the first fiber optics cables in the ground from Coast to Coast. It was done by MCI in the early 1980s. It was a competition to AT&T.
- 6.52** That couldn't have happened without the deregulation of the telecommunications space, which allowed for competition with AT&T for the first time. And of course, those cables ultimately would be the cables that are Internet. Now, they've added a lot of cables to that since then, but those are the cables which ultimately allowed for the creation of the Internet. Without that, you wouldn't have Internet.
- 6.53** Of course, from there, later in the 1980s, you get the innovation of cell phones, McCaw Cellular, and others who are innovators, entrepreneurs who go in and use some of their spectrum, which, again, they need permission to use, but they managed to get permission to use the spectrum, and they use the spectrum for telephony. We all know the story because we're all living it right now. How we use our telephones today is not how we used [telephones in the past].
- 6.54** Now, some of us still remember that long distance used to be really, really expensive. Calling overseas was something you almost never did. When I moved to the U.S. in 1987, I was a student. I didn't have any money. I never called my parents who were still in Israel because it cost a fortune. And if I call collect, it could cost them a fortune. You talked to them once every three months because it was just— and if you did talk, you talked very fast. It was short conversations. Today, you can do a video conference.

Anybody in the world, as many participants as you want, at a cost of zero. That's a consequence of this deregulation. That's a consequence of the fiber optics in the ground. It's a consequence of all the innovation that's happened because we freed up markets. It would have not happened if AT&T was still a protected monopoly.

- 6.55** Why did it happen? Partially, as I said, there was a lot of pressure from constituents. There was a lot of pressure from the public that America was not competitive, the economy was not doing well, inflation was driving up prices—something had to be done. Flying was something that only wealthy people could do; most people could not afford it. And there just was pressure on politicians to do it. But the pressure was not just from the public. The pressure was also from entrepreneurs.
- 6.56** Entrepreneurs like the founder of MCI spent real time going to Washington, pounding the table, demanding deregulation. FedEx. What FedEx does today was illegal once. Compete with the post office on parcel delivery? The post office literally had a monopoly, a government-protected monopoly. And yet, Fred Smith, the founder of FedEx, spent real time in DC lobbying for freedom to open up the ability to deliver mail—deliver packages originally, and then express mail and so on. UPS and FedEx were innovations that demanded deregulation. They demanded, and Congress abided by it. It's unusual that Congress moves in that direction, but it did in this period of time. These entrepreneurs were successful in actually lobbying Congress to open up markets. This happened in almost every industry that was deregulated. In the end, there were entrepreneurs who were pushing this, who were really working hard to get this done.
- 6.57** We're going into the 1970s. There's inflation, stock markets all over the place, but we're seeing some signs, positive signs—at least looking backwards, they look like positive signs—deregulation, and we've got competition, which is really challenging. Conglomerates are struggling. Their stock price is way down. They used to be high flying. Now they're way down. They can't compete. They can't compete because they're diversified— you can't compete with a Japanese company that's super focused on making televisions when you're making 150 different things. One division is doing televisions, but nobody's really that incentivized, that focused on maximizing productivity in that division. So conglomerates are not doing well.
- 6.58** And as we enter the 1980s, what we get is, if you will, a financial innovation. It's not really an innovation because it's always existed. But in the 1980s, it got primary focus and it exploded. And that is the ability to do a hostile takeover, to take over a company. To take it over and then extract from it, if you will, the value by selling off all the units, by focusing it on the thing that it does best and selling maybe the TV production company—well, the TV is gone, so there's no TV production company—the insurance company, let's say, to another insurance company.
- 6.59** By this point in the 1980s, the antitrust is less concerned about consolidation, horizontal and vertical consolidation. Now you can actually create that horizontal and vertical consolidation in a particular industry by selling units and consolidating units

from various conglomerates into one company that just does one thing. And indeed, what you get in the 1980s is a large number of takeovers, hostile takeovers of businesses. A lot of this is financed and funded because how do you do a hostile takeover?

- 6.60** You have to buy—these companies are big—you have to buy up their shares, and for that, you need a lot of capital. You need a lot of money to buy up the shares. Where do you get that money to buy all the shares? No one individual necessarily has that money to buy up a company. Well, you need to get financing to do that. By this point, also—it's fairly expensive to buy companies because there has to be a process by which you go.
- 6.61** In 1968, because of the conglomerations, because these conglomerates were being formed, a lot of companies were being bought up. Let's say I was ITT or LTV, I wanted to buy this other company and they didn't want to sell to me. So I did a hostile takeover of them. I could come in and say, "I'm going to buy all these shares at this price, and you got 48 hours to decide if you're going to sell it to me." They would come in, they would overwhelm, they would buy it up, and they would take over companies.
- 6.62** The managers of these companies were upset: "You got to give shareholders a little bit more time. You've got to let us negotiate. You got to save our jobs. Don't let us be taken over so easily." So Congress passed a law in 1968 saying, "Slow down. If you do a hostile takeover, you have to do it slowly. You have to make a tender offer. You have to give shareholders enough time to consider. You have to give managers time to negotiate or maybe find another buyer or maybe restructure and figure out how to avoid your takeover."
- 6.63** These takeovers in the 1980s are expensive. They're difficult to do, and they require a lot of capital. Here, the big innovation was— and I think this was an innovation; I don't think this had been really practiced earlier—what you call the leverage buyout. It's still called the leverage buyout. Basically, what somebody like T. Boone Pickens or Carl Icahn—these were individuals who were corporate raiders; they would go in and buy companies—what they would do is they would secure a huge amount of debt. It would kick in once they bought the company and be guaranteed by the assets of the company.
- 6.64** Let's say I wanted to buy an ITT or a LTV. I would go to bondholders, I would raise this debt, and this debt would kick in, it would be used—the debt would be used to buy the shares, and it would be guaranteed by the assets of the company that they now own. That would be the collateral. A lot of these companies were merged with very little equity and a lot of debt. We'll see in a minute why that's an advantage, not a disadvantage, in this context.
- 6.65** Where do you get the debt, though? We're talking about a lot of debt. This is debt not from one bank because banks are not allowed to do that and can't do that. This is debt

from a bunch of different debt holders, from different institutional investors, different places. How do you aggregate all this? How do you put it together? And then this debt, isn't it really, really risky? There's very little equity, and this debt has to pay a relatively high interest rate. It seems super risky. Who's going to provide it?

- 6.66 And this is the great innovation of Michael Milken. Michael Milken was a financier. He worked for Drexel, the investment bank. He had figured out that high-yield bonds or high-risk bonds, or what later became known derogatorily as junk bonds, were actually a good investment. Actually, they performed better on a risk-return basis than many alternative investments. He basically convinced people that these were good investments to make in these relatively risky bonds that paid a high interest rate.
- 6.67 He used high-yield bonds, he used what are called junk bonds, to fund MCI's first fiber optic cable across the country. Junk bonds were used to fund the first cellular network by McCaw Cellular. Indeed, if you look at the history of telecommunications, every single one of those innovations that required a lot of capital to be invested, to build something completely new, incredibly risky, were funded with Michael Milken's high-yield bonds.
- 6.68 As a consequence, he became the most known investment banker in the country. Drexel ultimately was the largest investment bank in the country until it went bankrupt. And what happened was when T. Boone Pickens and Carl Icahn and these guys wanted to buy a company, they would go to Michael Milken and say, "Can you arrange funding?" And Michael would go to all his clients, the people who bought bonds, and he would say, "Are you willing to fund this transaction?" He would usually easily get the funding.
- 6.69 T. Boone Pickens would go to the company and say, "Okay, I'm going to pay 25% premium over the stock price that it is today; I'm going to buy it up." On many occasions, the company said, "No, we don't want to sell." And they would do all kinds of shenanigans in order not to sell. But usually what would happen is, because now they were under pressure from somebody like T. Boone Pickens, even if they didn't sell, one of the ways they protected themselves from selling was by reversing the diversification by selling off their assets and focusing on what they did well. So instead of T. Boone Pickens buying them and breaking them up, they broke themselves up.
- 6.70 What you got during the 1980s were takeover after takeover after takeover, where companies were broken up. And then other companies looking at that and saying, "Oh, my God, I don't want that to happen to me. I'll lose my job. I'll do the break up myself." Some companies—an attempt to takeover occurred, they resisted it, and then they broke up. But generally, what happened in the 1980s on a massive scale—it's hard to comprehend the scale of this— American business basically shifted from heavily diversified, from a conglomerate business world to a heavily focused, single business-oriented business community.

- 6.71** Corporations completely changed within about a decade. Even LTV and ITT ultimately had to change. There was a hostile takeover attempt in 1985 on ITT. They resisted it. But in response to that, they shed tons of assets. They ultimately remained too diversified in spite of that. And in 1995, they split into three companies. So ultimately, they completely shed their conglomerate nature. LTV ultimately had to file for bankruptcy in 1986 and was split up in bankruptcy. So the conglomerate, as we know it—with two exceptions that I'll mention in a minute—basically disappeared, was gone by, I'd say the mid '90s, but really by the early '90s, most of it had been eliminated.
- 6.72** The market had reasserted itself. In addition to people like Carl Icahn and T. Boone Pickens—just two names that were famous for that, there were many others. By the way, if you want to understand what happened in the 1980s to American business, and you want to understand how hostile takeovers work and the math and why it makes sense and really how companies defended themselves and the whole dynamic of it, there's a movie, a really good movie that basically explains it all and lays it out in an unbelievably entertaining, fun way. The movie is called "Other People's Money." It's with Danny DeVito and Gregory Peck, of all people.
- 6.73** Gregory Peck plays the CEO of the diversified company. It's a conglomerate, and he wants to keep it. "What about the workers? We have to protect the workers. What about social responsibility? What about the responsibility to that community? We need to do this." And he gives a great speech—it's Gregory Peck, after all. He gives us great speech about the American way and helping our neighbors and keeping communities together and never laying anybody off and keeping the businesses together. "So what if they're losing money? One day they won't lose money. Things will get better." And then Danny DeVito gets up and gives maybe the best speech in all of movies about why that's all BS and why the company has to be broken apart and why he's the guy to do it and why it's all about making money. It's brilliant.
- 6.74** So it's highly recommended, "Other People's Money." It's entertaining. And in the process, they discuss so many different issues around corporate governance, so many different of the issues, partially that we discussed here, but also other stuff that is related to hostile takeovers and defenses of hostile takeovers. They each have a bunch of lawyers that are advising them on how to deal with one another. It's a good movie and very, very educational. But as I was saying, in addition to the Danny DeVito character who basically is playing Carl Icahn or T. Boone Pickens guy, somebody who goes in and does these—not a new, but a resurgent form of finance also emerges, and that is the private equity firm. What today are massive firms that everybody's familiar with.
- 6.75** Big firms in those days were pretty small and unknown. Maybe the most known of those is KKR—I don't think I can recount the names of the three partners that make up KKR. But KKR became the most well-known private equity firm in the 1980s, and it found its fame in doing a hostile takeover of R.J. Reynolds—no, RJR Nabisco, which was

a conglomerate that had recently bought Nabisco. Nabisco was a food company; they were a tobacco company, but they had other divisions as well.

- 6.76** I think their CEO was Johnson, and he was famous for the corporate jets and the fancy homes and all of that. It was the largest leverage buyout in history at that point. They paid a huge premium on it. They paid a lot of money to take over this company. Johnson and RJR Nabisco fought it significantly, but KKR won in the end. Looking back, they made good money on it. They went on to do a lot of leverage buyouts and made a lot of money off of it.
- 6.77** One of the things that happened during the 1980s was that if initially you had to go to Michael Milken and he had to go to his bond investors and get their permission, and then at some point—by R.J. Reynolds, this was already the case. All that happened was you'd go to Michael Milken and say, "Hey, I want \$20 billion to buy this company. Can you raise \$20 billion of debt for this?" And Michael Milken would say, "Yes." He didn't ask anybody. He'd just say yes. He would write a letter basically saying, "I, Michael Milken, guarantee that they will be able to raise \$20 billion for this transaction." No bondholders asked. They would take that, and that was as good as gold. Everybody treated it as a done deal.
- 6.78** That sped up the process. It made these transactions much easier, much faster. And it was Michael Milken behind R.J. Reynolds when they took over Nabisco. For all the good work that Michael Milken did during the 1990s, he was rewarded with a jail sentence of 10 years. One of the missions in life that Rudolph Giuliani had when he became the District Attorney at the Southern District of New York—which is an appointment by the Justice Department, he was a Deputy at the Justice Department—was that he asked for the position of being the District Attorney of the Southern District of New York. That is a district responsible for Wall Street. He figured he could make a political career out of going after Wall Street, out of going after finance people. Lots of other people in history have made careers by going after finance people.
- 6.79** Giuliani went to New York in explicit reason to do that. He was very explicit about the fact that he was going to make it his life mission to get Michael Milken. Throughout the 1980s, he arrested financiers over and over and over again for all kinds of supposed violations of the law. Many, if not most of these convictions, were overturned on appeal because most of them were just wrong; these people were not breaking the law. In the meantime, he had destroyed people's lives.
- 6.80** He's the first one, as far as I know, to use the RICO Act. The RICO Act was an act that was passed by Congress to go after the Mafia, where you seize their assets before they're convicted. So first you seize their assets, then you convict them, then you take them to trial. If they win, they get the assets back; if they don't, you get the assets. The idea was the Mafia has the ability to get the assets out of the country, so you want to seize them first. Well, he used that for the first time in a non-mafia-related case with regard to financiers, and he did that throughout the '80s, seizing assets.

- 6.81** For a financier, if you seize their assets, they're done. It doesn't matter if you give them back. They're finished. Their business is finished. They'll never come back. He destroyed a lot of lives during the 1980s. And ultimately, while he was not district attorney anymore, they did go after Milken, and they did manage in a variety of different ways to get him to claim that he was guilty on pretty minor counts. They initially convicted him of—they prosecuted him for 98 different violations, security laws, tax laws, and other laws. Ultimately, I think it was five that he agreed to plea guilty because they were harassing his family and they were doing all kinds of shady stuff.
- 6.82** He pleaded guilty for five. All of them would have generally resulted in a slap on the wrist, a fine, maybe, if it was anybody else but Mike Milken. But the judge in the trial said, "No, I'm going to make an example of you," and sentenced him to 10 years in jail. Ten years. He got out early because he got prostate cancer, and they let them out early. Anyway, that's an aside.
- 6.83** The point is, the private equity got its going during this period with these leverage buyouts. It then became a massive industry. We'll talk a little bit more about it in a future class. All of these companies basically became much more focused. The United States economically did quite well in the 1980s, did phenomenally well in the 1990s. And indeed, the 1990s are probably, to a large extent, a golden age, certainly in modern American history in terms of economic performance.
- 6.84** This is the era, of course, of tech, of the birth of tech, of the industry. This is the dot com era of the late 1990s. It's an era of significant economic growth, a lot of investment. An American industry coming back from, if you were beaten by the Japanese and the Germans, America reasserted itself in the 1990s. American businesses reasserted themselves as real productive giants of innovation and production. This is a renaissance in business during this period caused by all of this restructuring in the 1980s.
- 6.85** People used to ask me when I was teaching in the 1990s, people used to say, "This economy is doing great. It must be Clinton's. Clinton did it, or Alan Greenspan did it." And my attitude was no, "It was Michael Milken, or T. Boone Pickens," or whoever you want to pick, but it was basically the restructuring of American business in the 1980s. They made the '90s and the 2000s and even the world we live in today possible.
- 6.86** Think about it in another way as well. When these conglomerates were broken up and a lot of the unproductive, failed businesses shut down, as you'll see in "Other People's Money," if you watch, some of the assets were just shut down and sold off. This happened all over, primarily in the Midwest. A lot of manufacturing was shut down. It wasn't shut down because jobs moved overseas. It was shut down because America couldn't compete, because these companies couldn't compete. They weren't profitable. The assets were sold off. Where did that capital go?

- 6.87** A lot of capital was liberated, a lot of capital which was in plant and equipment and jobs that were not producing profits. So where did, in a sense, the employees or the children of the employees go? Where did the capital go? Well, it went to Silicon Valley. It's not an accident that when one area and one form of production is diminishing, another is rising. Capital flows there. And if you're a child of a steelworker in the 1980s, and you see your dad lose your job in the steel mills, you're not going to become a steelworker. You're going to go program and move to Silicon Valley.
- 6.88** You can focus on the loss of jobs, you can focus on all of that, or you can focus on the dynamism and the fact that jobs on net were created. So while many, many, many people lost their jobs in the 1980s, many more jobs were created than lost. Unemployment was all low at the end of the 1980s in the beginning, and just the sheer number of jobs was much higher in the United States in the late 1980s than in the beginning. It's just they were reshuffled and reshuffled overall in a much more productive way and in a way that actually enhanced and improved quality of life, standard of living, the things that really matter, and the products that people use.
- 6.89** Now, one other innovation during this period, there was a real realization in the 1970s that something had gone wrong in the '60s. This conglomeration should have never happened. These conglomerates should have never been formed. Something was off about the incentives of managers— we talked about this in a previous class about the separation of ownership and control. And it was a real realization we can't have managers running companies on behalf of shareholders if the manager is not a shareholder. One of the things that really changed in the 1980s is manager compensation. CEOs starting in the 1980s were not paid large salaries in cash. They were paid some salary in cash, but most of the upside was in stock options or stocks.
- 6.90** And now you get alignment: if the stock does well, they do well. If the stock does poorly, they do poorly. And indeed, CEO pay explodes. It goes up a lot because starting in 1982, the stock market only goes up. And because their wages are tied to the stock market, they do phenomenally well. Indeed, in the 2000s, people complained bitterly about CEO pay. But this is CEO pay that's tied to performance. This is CEO pay that is a consequence of them doing well.
- 6.91** Stock options have fallen out of favor for a variety of reasons, coming out of the early 2000s. But now, almost all CEOs own stock, so they get literal stock rather than stock options as part of their pay package. That is true today at Boards of Directors. So the alignment is much closer of shareholders and managers. Therefore, you'd expect it to be less likely to see that conglomeration phenomena again.
- 6.92** All right, quickly, I'll just note that since the 1990s, there's definitely been a backlash against this focus of American business, certainly a backlash against hostile takeovers, and there was a backlash against stock options. A lot of states didn't like the idea that companies that incorporated in their states were getting taken over, and maybe the headquarters moved out of state. A lot of states passed anti-takeover legislation that

made it very difficult for an out-of-state company to buy a company within a particular state. Pennsylvania is famous to having one of the first of these. That was a law that really crippled hostile takeovers. Hostile takeovers today are much, much, much harder to do than when they were back in the '80s. If we need hostile takeovers, if companies start conglomerating again, it's going to be much more difficult to fix the problem than it was back then.

- 6.93** In 2004, I believe, maybe 2002, Congress passed a law called Sarbanes-Oxley. Sarbanes-Oxley was a law that basically increased supervision of companies, internal monitoring, lots of regulations about how companies do their accounting and their internal finances. CEOs now had to sign financial statements with criminal liability, that if the financial statements didn't reflect the truth, they were criminally liable, which had never happened before. It became very expensive to run a publicly traded company because of all these internal controls.
- 6.94** As a consequence, the number of public companies, the number of corporations trading in the stock market, has declined dramatically over the years. I think the peak was in 1998, but it's really come down since Sarbanes-Oxley. If you're a startup today, you're much less likely to go public than to sell because going public is so expensive. It's so expensive to run a publicly traded company because of all the internal monitoring you have to put together, including a lot more controls of the board of directors, who sits on the board and so on, which makes boards a lot less effective than they used to be. So companies prefer to stay private.
- 6.95** This is one of the reasons we'll talk about why private equity is so successful. Private equity has grown so much because companies do not want to go public. They stay private. Private equity is the way in which they raise capital in order to stay private.
- 6.96** Ultimately, in 2010, we got Dodd-Frank, which is a whole other set of regulations this time on banks and investment banks, which again, constrains, I think, the market and hampers it. And we've seen a growing hostility towards big business again over the last 10 or so years. And as a consequence of that, we've seen a new, as we talked about, antitrust regime come about that is very suspicious of large business, particularly in the tech space, but generally of large businesses. And finally, we're seeing a growing interest in industrial planning and central planning from the federal government, both in the Trump administration and the Biden administration, an inclination to try to run things from the center, something we abandoned for the most part, with a little bit sprinkled in here and there after World War II.
- 6.97** So where are we today? That's what we're going to turn to in the next class. We're going to talk about the tech industry. We're going to talk about its evolution. This is the dominant industry of the 21st century. It was the dominant industry in many respects at the very tail end of the 20th century. It has a fascinating evolution in terms of how it came to be and the dynamics between the regulations, antitrust, and market dynamics that were involved. We'll turn to looking at that in our next class. Thank you.

Lecture 7: The Rise of Silicon Valley

At a glance — The 1980s restructuring and the rise of Silicon Valley.

- 7.1 So last class, we talked about how in the 1980s, American business was restructured. American corporations were restructured, conglomerates were broken up, and American business became a lot more focused. Also, a lot of capital, capital that was not efficiently used, was released and ultimately made possible, I believe, the boom in what has become the dominant industry in the United States, and that is technology, primarily in Silicon Valley. So today, let's talk a little bit about Silicon Valley.
- 7.2 Now, Silicon Valley has its start way back in really the '30s and the '40s as a place where a lot of eccentrics went to live— hobbyists, radio hobbyists, tinkerers, inventors. It was a place where a lot of people with strange, maybe, political views, utopian views, also went to live. A lot of hippies, ultimately. There were a lot of hippies in San Francisco in the 1960s. This started way before in the Bay Area. So the reputation it has, even today, is somewhat justified.
- 7.3 The first recognizable business that we would recognize today was founded in 1938 by David Packard and William Hewlett. HP, Hewlett-Packard, was an electronics company founded in 1939. And around it, a bunch of other small businesses were started all in electronics, electronics of the '30s, '40s, '50s, '60s. So even before we get the real boost of what became Silicon Valley, there was already a lot of businesses there tinkering with electronics, radio— a lot of hobbyists, a lot of people who like to tinker with this new technology.
- 7.4 These electronics firms, in spite maybe of the politics of the time, these electronics firms all did really good business during World War II and the Korean War. And really Silicon Valley in that period—Silicon Valley being south of San Francisco, the area that today is Mountain View and San Jose and that whole area—that became a defense area, an area where a lot of the electronics that they were building were ultimately going to use for defense.
- 7.5 Of course, the other huge advantage that area has, that positioned it well to become what it became, is the existence of Stanford University and the focus of Stanford, and Berkeley, on STEM, particularly on STEM fields. And in Stanford, in particular, a focus on STEM with an eye on commercialization. Stanford is probably the leading university in the world, or has been the leading university in the world, in terms of taking research done at the university, taking research done by professors, and actually getting it ultimately commercialized and starting companies. We'll identify a few leading companies that came directly out of Stanford.
- 7.6 You've got a place with a lot of small independent makers of electronic components, no big dominant company in the area. This really became a fertile ground for startups, for entrepreneurs, and for a lot of spinoffs. Companies would grow and immediately spin

off. And again, we'll see a couple of big examples of this, really, Silicon Valley was created from spinoffs, and we'll talk about that.

- 7.7 They also started to create a network effect. A lot of people doing about the same stuff in a particular geographic area. People get to know each other. They get to learn from each other. People can switch around. They can put teams together and start new companies. It creates an ecosystem. Just like Detroit did that for automobiles, Silicon Valley was doing that for electronics, electronics components, and anything to do with electronics.
- 7.8 Now, really, the beginning of the Silicon Valley we know today really starts with a company called Shockley Laboratory. Now, Shockley was this amazing engineer who had worked at Bell Labs. At Bell Labs, he had built the first transistor. This was a massive breakthrough, of course, changed everything. And initially, very difficult to build, very difficult to make. A lot of work had to go into refining the process and bettering the process. And ultimately, he left Bell Labs in order to found his own business devoted to semiconductors, which is what he had discovered in building a transistor. So he moved to California, started this up in Mountain View, I think, and opened up in 1956 and started a semiconductor business, the first real semiconductor business in the United States, Shockley Laboratory.
- 7.9 Turns out he wasn't a great boss. People didn't like working for him. And in 1957, just a year and something after he started the business, eight of his best employees left. They have gone down in history known as the Traitorous Eight. They turned their backs on Shockley and left. And they went on and they founded a company called Fairchild Semiconductor, not far from where Shockley was. They started it in San Jose, California. It was actually a subsidiary of, and it was actually not a subsidiary, it was actually financed by a guy named Fairchild, who had a company called Fairchild Camera and Instrument. And they set this up under the Fairchild name in San Jose.
- 7.10 And really, this is where the first real big breakthrough in semiconductor manufacturing happened. They discovered what's called the planar process. And the planar process is basically a way to create semiconductor chips through taking material out. I'm not an expert on this, but carving, in a sense, out of the silicon instead of adding to it. And this still today is the way in which chips are made—much more complex today than it was back then—but this became the process.
- 7.11 They became the dominant firm in producing semiconductors. For a while, TI picked up on the same process in Texas Instruments and started producing semiconductors, and I think TI is in Dallas, Texas. But Fairchild Semiconductor was the dominant. This is where the big innovations were done. This is where the big names, Robert Noyce, Moore, from Moore's Law. This is where they were working. These are part of the Traitorous Eight that left Shockley.

- 7.12** Now, Fairchild had an option in his investment in Fairchild Semiconductor, where after a few years, he had the right to integrate this investment into his company, into his camera and instrument company, which he did, which he did a few years after the company was founded in '57. So at this point, Fairchild Semiconductor became one unit in Fairchild Instrument's portfolio. Ultimately, as the '60s started, Fairchild, like pretty much everybody else in the 1960s, decided he wanted to become a conglomerate and started buying a bunch of other businesses. Indeed, for a while, the stock price went way up.
- 7.13** But the engineers at Fairchild were not benefiting from this increase in stock price. In those days, you didn't compensate your engineers with stock options or stock. They were salaried people at Fairchild. They were making these revolutionary discoveries. They were building the infrastructure for a brand new industry. And yet, financially, they weren't being rewarded really for it fully. And most of the return was going to Fairchild. And as a consequence, right—
- 7.14** Now, one of the things that happens in conglomerates, of course, is businesses succeed, fail. But Fairchild was making a lot of money—Fairchild Semiconductor, was making a lot of money—and Fairchild ended up using a lot of the profits from there to keep the rest of the businesses afloat. And that, of course, also irked the engineers who were building and making possible these immense profits, one of the great disadvantages of the conglomerate form. And as a consequence, people started leaving.
- 7.15** And a lot of people ultimately left Fairchild. And this is the beginning of Silicon Valley. I mean, pretty much every significant company in Silicon Valley has its roots in Fairchild. So people left. One of the chief engineers, at Shockley, left and went to revitalize a company called National Semiconductor, which became one of the largest manufacturers of semiconductors, ultimately. Applied Materials, which built machines to make semiconductors— the machines that etch out the semiconductor—was founded by people who were leaving Fairchild. AMD, which to this day is a major competitor to Intel, its founders came from Fairchild. Amelco, another semiconductor company that actually merged into others and is still around in some form or another today.
- 7.16** Probably the most well-known is Intel. Robert Noyce and Moore left Fairchild and founded Intel, maybe the most consequential, and really the most consequential Silicon Valley company, that really spurs the technological revolution and becomes, at some point, the largest manufacturer of semiconductors in the world. It has a rocky road. We'll talk about that. But all of this coming out of Fairchild. And in addition to that, the venture capital industry, as we know it today, spun out of Fairchild.
- 7.17** So two of the probably most successful venture capitalist firms in history, at least over the long run, and particularly in the past, of Sequoia Capital. And think about that: Sequoia Capital funded Apple, funded Cisco, funded Amazon, funded, I don't know,

hundreds of companies, most of the companies we know today. Most of the companies we know today as part of the internet or a part of technology were funded by Sequoia. Again, the guys at Sequoia came from Fairchild.

- 7.18 Second company was Kleiner Perkins. Pretty much any company that wasn't funded by Sequoia that you know today was funded by Kleiner Perkins. So between those two firms, they funded most of the basic cornerstones of what we today recognize as a technology industry in Silicon Valley. Both of them, spinoffs from Fairchild. Kleiner Perkins also does, I think Sequoia does, too, quite a bit of biotech. Even the biotech industry is a product of these companies.
- 7.19 Overall, there were 29 spinoffs from Fairchild. There are maps that have been drawn in terms of starting from Fairchild, and all the spinoffs, and all the children of the children of the children and where it gets, and it basically dominates technology. It dominates. It all starts there. They're still called the "Fairchildren," the children of Fairchild, and the grandchildren and the great grandchildren, and they're all there. You can find online; you can find these maps out there.
- 7.20 What we see here is the dynamism, I think, of a process here, of American corporation, of American business. The ability of people to leave, start new companies; the ability to do these spinoffs, the ability to reject, let's say, a failed business model like the Shockley or failed business model like the conglomerate under Fairchild. You're not stuck in that model.
- 7.21 Again, I think as compared to other countries, particularly Europe, where there's a lot less of this dynamism. There's a lot less of this entrepreneurial spirit that leads an engineer, who's got a pretty good job doing amazing things, really cool, it's interesting, and so on. Why leave? Why risk it all on a venture in a sense? Why become a manager? Why become an entrepreneur? And in the U.S., particularly in Silicon Valley, the spirit of entrepreneurship is really still alive.
- 7.22 And a lot of companies, even today, that are new companies that are created are often spun off of people who are leaving other companies to start these new companies. A lot of the AI today, if you look at AI, if you draw a map of all the AI companies, and there are a lot of them now, how many of them start out with OpenAI, or how many of them start out with one or two or three of the key AI companies? How many of them have generated all this flurry of AI companies? Usually, it's an employee from one of these companies leaving and starting something else.
- 7.23 One of the, this is for later on in a sense, but one of the other big spinoff companies, big dominant spinoffs, ultimately was from PayPal. The PayPal Mafia, as they're known in Silicon Valley, is a huge group of people who worked in PayPal in a particular period of time in the 2000s, when its founding through the early '20-teens, then went out and created what? Tesla and SpaceX and everything else Elon Musk has made. Elon Musk was at PayPal. Peter Thiel. Everything Peter Thiel has funded— Palantir is probably the

most well-known company right now that he funded— but he has funded hundreds of companies, and he was CEO of PayPal.

- 7.24** LinkedIn, Hoffman was one of the PayPal Mafia's, but there are lots of others. There are dozens and dozens of these people who all trace back— many of the internet companies or the tech companies that were created in the '20-teens, you can link back to PayPal. And what it is about the PayPal culture that produced this—because there are lots of other companies—it's hard to tell, but it's a reality. A lot of the most successful people coming out came from PayPal.
- 7.25** So there's something about the dynamism. There's something about the ability to leave, the ability to raise capital, the ability to start a business, that is unique to Silicon Valley, in a sense, unique to the U.S., much more difficult to do elsewhere, and we don't see the same pattern.
- 7.26** By the way, the same thing happened in the auto industry in Detroit. An engineer would work for an auto company and get frustrated because they didn't like his design, or they didn't like his newest engine, or didn't like this. He'd leave, start his own auto company. And this is why there were dozens of auto companies, but many of them—again, the spinoff, going and doing your own thing and starting your own thing. And then ultimately, there was consolidation in the auto industry, and we got ultimately three firms, but the way it was created was, again, through these engineers leaving to do their own thing.
- 7.27** Intel, of course, is a dominant player here. Intel, starting in the late 1960s, is spun off, and then in 1971, they produced their first microprocessor, and they become the dominant player in the microprocessor space. The first microprocessor as one that is programmable. So before that, microprocessors could do one thing, and you'd have to design each microprocessor for different functions. You'd design it in the hardware to do different things.
- 7.28** Intel produced a programmable, cheap processor that could be programmed. And of course, this becomes really crucial for the next big innovation that's coming, which is the personal computer. Now you have something cheap. At some point, you could buy them for \$75, an Intel chip with quantity discount. And of course, you need that if you're going to sell personal computers to large numbers of people.
- 7.29** Let's talk about personal computers and how this industry came to be and how it evolved. Again, personal computers started with hobbyists, people just tinkering, people just playing around. And there were a lot of them already in this part of California and Silicon Valley. There were already a lot of people playing around with electronics and playing around with computers. The basic Intel, the fact that they could buy Intel chips and connect them and buy other pieces from different companies and play around with them and create their own designs. You know, it was was a lot of creativity, a lot of people tinkering and a lot of people creating new things.

- 7.30** The first personal computer, it's hard even to call it a computer, but I guess it was. It was the Altair 8800. It was basically a box with lights that turned on and off. It was a "yes/no" thing. You programmed it with little switches. You'd up and down toggle switches, that you would do. And that's how you asked it questions or programmed it, in a sense, and it would give you a solution. It had no monitor. It had no keyboard.
- 7.31** I remember I was in high school, and I was in Israel, so we were a few years behind. So this would've been '75, but by '78 maybe, one of these machines showed up in a classroom in Israel, and it was not impressive. I mean, it's sad because maybe if I'd been really excited about it, I would have gone into computer science and been a billionaire today. But it just wasn't that impressive. But it was interesting.
- 7.32** So from there, a lot of these, there were a lot of these, again, hobbyist clubs. One of them was called the Homebrew Computer Club. In a sense, out of Homebrew came Apple; Wozniak and Jobs presented there their ideas. But in 1977, RadioShack came out with a computer called TRS-80 and Commodore. I don't know if anybody—I remember the CommodorePET. Now they had a keyboard, they hooked up to monitors.
- 7.33** Now you started to have what we would recognize more as a computer. But these were still very primitive. They could do very little. They didn't connect to anything. I mean, they could connect to maybe a printer, but there were really no printers, not as we recognize them today. It could connect to peripherals, but the peripherals had to be produced in the industry—peripherals were just starting out. There wasn't a lot of that.
- 7.34** This is also the period in which Jobs and Wozniak, the founders of Apple, got together and started tinkering with building a computer. Wozniak was working at HP at the time, at Hewlett-Packard. Jobs was I was working at Atari, which was a gaming company. And they'd got together and they had put together some basic computer. Wozniak was really the tech guy, and Jobs was the business marketing vision guy, very much a vision guy in terms of where they could take it.
- 7.35** They presented this at the Homebrew Computer Club with all these hobbyists around, and people got pretty excited about it. They went to HP and they said, "Hey, would you like to produce it? Team up with us to do this?" It was just two guys tinkering, really, in the garage. By the way, Hewlett-Packard, the whole idea of starting your business in a garage, Hewlett-Packard started their electronics business in the garage, so it became a theme in Silicon Valley. People lived in single-family homes with a garage, and that's where the business was started.
- 7.36** HP said, "No, I mean, this is not our business. We're not interested. It seems like a small nichey thing. Not that important." By 1977, they had raised some capital; Sequoia had raised capital. And Don Valentine, who had been at Fairchild, met Steve Jobs, was not particularly impressed by him, except, I mean—this is a period where Steve Jobs was

going through a phase where he didn't believe in showering, so he wasn't particularly impressive.

- 7.37** But he was impressed by the guy's vision. He was impressed by his passion, by his motivation, and the vision he had to what it looks like, what a computer could look like, and how it could change the world. Sequoia ended up writing a \$500,000 check, which made all the difference in those days. Other investors came in. They brought in professional managers. Again, at the suggestion of Don Valentine, they hooked up with a professional manager, brought them in, and ultimately built what came to be known as the Apple II.
- 7.38** The Apple II made its debut at the first West Coast Computer Faire in spring of 1977. It soon became quite a hit. People started buying it. Now, the Apple II was a plastic case with a built-in keyboard. I don't know if you've seen pictures of this. It could be expanded—think about this—it could be expanded from 4K to 8K memory. 4K. You can't do anything with 4K today. 4K to 8K memory. It actually could drive a color monitor, and it connected to a cassette recorder. There were no drives—and it had on a chip—it had a version of BASIC.
- 7.39** I don't know how many of you have ever programmed in BASIC, but BASIC is a relatively simple, straightforward programming language. I used to have a— when did I have this—probably in the early '80s. I had a Casio calculator. It was about like this big with a nice screen on top, completely programmable. I could basically program my entire engineering exams on the thing and professors never knew because they didn't realize these things were so programmable. They just thought it was a calculator. But you could program the whole thing. You could program a really complex, hundreds of lines of code, into one of these Casio programs. Well, they were using BASIC on the first Apple, too.
- 7.40** Sales took off. In a sense, the rest is history, although the history of Apple is more complex than that. But sales took off. And at the beginning, in this stage, they basically relied on outsiders for almost everything. So every piece that went into this computer was brought in from the outside and it was assembled and delivered by Apple.
- 7.41** The "killer app" ended up being, the killer app for the personal computer and the killer app for the Apple II, ended up being a program called VisiCalc. VisiCalc was the first spreadsheet. And spreadsheet is a super cool application. It's uses of—there's no end to the number of uses you can put to a spreadsheet. Of course, today this is dominated by Excel, and Apple has its own spreadsheet called Numbers. But VisiCalc in 1979 really became the app that everybody needed to have, and it became quite successful on the Apple platform.
- 7.42** Surprisingly, Apple never really took advantage at that point of its success to break into the business market, and this led to a significant potential for somebody else to come in and take some of this PC market. Apple IIs were primarily bought by individuals,

maybe small businesses, but never really went into the big business market. That, in a sense, was left for IBM to pick up, and ultimately for the IBM clones to pick up.

- 7.43** IBM, of course, was a dominant company in the mainframe business. They had built these big computers that filled rooms, and they had indeed dominated that industry to the extent that they were prosecuted by antitrust starting in the late 1960s, a antitrust lawsuit that wasn't really resolved until 1982. This took well over 10 years in the courts, fighting back and forth for, ultimately, a deal to be cut between IBM and the Justice Department. But IBM dominated the mainframe business. There were other companies that were also making mainframes, but IBM had a huge percentage of the business.
- 7.44** IBM was also competing with companies making supercomputers. So computers that were even faster, more specialized than the mainframes. In the 1980s, a company called Cray was making what was called a supercomputer. By the way, your iPhone today is significantly more powerful than a room of a Cray supercomputer. And think about this. What are we, 40 years later? In 40 years, it's just unbelievable the degree to which what we have today is so much more powerful than what we had just 40 years ago. We'll get to how that happens. That's Intel's story in a minute.
- 7.45** So IBM was dominant, but it also became a very centralized corporation. A lot of control by senior management. There was a senior management committee that basically oversaw everything at IBM. IBM also primarily viewed itself as a computer servicing company, at least in its origins. It leased the machines. It helped people set them up. It helped people figure out how to use the computer for their business. They were really there—they didn't sell the machine. They didn't make the money by selling it. They made the money by servicing it over time. Not just the repairing, but also helping companies use it and figure out its best uses.
- 7.46** And because of the antitrust lawsuit, they were forced, ultimately, to start selling machines. The culture in IBM changed significantly. IBM almost went bankrupt and had to really change its model—a complete new business model. In the 2000s, they had a—maybe it was before the 2000s—they had a CEO come in who basically completely restructured the company and focused it back on services, which is what it does today. IBM is primarily today not a hardware company, but a computer IT services company.
- 7.47** Anyway, they had created—so they were highly centralized. And when engineers came to them with this idea of, "Personal computer, it's going to become a big thing. We should be in on this. We have skills that match up with this." It was very difficult for IBM because—it was hard for them to figure out how to do this within the context of what they were doing at the time.
- 7.48** And indeed, they set up a little committee to investigate this personal computer. The committee came back and said, "Look, you can't do it. To do this, you would basically have to create a startup mentality, a new company within IBM. You can't go through

the bureaucracy of IBM and get this stuff done fast enough to make a dent in the personal computer business." But to their credit, that's what they did.

- 7.49** They basically created a unit within IBM that reported directly to CEO, not to the committee, not to the bureaucracy, a unit within IBM. It was given autonomy in a sense that they could build what they wanted. And then when they built it, they didn't have to rely on IBM to supply it. They could buy outside of IBM. They could create a completely independent business unit like this. So very innovative and very difficult. They operated like a startup, no interference for management, and no bureaucracy with an expanded mandate.
- 7.50** They started to build this computer, and they built it basically on certain standards that were, in a sense, being determined from the outside. They built it on a bus that was already being used. A bus means the way, I guess, the computer communicates with its various parts. They built it on a chip that was made by Intel. Then they needed an operating system. They needed software.
- 7.51** And they went to a small Seattle company by the name of Microsoft, who was at the time building a BASIC operating system—BASIC is the programming language for Atari. They came to them and they needed this new operating system partially because they were using a new Intel chip that nobody had written an operating system for. Nobody was using it yet.
- 7.52** Bill Gates, CEO of Microsoft, went out and he actually found another small company in Seattle that had actually already started, and almost completed, an operating system for of the Intel 8088, which was the new chip, and he basically bought it from them. He bought the operating system from this other company. He put the finishing touches on it. He completed the project, and he sold it to IBM.
- 7.53** Now, smartly, he sold it to IBM, but he insisted that he could lease it to other companies. This is MS-DOS, what we know as MS-DOS. And so he sold it to IBM. IBM could use it, but then he could lease it to other companies using the same, basically, Intel platform and basically would use the same software.
- 7.54** IBM, this big corporation and incredibly successful, was not building a PC where all its key components, I mean, the real key components, the bus, the chip, and the operating system, it didn't really own. Every one of those was owned by somebody else. Intel could sell a chip to anybody. The bus was a standard, and now the operating system was really owned by Microsoft.
- 7.55** IBM put out the machine and priced it fairly cheap, and it started selling it for the first time, setting up to sell it retail stores, Compuland and other electronics chains. To build the machine, it used competitive bidding. It got cheap good parts from out there. But again, if IBM divisions could bid on it, but there was no commitment to actually going with an IBM bidder.

- 7.56** And it was a huge success. By 1983, IBM PCs dominated, well, had 23% of the market and were becoming the biggest personal computer in the U.S.. I remember in '83 or '84, I was a student at the Israeli Institute of Technology. We had no PCs at that point. The university put out a bid out there to buy computers for all the professors and to set up computer labs at the university. This makes sense.
- 7.57** They got three bids. They got a bid from Apple. I think it was still the Apple II. They got a bid from IBM, for the IBM PC, and they got a bid from Digital. Digital had been a producer of microcomputers, which were not mainframes, smaller, and targeted for particular industries, particularly for research. They were in a lot of universities,. And there were DEC, D-E-C computers, and they also had entered into the PC market.
- 7.58** And of course, Digital used a completely different operating system. They didn't use the Apple operating system, and they didn't use MS-DOS. They used their own proprietary operating system, and their computers were—their PC was, again, keyboard and screen and everything was nice, and it was a good PC on their own operating system.
- 7.59** And of the three, the Israeli Institute of Technology chose Digital. By the time the computers arrived in Israel, Digital was out of the PC business; it already shut down. And basically, there was no support for the operating system because nobody else was using this operating system. And by the time they got the PCs, they were obsolete. They could have chosen IBM, and ultimately, they used them for a few years, and ultimately, they replaced them all with IBM clones.
- 7.60** This was an incredibly dynamic, fast-moving industry in those days. There are constantly competitors, constantly people coming and going, and people shutting down. Dell, ultimately, different units of it get sold in different ways and ultimately it disappears. It doesn't exist today as a computer company.
- 7.61** By 1988—so '83, they have 26% of the market. By 1988, IBM clones, companies that are basically using very similar components to what IBM is using—the same bus, same chip, and same operating system—dominate IBM. They dominate the market. By 1988, you're buying a computer from Compaq. I don't think Dell has started yet, but you're buying from a whole array of companies that are selling basically IBM machines.
- 7.62** There was a story that in Asia, where some of these IBM computers were being manufactured, or assembled, really, they would, during the day, make the ones with the IBM logo, and then in the evening, they would make the ones with some clone logo on it, basically same machine, and distributing it out there. IBM did not control the essential parts of this. They had no control over the IP. They had no control over the market, and they basically lost it completely.
- 7.63** During this period, of course, use of PCs explodes. In 1984, about 8% of households in the U.S. had a PC; by '93, 22%; and by 2000, 50% of households have a PC. So this is an industry that grows really, really fast. And what we see here is that—the 1970s, the 1980s, really starting in the late 1960s—this process of slowly building momentum. And

what we get in the United States is the incubation of a totally new industry, a completely new industry formed out of nothing.

- 7.64** People love to bemoan the loss of American industry, the loss of the wonderful 1950s, and what was an American business so much better in the 1950s and 1960s. No, it wasn't. A lot of that manufacturing, a lot of the jobs that went away, a lot of the capital that got redeployed, made possible this completely new industry that at least at this stage in the early 1980s is dominated by the U.S., is dominated by American tech companies.
- 7.65** And to this day, there's no equivalent of the Big Five tech companies in the U.S. anywhere in the world. Maybe in China, but China has created a tech ecosystem that's somewhat isolated from the rest of us. But there's nothing like this in Europe. Everybody in Europe uses American products. There's nothing like this in Japan. There's really nothing like this anywhere in the world. All right.
- 7.66** So let's talk now about just another layer here, and that is the chip, the microprocessor that is so important for this industry. Coming into the 1980s, the U.S. dominates microprocessing. And the moneymaker at this point for Intel, which is the largest chipmaker in the world at this point, is the memory chip. You remember that 4K to 8K? Right now they're getting up to, in the 1980s, they're getting up to 1 megabyte.
- 7.67** But Intel dominates this, and it's the largest chipmaker in the world, and it's making a lot of money, and it's where all the innovation happens. So going and getting more and more and more memory is happening at Intel. Of course, this is partially driven by Moore's law, which is a law— I guess it's not a law of physics, but it's a law of business, a law of industry, a law of technology— articulated by Moore, who left, of course, who is one of the Shockley Eights, and—I think he was one of the Shockley Eight—but he was one of the people who left Fairchild.
- 7.68** Moore's law says the number of transistors on an integrated circuit will, or should, double every 18 months. Now, if you have a calculator, you can pull it out and calculate what that means. But that is, talk about exponential growth. That's why you can have hundreds of millions of transistors on a semiconductor right now. It's a consequence of this Moore's law. And Intel, where Moore was working, was driving this. They were driving the innovation and driving it forward.
- 7.69** But starting in the mid-1980s, Japan, Japanese companies, had figured out— Intel had become so good at basically iterating to the next Moore's law, to the next level of these DRAM chips, to these memory chips, that the Japanese figured out how to do it themselves. And what the Japanese brought that Intel didn't quite have, to the degree the Japanese had, was quality control. So the Japanese specialized not in innovating—at this point, they were just doing what Intel had done—and now they could produce better, faster and faster, more transistors on a chip because the process had been determined by Intel.

- 7.70** But what they brought was very little, very few duds, very few problems chips. The quality control was by far superior to anything the American has. And if you think about it, that's what made ultimately Japanese cars so much better than American cars. It's the reliability. They were cheap to maintain. They didn't break down often. That manufacturing mentality that the Japanese had, that focus on detail and quality, they now brought to the chip industry.
- 7.71** And by 1986, Japan was manufacturing more microchips than the U.S. was. And there was real panic in the U.S.. The U.S. was losing its edge in chip manufacturing, and Japan was taking over. Japan will ultimately, supposedly become the largest economy in the world. Sound familiar? And they were beating us at everything, cars, televisions, and now microchips. The American industry, microchips, they were now beating the U.S. even at that.
- 7.72** And of course, Intel's business had been these memory chips. And they had relied on it, they basically invented the business. And they were in a quandary; they felt like they were in real distress at this point. And Moore and Noyce had to go away and re-strategize, rethink what it is that they were doing. So Japanese are better than Intel, at actually producing the chips that Intel is innovating and that Intel, invented, if you will, came up with originally.
- 7.73** Part of what the semiconductor manufacturing industry started to do in the early 1980s, as a consequence of this Japanese competition, is lobby Congress. They started lobbying Congress for protection. And there was this belief, turns out false, but there was this belief that the reason the Japanese were so good at what they were doing is because the Japanese were getting help from the Japanese government. Again, very familiar, if you follow debates about China today, it's very familiar.
- 7.74** Japan had a industrial planning board called MITI, M-I-T-I, that allocated resources to different companies, both in R&D but also in manufacturing, that helped joint research and all kinds of other stuff. And there was a lot of envy of the Japanese model. The Japanese model seemed to be beating the United States. Of course, looking back, we look at media and we look at what they did, it's hard to attribute much of the success in Japan to that. They allocated money to some of the industries.
- 7.75** The reality is that American industry was selling to the Defense Department—in a sense, getting money from the government in that way—far more than what MITI was allocating to the Japanese companies. In a sense, and in a much healthier way, because a lot of the money the MITI distributed was wasted; it went nowhere. Whereas if you're selling a product, the product has to be done already. And you're getting resources in order to continue producing products that you're already good at.
- 7.76** It turns out that Japanese companies didn't really want to cooperate with one another. They were just as competitive as American companies. Even though they created these consortiums of cooperation under the framework of the Japanese government, they

didn't do their cutting-edge research there. They did it at home. So it turns out that the Japanese model wasn't quite as lucrative, and it wasn't quite as the driver of success that people attributed to it or people assumed that it was.

- 7.77** But the American semiconductor industry was eager to try something because they were being beaten. They were being beaten in a competitive game. So the first step was Micron, which is a semiconductor company that was founded, of all places, in Boise, Idaho, with money from the french fry king. I don't know if you're familiar with this billionaire who basically figured out how to provide potatoes for french fries to McDonalds and became a billionaire by being the biggest manufacturer of potatoes in the United States. Anyway, he was the investor in Micron semiconductors that became a big manufacturer of semiconductors.
- 7.78** They filed an anti-dumping lawsuit against the Japanese companies. Anti-dumping, of course, is you're selling it at too low of a price. Remember, we talked about antitrust and the fact that too low is problematic because it keeps competitors out. Now, how the company can survive if it's selling too low, below cost? Well, the idea is: all the competitors fall away and then it raises prices and makes the money again. But then why can't the competitors come back in? It just doesn't make any sense from economic theory perspective.
- 7.79** And there was no evidence the Japanese companies were selling below cost. They were just cheaper producers because they were doing a better job, partially because they had fewer flaws, fewer bad chips in any run, they were a much cheaper producer of chips than American companies were. So they were anti-dumping.
- 7.80** There was significant pressure on Congress. And ultimately, Reagan, who had negotiated a deal with the Japanese to limit the number of cars coming into the United States to save the American auto industry, now negotiated a deal—or basically didn't negotiate a deal—basically imposed 100% tariffs on chips above a certain level, above 300 million of imports from Japan. So 100% tariff in order to hold back the Japanese dominance of the industry.
- 7.81** In addition to that, Intel and a number of the other, about a dozen other chip manufacturers in the United States, got together and formed a consortium, a research consortium, to try to mimic the industrial planning that was happening in Japan. They got together. It was called SEMATECH. SEMATECH was based in Austin, Texas, and it got about \$100 million from the U.S. government, from the Defense Department, I think, to start. Over five years, they got 100 million each year, \$500 million they spent. Primarily, it ended up being focused on improving the machines that make the chips themselves.
- 7.82** Nobody really thinks this was a great investment. Did they make some marginal improvements during this time? Maybe. But nothing really came of SEMATECH. And it certainly wasn't the thing that resurrected chip manufacturing in the United States.

- 7.83 And ultimately, today, it's just like a trade organization. It doesn't really do anything. It doesn't really do anything substantial. After five years, most of the participants left. Again, you had all these competitors coming together to do research together, and most of them were like, "Wait a minute. We'll do our own research, thank you, and we'll compete with you guys."
- 7.84 What led to Intel's resurgence? Because Intel, by the late 1980s and early 1990s, is the biggest chip manufacturer in the world again. What led to it are two things. One: significant improvement in manufacturing. They learned from the Japanese. They figured out how to make chips without flaws, with fewer flaws, how to just do them better. And they started to match the Japanese in terms of just quality control.
- 7.85 But second, and much more important, was a strategic shift. Intel basically shifted from making DRAMs, from making memory to making logic chips, to making real microprocessors that go into computers, that go into servers, that go into basically so many of our goods. They moved away for something that was relatively a commodity. It was relatively easy. Yes, you constantly had to innovate to get more and more transistors onto it, but it was a pretty static thing, memory.
- 7.86 Versus logic chips that were far more complex, required a lot more engineering and a lot more thought and are the things that drive our computers. And they turned to that and became dominant in that. Really after a while, the only competitor they had was AMD. Remember another one of the Fairchild children? AMD became a competitor, still is to this day a competitor of Intel's.
- 7.87 So interestingly enough, because of these tariffs imposed on the Japanese companies, and because of now the intensive competition from American companies who are matching the quality of Japanese, the Japanese started to decline in memory chips. And the focus shifted from Japan to where would it go—in terms of making DRAMs really, really well and at cheaper and cheaper prices? South Korea.
- 7.88 So South Koreans became the largest manufacturers of DRAM chips. Samsung, by the early 1990s, was the largest DRAM manufacturer in the world. Think about that, too, in terms of TVs. Think about that in terms of automobiles. Japan becomes, again, a high-end—just like the United States—shifted from manufacturing, to services or to design at a high-end. Japan went through exactly the same process, shifted manufacturing to Korea. And then, of course, some of that is shifted now to China. And where does it shift next? But a lot of the simple manufacturing stuff is constantly moving to where it can be best and cheapest manufactured.
- 7.89 Go back to the PC. One of the things in the story of Apple that is one of the major events in the story of Apple is a visit Steve Jobs makes, I think in the very early '80s, to the Xerox Palo Alto Research Center. This is Xerox, the copying machine. But it turns out that the research center, the R&D part of Xerox, was doing amazing stuff, was coming up with amazing ideas, amazing products, but nobody at Xerox appreciated it.

They were a copy machine maker, and they did copy machines. And this whole PC stuff was not of interest. And nobody could see the potential of the things that Xerox had developed.

- 7.90** And Steve Jobs did this tour. He was reluctant to go, he didn't see what the point was. And yet what came out of it is a modern PC that we have today. What did he discover at Xerox? Bitmap graphics, the graphic interface that we have today on a computer screen. Overlapping windows. Computers in those days had one window. You had to close it, get rid of it. Overlapping windows, huge breakthrough. And shockingly, the mouse. The mouse was a creation of Xerox.
- 7.91** Jobs, who recognized the value of these things immediately, as soon as he saw them, immediately went to Apple and started building a computer based on these ideas. Changing the operating system, introducing the mouse, integrating graphics into the screen, overlapping windows, all of that—which Apple II did not have—he now embraced fully.
- 7.92** The result of this was a massive flop. It was a computer called Lisa, L-I-S-A, some of you might remember this. It was a \$10,000 computer. It was brilliant. It could do amazing stuff. But nobody wanted to pay \$10,000 for a computer. Of course, all of that, ultimately, Jobs took and built into the Macintosh computer. And the Macintosh computer was a revolutionary computer. It had all of this. It was also relatively inexpensive. It had a built-in screen. It was this model of industrial design. At the time, it was considered breakthrough and beautiful and amazing.
- 7.93** My first computer was probably 1990, '89, '90, '91, somewhere around there. And it was a Macintosh SE/30. So this was a more advanced version of that original Macintosh. Tiny little monitor right in the middle of this box and a keyboard and a floppy disk, which was again a brand new technology. And I thought it was like magic. I thought it was the greatest thing ever. It was just amazing. I wrote my dissertation on that one, did all my statistical analysis on that.
- 7.94** Macintosh doesn't go on to dominate the PC world, right? The IBM clones do. But Windows operating system, which ultimately is the operating system that dominates the world, is basically an attempt to copy the Apple operating system. I don't know that Bill Gates will completely admit this, but it's obvious. The graphic interface, again, the way, the look and feel of it, is an attempt to copy Apple.
- 7.95** I remember the first time I was getting an MBA and we were doing a group project and we were struggling on a PC to get the stuff to work, and working on a spreadsheet and everything. One of the groups said, "Let me show you something." And he took us over and the university had a Mac lab with Macintoshes. We sat down and I fell in love. Everything was so easy, so straightforward. Everything was easy to manipulate. In those days, you had to write commands. In MS-DOS, you had to write commands. This was before Windows. And the Macintosh, you didn't have to write any commands. You just

did it. You just did what you wanted to do. And that was it. Since then, only Macs in my house.

- 7.96** What happened to IBM? What is it that made it possible for IBM not to become the dominant in the computer industry? I mean, a lot of people expected it to be. It was a dominant player. It was a big company, a big corporation. It had infinite resources, if you want. It had no limit to how much money you could throw at this. And it had lost the PC.
- 7.97** And some people say it's the antitrust lawsuit. They were busy with the antitrust lawsuit. They were afraid if they got to dominate the PC industry, that would really kill them from an antitrust perspective. And there might be something to that. Certainly, there might be something to this idea of distraction. We'll get to that again when we talk about Microsoft lawsuit. But it's probably not the main cause.
- 7.98** Probably the main cause is its bigness, its bureaucracy. There are, as I said in one of the previous classes, there are definitely diseconomies of scale. You get to a certain point that you're big, it's just hard to manage. And it's certainly hard to innovate, and it's hard to see new opportunities, and it's hard to take advantage of those new opportunities. Big is not always good.
- 7.99** Corporations can get too big. The market has great ways to fix that. One way is to compete them out of business, which is what happened with IBM. Other ways, as we talked about, is to break them up, is to take them over, is to replace management, is to spin off and to shrink.
- 7.100** I mean, that is part of the beauty of the marketplace. It self-corrects. Human beings are fallible, and mistakes are going to be made. But in markets, those mistakes are corrected fairly quickly. In free markets, where there are no barriers to correction, that is corrected fairly quickly. And you can see that over and over and over again in the history of technology, constant correction because the market is relatively free and relatively left alone.
- 7.101** IBM continued to struggle through the '80s, trying to build PCs. In '85, they even designed their own operating system to try to break away from Microsoft. It was called OS/2. This is about the time that Gates pivots towards Windows. He sees Macintosh, he pivots towards Windows, and OS/2 just doesn't have a chance at this point. And by the early 2000s, IBM basically sells its PC division to Lenovo, a Chinese company. IBM does not make PCs anymore.
- 7.102** At some point in the late 1980s, IBM looked at this PC division and said, "Why do we need this stepchild all independent over there, all by itself?" They integrated it into the big bureaucracy of IBM, and part of its death was the fact that now it's just part of this big company. It was constantly now being restructured because mainframes were gone, or the market for mainframes had shrunk dramatically.

- 7.103** And PCs, they couldn't catch up and they had to re-imagine their business, which they did to their credit. They did, but they had a complete management overhaul, a complete shakeup of the company that happened in order to achieve that in the '90s and to turn it into, really, a service business. A business that helps companies set up their IT infrastructure, rather than a company that sells hardware, or leases hardware.
- 7.104** All right. Of course, the big thing that happens once we have PCs is the Internet. And the Internet is also probably the one tech that people say, "Well, see, government investment made it happen. The government made it happen. This is the best of industrial planning." So we will turn to talk about the Internet in our next class. Thank you.

Lecture 8: The Internet Revolution

At a glance — The internet revolution and the modern corporation.

- 8.1** Last class, we talked about the early evolution of Silicon Valley and the growth of the personal computing market, but we haven't really gotten to the big story. Because what the chip revolution, the microchip revolution, what the PC revolution ultimately leads us to is the Internet, which is a complete game changer in human history.
- 8.2** Now, the Internet has its origin in funding that was provided to scientists at universities by the U.S. government. And a lot of people give credit for the Internet to either Al Gore or the U.S. government more broadly—kidding, but Al Gore is taking credit for the incident.
- 8.3** In the late 1950s, the Soviet Union put Sputnik up into space, and that immediately created a panic in the United States. The idea was, supposedly, we were falling behind technologically, behind the Soviet Union, and the government got heavily involved in starting to fund science and technological projects. One of those projects originating at the Defense Department in 1958 was called the Advanced Research Project Agency, or what came to be called ARPA. ARPA was a project to fund basic research, almost all of it, at universities. It turned out that it focused mainly on computer science. Most of the money went into various computer science projects.
- 8.4** In 1966, there was this idea: what if we try to connect different computers that are being developed in places where ARPA is providing funding? What if we connect them, even though they're different computers made by different companies and are not really compatible? Can we do this? ARPA funded this project of trying to connect these computers, and it became known as—the computer network became known as—the ARPANET.
- 8.5** The ARPANET used something new called packet switching. Packet switching is something—when they went to AT&T with the idea of communicating between computers using packet switch, AT&T said, "Can't be done." AT&T basically converts whatever speech you have into, in a sense, one packet, into one message that gets delivered. In packet switching, the message is broken up into units. The units might go different routes, and they're all reassembled on the other side. This is one of the innovations that made the communication between computers possible and the transmission of large—relatively speaking, for those times—large sets of data. As a consequence, you need to deploy a router in front of every computer. Every computer had to have a router to basically create the package, send them, and then reassemble them on the other side.
- 8.6** Today, we think of routers like your WiFi router, these little boxes that sit next to a computer. In those days, a router was basically a computer that was the size of a refrigerator that sat next to the computer, it was probably the size of your room, and

they got to communicate through this. By 1969, the first computers had been connected, and you now had a network. Networks like this between incompatible computers were also being developed at the same time in Europe, and it was also an equivalent to an ARPANET in Europe.

- 8.7 The next big thing was, okay, we've got networks of computers, we've got this computer in different locations, networks of computers. How do we connect them? Here, a major innovation came out of UCLA, which was the design of a TCP/IP protocol. We still use it. The thing about these early days of the Internet is we use a lot of the stuff—the protocols, the basic standards are still being used today. So TCP/IP became a protocol where different networks could now talk to each other. Of course, there weren't that many networks to connect, but this was started.
- 8.8 In 1972, this network got their first killer app. The killer app for networked computers was email, was messaging. Nobody planned this. Nobody thought this is how they were going to use these networks. They had all kinds of other ideas. But it turned out that these networks were primarily used for transmitting messages from scientists to scientists, from academics to academics, to people all around.
- 8.9 In the 1980s, they started connecting these computers to university computers, to university networks. So it started to grow, and it went beyond computer science. Now, they were connecting other departments, libraries. You now started to get a real network effect. Remember, these were still the days when to program one of these, you were using punch cards. I don't know if anybody ever saw a punch card in my program. I took a programming class in—I don't remember the year—probably '83 or '84. It was the last computer science class that used punch cards.
- 8.10 We learned assembly, which is a very foundational language for computers in base eight. Everything is base eight. Instead of base two, which is zero one, this is one level above that, but not a lot above that. And we did programs in punch cards. You'd punch your cards, you'd go in, you'd put it into the machine that fed it to the computer, you'd go home, you'd come back the next day and figure out whether the program worked or didn't work. And if it didn't work, go figure out what you did wrong. It was quite something.
- 8.11 But these computers are now being connected. The National Science Foundation funded some of these connections. Again, a government-supported entity funded these connections. And in 1985, there were 2,000 computers connected with one another in the United States. By 1987, two years later, there were 30,000 computers connected, and by 1989, there were 159,000 computers connected. Again, mostly at universities, businesses, some government facilities, particularly related to the Defense Department.
- 8.12 Of course, the rise of the PCs—suddenly people who own PCs are thinking, "We'd like to get connected," particularly academics who are working on their PC. They'd like to

be able to connect to their own computer and through there to other computers in other places so that they can do a search from their computer into a library that might exist at a different university. So now we're connecting PCs into this system. Of course, PCs are being connected to computers and to other PCs using another standard, which we use to this day, that's Ethernet. Ethernet was another one of those inventions that came out of that Xerox lab that Xerox had neglected and left to the side, and that others had capitalized on the stuff that they had created and invented.

- 8.13** In 1990, people at ARPANET are starting to think about privatizing this. They've got this network, they don't know what to do with it—communicate messages, maybe do some searches in libraries, but it's not exactly clear what they can do with it. And yet many of them have these visions of a future where everything is connected and they have a science fiction vision of what turned out to be the actual Internet. But they don't know how to create it themselves. So the idea is to privatize it and see what the market can do with it.
- 8.14** And indeed, by 1995, the entire network is privatized. The backbone is provided by MCI and Sprint. The governance is provided by a nonprofit organization that is created to manage addresses and things like that. But basically, the computer is owned by whoever owns the computer, and to some extent, it's a free-for-all. There are very few limitations on what actually happens on the Internet.
- 8.15** Could this have happened without government? Well, the answer to that is clearly yes. These investments are not huge. These are not investments that private companies couldn't have done. It's true that it probably wouldn't have happened when it happened. That is, it probably wouldn't have rolled out as slowly as it did. It probably would have started a little later and been a lot faster when it actually happened.
- 8.16** As I said, there were other efforts to do this in other places around the world. It wasn't uniquely American. Probably the thing that ARPANET contributed most to the Internet rather than the networking itself or even the idea of networking, which I think would have evolved from the private sector anyway, is the open modular standards that we still use today. Open modular standards were developed for the PC, so it's not like the private market couldn't have done that and probably would have done that, but if you had to point at one real benefit that came out of this, partially because it came out of universities, it had these open standards.
- 8.17** The funding that the universities received—if government hadn't funded it, it could have been private funding, it could have been professors doing this anyway. There's no reason to believe that you need government investment in order to get projects like this done. Indeed, nobody in government imagined where it would go. They had vague ideas and science fiction stories that they told themselves about what it would be, but they, of course, couldn't execute that. That had to be transferred to the market to make it a reality.

- 8.18** So we had MCI Sprint providing the backbone. We had this governance agency. People wanted to get onto this Internet thing. So you started to get services that provided you with connectivity and with some framework of how to access information online. Remember, there are no browsers. There are no browsers yet. That technology is yet to come. What you get is companies like CompuServe, Prodigy, AOL, even Microsoft gets in the business with MSN, with the idea of creating a way for you to communicate.
- 8.19** You get on the internet using a modem through your telephone line, you pay for it, you have a monthly fee, and then you pay per minute used. You actually get billed depending on how many minutes. I can still hear that noise that the modem used to make—some of you might remember that—as it connects to the Internet.
- 8.20** While this is going on, in Switzerland—again, as I said, Europeans are working on this as well—a researcher there develops the HTML hypertext protocol, which we still use today, and HTTP, which is a transmission protocol, which again, we still use today. And the first simple browser that allows you to start interacting with the web. He packages this all up, HTML, HTTP, and the browser, and he calls this package the World Wide Web. So the web is really a software package or a protocol package rather than anything specific.
- 8.21** In '91, this package is distributed around the Internet. And of course, all these Internet online access points are using this World Wide Web to get on. For a long time, people talked about the World Wide Web and the Internet as if they were interchangeable, because that was the only access point they had onto the Internet. But the World Wide Web actually represented these protocols.
- 8.22** The University of Illinois had received some money in the Computer Science Department to work in computing, and a group of undergraduate students split off and decided to design a browser for the Internet, an interface with this new thing that was coming about using the new hypertext and the transmission protocol. This group ended up designing what is called Mosaic. Mosaic is still around out there, floating around, or certainly pieces of it are being used in a bunch of different browsers.
- 8.23** Leading that project was an undergraduate by the name of Marc Andreessen, who is now a prominent venture capitalist, and actually quite prominent in the Trump administration. Marc leaves the university, goes to Silicon Valley, joins a company there, but ultimately meets up with some other people and decides that he is going to turn this browser that he had helped work on into a company.
- 8.24** Marc, with a partner, starts Netscape. He hires a bunch of the undergrads who worked with him on Mosaic, and they sit down and develop what became known as Netscape Navigator, which is really the first stable feature-rich commercial browser that people started using. In 1994, he released a beta version, made it available for people to use. In '95, there was a full version. And by the end of '96, 45 million copies had been downloaded, 45 million copies. It was the way to get on the Internet; it gave you access

to the entire Internet in ways that you couldn't have done beforehand. He had 85% of the browser market at that point, 85%.

- 8.25 In 1995, Netscape IPOed. It was one of the biggest IPOs ever. Immense excitement. The stock just took off, and the company at the end of the first day of trading, was worth \$2.7 billion. The world shook. Suddenly, people noticed what was going on in Silicon Valley. Suddenly, people noticed that a revolution was happening around the Internet. Up until that point, people were tinkering, people were playing around, but for a company to be valued that much? I mean, today, maybe \$2.7 [billion] doesn't sound big given the size of many of our tech companies. But in those days, this was a big deal.
- 8.26 And what you saw is people start focusing on the Internet, people start thinking about the Internet, venture capital is starting investing in it, and you saw a real boom. Netscape's business model was they gave away the browser for 90 days, and you were expected to pay after 90 days, although enforcement of that wasn't always that great.
- 8.27 What they also did was charge PC makers. They charged them, they gave it to them, and the PC makers then bundled it with the computer. You would get your Compaq computer, and there would be a Netscape browser loaded up already for you to use, and Netscape would pay for it—sorry, whatever the company was—Compaq would pay for it.
- 8.28 Of course, browsers now allowed you to put some of the applications that sat on your desktop into the browser. A couple of guys who left Apple started a company called Hotmail, which was the first email app to sit on your browser, where you could get email on your browser rather than as a separate app. We know now that you can do a lot of stuff in the browser. Browser, in some sense, is its own operating system that sits in the background.
- 8.29 We have a browser and people are building websites and the internet is growing. Now the big challenge is, how do I find the stuff that I need? So search becomes a big deal. And in '96, two Stanford PhD students, instead of writing their dissertation, start Yahoo, and Yahoo becomes the dominant search engine. Yahoo is a search engine built on human beings cataloging the Internet. They're basically creating catalogs of every website, categorizing them, giving them code words so you can search through them and you can hit on a web page. Yahoo dominated search in those days.
- 8.30 In '98, another pair of Stanford PhD students figured out you could do it differently by creating an algorithm that actually doesn't—you don't have to actually categorize them. It actually searches the websites themselves and identifies the keywords on the websites themselves. This is, of course, Google. But Google was tiny in '98. Nobody thought Google would be what Google became, ultimately. So search becomes a big place.
- 8.31 Also in the mid to late '90s, you start getting marketplace. People figured out, "Maybe I can sell stuff online." The biggest marketplace in those times was eBay. eBay was a

monster when it came out. It was the first place where you could go and buy all kinds of stuff. It was a place where it wasn't merchants and consumers, it was basically people, instead of listing their old television on the classified ads in newspapers—you remember newspapers?—now they listed it on eBay, and consumers could buy it. So it was consumer to consumer, really, buying stuff on eBay.

- 8.32** One of the big problems they encountered was trust. How do I know that if I pay this guy, I'm going to get the goods. Ebay was really the first website to introduce customer ratings as a means of trust, establishing trust. There are sellers that get a high rating, sellers who get a low rating. I can evaluate who I'm going to buy from. The same with customers—how do I know I'm going to get the money if I ship them the TV? Customers are being rated. This is a massive innovation on the part of eBay. Ebay was one of the best venture capital investments ever, given how much money was put in originally and how much money came out on the other side. It was a huge win for the venture capitalist. In '98, when eBay went public, it was valued at \$2 billion. So again, massive valuations for these internet companies.
- 8.33** In 1994, Jeff Bezos has the idea of selling books online. I think he already knows that one day he'd like to sell everything online, but he starts off with books; books are easy. All books are the same. Any particular book is always the same; there's no variation in the units. You can offer a huge variety of it. And of course, he launches that in '94 and really becomes, if you will, a stepped-up version of one of the great innovators of the early 20th century, which is Sears. So he's replacing the Sears catalog now with a real catalog online. And as we know, Amazon goes on to become one of the big five.
- 8.34** The largest Internet vendor in the very early 21st century— 2001, 2002—the largest vendor online was actually not Amazon, and it was not eBay. It was Dell. Dell was the first to start selling computers online. And the beauty with what Dell did is you could design your computer: what hard drive, how much memory, what processor, what monitor, and you could design it. What Dell had done, what Dell ultimately did is, he would link that up directly to his subcontracts, directly to who was providing the goods.
- 8.35** So you didn't have to go to Dell, and then Dell would go out and call up all its people, "We need this and we need that." No, it went directly. They would already ship it. It would be assembled. This was what's called just-in-time inventory. Computers were assembled for particular people. There was no inventory involved. There was no keeping of inventory at Dell. All Dell was— all they did was assemble the stuff that already been ordered through the Internet from the subcontractors.
- 8.36** All of this could happen because Netscape had introduced something called a secure socket layer. Don't ask me what that is, but all I can tell you is it makes financial transactions possible and secret. That is, you put your credit card in and people can't steal it. It secures the actual transaction. It makes sure that the vendor knows that it's a real payment, and it makes sure that you are secure in that this is not going into the Ether, and that anybody could pick up your credit card number at any time. So lots of

little things that have to come together in order to make this possible, but the entrepreneurs, the engineers, at these companies, are making it happen.

- 8.37** Up until about 1995, all this computer stuff that was happening, the PC and everything else, it was a real mystery among economists about what was going on because the reality was that when you looked at productivity numbers, they weren't increasing. That is, in spite of the introduction of computers and everything, you couldn't see it in the data. There were a lot of people—I remember there were stories in the Wall Street Journal about this, about, "Is this wasteful? Are people using it just for fun? This should manifest itself in us becoming more productive, this computer stuff."
- 8.38** Then suddenly in '95, it started to appear, and it appeared in the form of massive, significant increases in productivity. The same phenomena had happened in the early introduction of electricity. It took a while in order for people to figure out how to use it, how to apply this new technology to actually increasing productivity, to actually making companies more efficient in the production of products. But once it happens, you get real takeoff. And from 1995, you can actually see the increased productivity.
- 8.39** Of course, this has to do with the introduction of computers, but more than that. I think for 1995, the economy takes off, at least in the late '90s, as we've said already, to a large extent because of the restructuring that had happened in the 1980s. American companies were leaner, more focused, and much, much more productive. We see that in the numbers. And I think ultimately, one of the stories that is very relevant to today is while we see it in the numbers, people really have no appreciation for it.
- 8.40** Right now, we've got stories going on all over this country and in the press and all over the place, how wonderful life was in the '50s and '60s, and wages have stagnated since the 1970s. None of that is true. Just empirically, none of that is true. Indeed, wages have been going up since the early '90s quite dramatically at every level, from at the top, which everybody acknowledges has been going up, but all the way to the bottom. Every single category of decile of wages has been going up significantly since the early to mid-1990s, and nobody seems to recognize it. And I would argue—I'll argue in a minute—that that is not reflective even of a fraction of how much our life has actually improved in terms of standard of living and quality of life. You can't measure that in terms of earnings because the question is also, what do those earnings buy? What can we buy with that stuff? And what we can buy today, I don't think people 50 years ago could imagine what we can buy today—maybe they're science fiction readers, but not very many other people.
- 8.41** To wrap up the story of at least the 20th century—and we'll get into the 21st, but the 20th century—one has to talk about Microsoft and Microsoft's antitrust case of the late 1990s. Microsoft noticed Netscape and realized, Bill Gates at some point realized—he had always had a dream about the Internet. He always had an understanding of where the Internet could take us. But he had never really invested in it. And then when they saw Netscape, actually, finally saw the product that could launch Microsoft and the rest

of the world into that Internet age. And what Gates did is very similar to what IBM did with the PC: he basically took a team and gave them this project to create a competitor to Netscape and do it on the side without the bureaucracy of Microsoft and reporting directly to him.

- 8.42 The result of that was Internet Explorer, which Microsoft distributed basically for free, integrated with Windows, with the operating system. It also did a lot contractually to pressure its partners to use Internet Explorer. They made using Internet, then adopting Internet Explorer, contingent on them receiving other deals from Microsoft. For example, they invested \$150 million in Apple, and part of that deal was that Apple makes Internet Explorer the default browser on the Apple.
- 8.43 Now, the \$150 million was not just for that. The \$150 million was basically to bail Apple out. Apple was almost dead. Steve Jobs had just returned to the company—Steve Jobs was fired in 1985 from Apple, went on to build a company called NeXT, which was never successful, but built a really cool operating system. In 1997, he was rehired by Apple to become its CEO, and one of the first things he did was cut this deal with Bill Gates, where Bill Gates invested \$150 million in Apple, IE [Internet Explorer] became the browser, and Bill Gates committed to providing the office suite for Macs. There'd been a dispute about whether Microsoft would continue to provide the office suite on a Mac or not, and he got him to secure that.
- 8.44 Microsoft also got AOL, which was at that point the number one way in which people access the Internet to make Internet Explorer to replace Netscape with Internet Explorer as their default browser to get on. Netscape initially responded by making its browser free, but it was really all over at that point for Netscape; they couldn't really recover. Ultimately, they sold, I think to AOL, which bought them, shut them down, still used the Internet Explorer as their browser, not clear exactly what happened with the other assets of Netscape. But Marc Andreessen left, made quite a bit of money on the sale, but had already made money on the IPO, and left and became a familiar venture capitalist.
- 8.45 Gates viewed this as really important because he thought that Netscape could compete with the operating system. He thought a browser could ultimately become an operating system, and he wanted to dominate this field, and he wanted to make sure that Microsoft was a player.
- 8.46 The Justice Department had been eyeing Microsoft as a monopoly because of its dominance for a few years. And indeed, in 1995, before the whole Netscape Internet Explorer thing, there had been an agreement between Microsoft and the Justice Department, whereas Microsoft agreed not to bundle software with the operating system.
- 8.47 The way the Justice Department looked at it was Microsoft had 95% of the IBM clone PC market. That's how they defined the market. Apple wasn't included. They just

defined it as that. That sounds like a monopoly. Could they show any harm to consumers? No. Any other inefficiencies? No. But bundling, going way back in antitrust, that was not allowed. It was okay if the apps were integrated into the operating system; that was okay. But if they were just bundled with it, if they were separate entities, that was a violation of antitrust. So Microsoft agreed no more bundling in '95.

- 8.48** By the way, there's a good reason to think that Bill Gates invested \$150 million in Apple to keep it alive so he could always say there's competition. One wonders, if not for the antitrust suit, if Microsoft had invested in Apple. We don't know, but there's a good reason to believe that that was part of the rationale for doing that.
- 8.49** Anyway, in 1996, a year later, at the instigation of Netscape and Marc Andreessen, the Justice Department came back to Microsoft and said, "You're in violation of the decree. You're bundling Internet Explorer— they're bundled together, and that's a violation." A district court found against Microsoft. It was appealed. An appellate court reversed the district court finding and found for Microsoft.
- 8.50** But in the meantime, the Justice Department decided to go all in against Microsoft, and they filed a big antitrust suit under the Sherman Act, both Section 1 and Section 2, basically defining Microsoft as a monopoly. Initially, they tried to claim that there was tying involved, that they were leveraging market power in one area to dominance for another, but the judge dismissed that very quickly.
- 8.51** It became basically this idea that Microsoft was using its operating system as a barrier to entry, that it controlled the bottleneck to all computing. It had a network effect going for it. Because it was 95% of all the PC clones, everybody made applications for it, for the PC clones that worked with their operating system. Nobody could introduce a new operating system because of the ecosystem Microsoft had created, because of all these other applications. So the success of the operating system was a barrier to entry from anybody else coming in. This was the primary rationale.
- 8.52** They also came after it for what they call predatory behavior, these aggressive moves contractually to get people to use their operating system or to get people to use Internet Explorer as part of the operating system. Ultimately, the appellate court had ruled that the Internet Explorer did not violate the original decree because it was not bundled, it was integrated. Internet Explorer somehow integrated in the operating system, and the judge called it. This is why they found this other suit that was not dependent on that.
- 8.53** The company, of course, denied all charges, claimed that it existed in a dynamic, competitive market, that it constantly faced competition. Indeed, Bill Gates described in one of his, I think in one of his testimonies—he described a future world in which people would have small computers in their hand and the PCs would basically be dead. He completely described the world in which we live to the court and in which Microsoft was not necessarily a part of that.

- 8.54** In April 2000, Judge Jackson ruled for the government pretty much on all the charges. The ruling was that Microsoft—the remedy was that Microsoft should be broken up. The operating system and what we call Office, and all the other apps would be two separate companies and unrelated. Of course, this was appealed, and an appellate court reversed some charges, agreed with other charges, but rejected the idea of breaking Microsoft up. Microsoft still, of course, is a company that has both of those.
- 8.55** The remedy that the appellate court dictated was the creation of an oversight board that consisted of three members, experts in computers, who would now oversee all contracts and all decisions at Microsoft on behalf of the Justice Department for a period of five years. So starting in 2001 or 2002, for five years, Microsoft was overseen by this committee, if you will, of experts.
- 8.56** What effect that had on Microsoft's inability to create its own operating system for a phone and to get into the phone business? We will never really know. The fact is that during this period also, Bill Gates left the company, at least left the CEO. He became the executive chairman and then chairman. So the mind of Bill Gates was now extracted. One wonders, if not for this ruling—if Bill Gates had stayed CEO of Microsoft, if it had lagged so badly when it came to phones and when it came to other parts of the new Internet. Microsoft ultimately has done very well, but it took a long time for Microsoft really to recover from those five years of oversight where they dropped the ball on many potentially new technologies and new abilities that they had.
- 8.57** One of the things that this did—and this is reminiscent of what we talked about when we talked about antitrust, generally—really, this has increased the incentives for Microsoft rather than to contract with other companies and to, in a sense, try to get them to use their products, it's much easier just to buy them, and then they can use the product. So one of the things that this induces, which antitrust generally has induced in certain periods of history, is more vertical integration, and Microsoft trying to do more things itself rather than relying on other market players. So antitrust has multiple effects at multiple levels.
- 8.58** During this period of the late 20th century, last few years of the 20th century, as we said, we saw a huge explosion, huge increase in investment in internet companies. Venture capital investment, for example, in 1995, about 1,864 investments, 1,864 companies, \$7.2 billion invested. In 2000, they made six times that—7,974 investments, \$98.6 billion. More than 10 times the amount of money invested. Huge amount of investment, primarily into technologies that were going to take advantage of the Internet.
- 8.59** The Nasdaq, where most of these companies ended up listing, where Netscape had listed, went through the roof. In the five years leading up to its peak in February 2000, the Nasdaq had gone up 6x. In '99 alone, the Nasdaq was up over 90%. This is about three times as much as the broader financial index of the S&P 500. This is called the dot-com bubble. It was a massive amount of investment, a lot of companies going

public, and the Nasdaq just taking off with the excitement of what potential the Internet held.

- 8.60** No question, it was a bubble, and the bubble burst when the Fed started increasing interest rates in 2000. Ultimately, the Nasdaq dropped by 75%. By mid-2001, it was down 75%. Of course, the U.S. economy, as the Fed increased interest rates, the U.S. economy ultimately entered into recession, and then 9/11 happened, which only intensified that recession. The recession was pretty mild, maybe the mildest recession U.S. has ever had, but obviously the Nasdaq, what happened was this was an opportunity to clear out and figure out who the winners and who the losers were.
- 8.61** A lot of people look down on bubbles, but the reality is that a bubble is just a lot of people getting very excited and investing in a lot of different things. It's a period where there's a lot of capital and a lot of human resources available for experimentation, available for new things and new exciting things. Some of those things are going to be stupid: pets.com. But some of them are going to be Amazon and Google. Nobody knows in advance what is what, or very few people know in advance what is what. Indeed, the venture capitalists probably invest in pets.com and in Google, and they're willing to let pets.com go because they make so much money in Google. It doesn't matter.
- 8.62** If you look at the value that these companies created, the value of the winners, the ones who survived, it far exceeds the losses of the bubble bursting, ultimately, if you look at where these companies have gone. Nasdaq doesn't really recover to its peak until 2013. It's bottomed in 2001, and it took 12 years to make that level again. So at least in terms of market valuations, this was a big deal. But overall, there's no question that the dot-com bubble increased value overall.
- 8.63** From a macroeconomic perspective, probably the biggest consequence of the bubble was the response it elicited from the Federal Reserve and what that response ultimately led to. Coming out of— as a result of the bubble bursting, 9/11, and the recession, the Federal Reserve engaged in some of the most aggressive interest rate cuts in American history. Interest rates went down to historical lows, 1%.—that was the Fed funds rate. But really, in terms of real rate terms, they were zero and maybe even negative.
- 8.64** This resulted in people borrowing money, a lot of money. If you can borrow money at zero, basically, you're going to do it all day. The bubble had burst, so the money was not flowing into investments, the money was not flowing into the Internet— people were skeptical about the Internet at this point; they were cautious. There was another industry that was heavily subsidized by the government, there were lots of goodies to be had if it went there, and that was housing.
- 8.65** What happened was that money flowed into housing; cheap money flowed into housing. You get a tax deductibility for housing. Fannie Mae and Freddie Mac, which

are government institutions, subsidized the cost of your mortgage. Mortgage rates were historically low, and home prices just took off. They took off not because of shortage of supply or imbalances between supply and demand— put it this way, not [because of] natural imbalances, but because massive amounts of money were flowing in.

- 8.66** I remember the woman who cut my hair in California, all her income came from cutting hair, and she owned three homes, four homes. I mean, why not? Prices only went up, and you could get basically zero down loans. So why not? We're not going to get into all the causes of financial crisis, but basically, there's no question, the financial crisis of 2007, 2009 was caused by the Fed, the misbehavior of the Fed, both leading up to the crisis and then within the crisis making it worse.
- 8.67** Just like the Great Depression, all over again, the same story. Just slightly different— Ben Bernanke was the chairman of the Fed this time, so he was a student of the Great Depression. He wrote most of his academic papers on the Great Depression, so he didn't make quite the same mistakes they made in the Great Depression, but different mistakes were definitely made. The 2007-2009 was definitely the worst recession since the Great Depression. It was quite devastating and quite bad. This is, I think, a recession that many of you lived through.
- 8.68** In spite of it, though, innovation continued. Innovation in tech continued. Innovation in the Internet continued. In 2000, 52% of Americans were connected to the Internet. By 2008, 75% of Americans were connected the Internet. Today, almost everybody is connected to the Internet, certainly through their phone, if through nothing else. By 2008, 45% of connections to the Internet were made wirelessly. The whole wireless was changing.
- 8.69** There was the beginning of widespread use of a high-speed Internet—maybe not high-speed then, as we categorize it today. But relative to those modems that made those noises, It was high-speed Internet. You could now get it through your cable, you could get it through your telephone line, but it wasn't quite with those funny modems. You could now actually browse the Internet.
- 8.70** Of course, all of that, ultimately, would make cloud computing possible. In the early 2000s, Amazon introduces cloud computing—we talked about this in a previous class—as a byproduct of the fact that they were creating a cloud for themselves to build this massive commerce website and now realized that they could lease out space to other companies. Think about what that does—that lowers dramatically the cost of starting a company. In the past, you'd have to buy computers and networking equipment and all this stuff. Now, you don't have to buy anything.
- 8.71** You buy basically a laptop and you can connect to the cloud and you can build your entire website and everything you want to do on the cloud and you just lease it. You just pay for what you use. You don't have to become an IT person. Amazon does the IT

for you. Since then, of course, Google and Microsoft have also entered that space and made a lot of money from their cloud services.

- 8.72** The big product innovation, of course, of this period of the late 2000s, was indeed the iPhone. The iPhone is a product that came about because Steve Jobs returns to the Apple, really. There'd been other attempts. I had a PalmPilot. I don't know if anybody else had a PalmPilot. Those were cool. They had a stylus and all of that. Steve Jobs hated styluses— this is why we get a touch screen. Steve Jobs thought styluses were a complete cop-out. You wanted to make it organic. Why aren't you using your finger rather than using something that works like a finger to manipulate images?
- 8.73** There was the Blackberry—I loved my Blackberry. Blackberry was an amazing little machine, but it was basically an email machine. It was basically there to provide email. It was great for email. It had this keyboard, this mechanical keyboard—Steve Jobs hated mechanical keyboards. It defeats the whole purpose. And if you think about the way the iPhone comes, the keyboard pops out. It's on the screen; everything's touch screen. There's no mechanical. The few mechanical buttons at the sides were big compromises he had to live with. He had to accept it because the technology required it.
- 8.74** When Steve Jobs returns to Apple in 1997, the company is on the ropes; the company is on the verge of bankruptcy. It's doing very badly. This infusion of capital from Microsoft is essential for it, and a good relationship with Microsoft is essential for it. Part of the reason Jobs joins is that Apple has made the decision to take the operating system from its previous company, NeXT—which never went anywhere—but to take that operating system and integrate it now into the Apple and to provide a next generation operating system for Apple, which is ultimately what happens.
- 8.75** One of the first product launches or the first product launch of Steve Jobs, is the iMac, which is released in 1998: this gorgeous, rounded machine, which is in all different colors with a screen and everything integrated into one package. It becomes a huge hit and really revitalizes the Apple name and presents Steve Jobs' focus on esthetics. And iMacs do phenomenally well.
- 8.76** But the real interesting thing is what comes next, which is not a better computer, not a revolutionary in computers, but a music player. We're living through an era in the 1990s where CDs have come out, so music is now digitalised. MP3 technology is out, so people can download the music and they could put it on the computer as an MP3 file. And then somebody comes up with the idea that, "Hey, we can all share our MP3 files among each other. I can play it off of your computer. We're all networked now." Some of you might remember Napster, which was this software that allowed you to basically access other people's MP3s and play them on your computer, and there was no limit.
- 8.77** The music industry flipped out completely. They were losing an enormous amount of revenue. "Why buy music if I can just access it over the internet and get it from anyone, anywhere at any time?" And there's no way to get royalties on that because nobody

knows that that transaction is happening. Ultimately, Napster becomes hugely popular. It's a huge success. Ultimately, the court shut it down as a violation of intellectual property, and Steve Jobs sees a real opportunity there.

8.78 First, some of his engineers identify—I think it was Toshiba— a small hard drive that can actually carry a significant volume. One of the problems with music players before that was you couldn't get enough songs onto the drive. He basically had them buy the entire supply, all the hard drives that they could find, as they're working on a music player. A music player that has the aesthetic styling of Apple. And that comes out in the early 2000s in the form of what's called the iPod.

8.79 A year and a half later, the real innovation happens with the introduction of iTunes, which is the music library that feeds the iPod. Steve Jobs, I think in the power of his personality, basically went to all the big music companies and got them to sign deals with him to provide him with basically MP3 files of all their music, and he would pay them a royalty on every download, which completely changed the model of the music industry, revolutionized the music industry. Now they were getting paid for the MP3s, whereas before, they weren't.

8.80 This completely changed everything for them. At the same time, Steve had embraced the idea of getting the cost of all these things down, making them more effective, more efficient. Tim Cook, who today is the CEO of Apple, became the guy in charge of all supply chains for Apple, cut the number of subcontractors from 100 to 24, made it a lot more efficient. As we know today, Apple actually builds very little; almost everything is outsourced. All the design, all the planning, all the servicing, all the engineering happens at Apple, but really none of that raw manufacturing. Jobs, of course, was responsible for product and design, while Tim Cook did the supply chain.

8.81 Now you had a music player—Jobs hated phones, thought cell phones were really cumbersome and ineffective. As I said, he hated the mechanical keyboard, he hated the stylus, so he came up with the idea of taking this music player and integrating it with a phone. The idea originally was a music player and a phone. That's it. And having a touch screen and doing all this, and, of course, that was maybe his most famous "one more thing" moment at the Apple presentation, where he reveals the iPhone and basically revolutionizes all of our lives.

8.82 And of course, even for the iPhone, what really makes the iPhone special and what really changed our lives is the fact that there was an app store. Now you could have third parties writing all kinds of apps to make the iPhone not just a phone, not just a music player, but everything, a Swiss knife, everything. There's nothing that this iPhone cannot hold. Whether it's maps or all these other things, that all comes from third parties. It's pretty easy to put a GPS chip inside this platform once you have it built.

8.83 Microsoft had lost focus by this point. Again, Gates was out. They were being supervised by these people. They tried to design an operating system for the phone,

which is basically the operating system from the PC. It never really took off; it never really worked. They even bought Nokia because Nokia was floundering; they couldn't compete. They thought they'd bring Nokia in with their ability to build phones, Microsoft would make the operating system. It never came together; it never actually worked.

- 8.84** Instead, the competitor to Apple was not Microsoft, the people you'd expect it to be. Markets are always full of surprises. The competitor to the iPhone was Google. Now, Google, of course, had made its name with search and advertising, and it had figured out how to monetize search with advertising and become incredibly good at it and become incredibly good at creating marketplaces for advertisers and placing the advertising right on the page and doing all the things that would make it the powerhouse that it is today. That required a lot of innovation and a lot of thinking.
- 8.85** In 2004, Google had gone public. In 2005, it had bought a small company called Android that was building an operating system based on Linux. I think Google was thinking, "We want to get into the operating system business, and we could integrate this ultimately with some browser, and we can compete with Microsoft and everybody else." And indeed, at some point, if you remember, there were Chromebooks—maybe they're still around today, very cheap laptops that run on a Google Android operating system that were used in schools and places like that.
- 8.86** So they got the Android, and then when the iPhone came out, they immediately had the idea of pivoting Android to being a phone operating system. They got a consortium of 30 something companies together, 34 members, I think, an open alliance for this. And they said, "Let's use this as the standard." Other companies that were ambitious with regard to phone technology, jumped in. And the first phone based on Android was released about eight months—eight, nine months—after Apple released iPhone. So very quickly, they did it all super fast, super effectively, and now they dominate the non-iPhone market.
- 8.87** Indeed, the non-iPhone market dominates the world. The only other operating system, I think, exists—there might be several others, but the only big one exists in China. China is trying to separate itself from depending on the U.S. I think now, Huawei has its own operating system for their phones, but generally, Android is on everything. Google also created an app store, Google Play, which mimicked the iPhone, and made it into this ability to do everything.
- 8.88** In 2008, Google launched its own browser. That browser, Chrome, has come to dominate browsers. The integration between the browser and search, the integration between browser, search, and advertising is dominant. Of course, it's one of the reasons the antitrust division of the Justice Department as well as the FTC are constantly going after Google. Indeed, right now, they are asking a judge, who has determined Google to be a monopoly in violation of the Sherman Act, to break it up. So

right now we're facing a situation where in the next few months, maybe by the time this course is out, a judge will decide whether to break up Google or not.

- 8.89** Once we have the Internet, we're going to get things like social media. Social media started really in the old days in little chat groups where people connected in a variety of different ways through different message boards and chat groups. Ultimately, you got something like MySpace, creating a place where people could create profiles and friends and communicate with others. Part of the big problem with MySpace was there was no quality control, so pretty much anybody could put up stuff. There was a lot of fake stuff. There was a lot of unpleasant stuff going on on MySpace, and people rebelled against it.
- 8.90** Probably the biggest innovation Facebook did was that it policed what people did on Facebook. It created certain quality controls around Facebook. Think about it. It was launched with just Harvard students, just students at Harvard to connect with one another. Then alumni of Harvard, then other Ivy League universities. And only once they'd figured out how to run it all and how to do all that and how to make sure the quality control and everything, did they open it up to the public. Of course, once it opened up to the public, it became the massive network that it is today with hundreds of millions of people on the platform.
- 8.91** Another application for the Internet was, of course, streaming. Netflix, which started out—I remember signing up for Netflix and getting my DVDs in the mail. You'd pay a monthly fee and you could order as many DVDs—I think you could get four at a time. You'd get four, then you'd send them back, and you'd get the next four. That was Netflix. Of course, as soon as bandwidth got wide enough, Netflix understood something Blockbuster did not. It understood that the future was streaming, that it was not in the DVDs. I mean, Blockbuster had even a hard time going from VHS to DVD. They really were committed to VHS. And by 2010, as a consequence, Blockbuster was bankrupt.
- 8.92** And Netflix—streaming is pretty simple. And once you digitalize movies, they can live anywhere. So competition is very easy on these platforms. It used to be the Netflix was it, but today you have, I don't know, 10 different streaming services or 20 different streaming services that you can go for, including Apple and Amazon that have massive streaming services of their own. So there's massive competition. Because of the barriers to entry are so low, competition is very intense in all these services.
- 8.93** And now the competition is not so much about streaming to you; the competition is about content. This is why Netflix makes their own movies. Apple makes their own movies. All these companies are making their own movies because that is the way we decide. "I want to sign up to Apple because they've got this new, crazy, great series that I really want to watch. So I sign up for their streaming app. If Netflix, the quality goes down, I'm going to cut them off. I'm going to stop streaming over there." So massive competition on streaming.

- 8.94** And then just generally, what the Internet does is it disintermediates. It gets rid of the middleman in everything. It gets rid of the bookstore. It gets rid of the travel agent—does anybody use a travel agent? I was just thinking about this the other day. I used to have this great travel agent who I really liked, and she would book all my stuff. And that's ridiculous today. Who would you use that? You do it all yourself, completely disintermediating the middleman in all of this.
- 8.95** Maybe the biggest disintermediation was with Airbnb and Uber, this idea that I can use a room in my apartment and let it out, and I can use the Internet to find somebody. And again, the rating system can figure out who's reliable and who's not reliable and reviews. People write reviews about the apartments they rent and don't rent. And of course, Airbnb has competitors, and it's blown up. It's much of how people today travel and stay places—through Airbnb. It's amazing how much more travel we do today. I think partially that is the deregulation of the airlines and the cheap prices. And partially it's Airbnb. It's the fact that we can get into nice little apartments at a relatively good cost without having to go through hotels.
- 8.96** Uber completely disintermediated the taxi industry. Uber had to face a lot of regulatory barriers. It was very innovative in how it dealt with those regulatory barriers, primarily by ignoring them, but building up a constituency that would then lobby on their part when the government came after them. In New York, when the city council was thinking of banning Uber, every time you used Uber in New York during those months where this was under consideration, the app would drop down, you'd get a picture of the mayor of New York, if you were in New York, and it would say, "Your current ride is going to be \$10. If this happens, you're going to have to pay \$20" or whatever. People got upset. People started phoning the mayor and lobbying them, and that proposition of banning Uber went away.
- 8.97** Uber completely changed the taxi industry. The number of people who lost fortunes on their taxi medallions is truly astounding. The whole taxi medallion process was ridiculous, but there was a bubble. Before Uber had arrived, there was a real bubble in taxi medallions. A taxi medallion in New York would cost over a million dollars. And suddenly, within a few months, that taxi medallion was worth less than \$200,000. A lot of people had borrowed a lot of money to buy those medallions, and they lost everything. It was brutal.
- 8.98** But that's what happens when you get inefficient market and you get these technologies coming in, they completely disrupt it, they completely change it. And of course, the beneficiaries of that is everybody who uses transportation. Uber has changed our lives in profound ways, I think, as we travel really around the world.
- 8.99** You can go area after area after area. The Internet is changing our lives. It's disrupting existing—where are the travel agents? They're all unemployed. We should start a pressure group to bring back travel agents. It's not fair. Why only manufacturing jobs? We should also bring back travel agents. The economic impact here is massive.

- 8.100** It's hard, really. This is a problem that's hard to put numbers on. The fact that we buy stuff on the Internet, how much money does that save us? Relative to if we didn't have an internet, we just had the regular stores that we always used in order to buy stuff. By some estimate, we save about 25% because of the existence of internet and the existence of internet commerce. So even if wages are static, you're paying 25% less than what you would have if the internet didn't exist.
- 8.101** The standard of living in that sense is better because of these technologies. They've disrupted certain parts of the economy, but the net result is a dramatic improvement in quality and standard of life. Again, the amount of travel people do today, which was unimagined—in the 1970s, when you went on a plane, it was a special occasion. It was a big deal. It cost a lot of money, you dressed up, and it was an event. Today, it's like getting on a taxi. Nobody thinks twice about flying somewhere they need to fly. And it doesn't cost that much. It could cost even less if they actually deregulated properly the airline industry.
- 8.102** In Europe, you can get on a plane in London and fly to Lisbon sometimes for 15 bucks because of some of these discount airlines, their leftover tickets, they sell to whoever is willing to pay, whatever they're willing to pay, to get from point A to point B. It might mean you're stuck in a middle row at the back of an easyJet plane, which is not fun, but you get to where you want to go, and it's pretty amazing. People who could never afford to fly suddenly can fly all over.
- 8.103** Somebody tried to calculate what is the value of search— can we put a number on what value we get from search? And I don't know how they did this, and I don't know if it's legit, but I thought it was interesting. The median individual gets a value of \$17,000 a year from search. So whatever income you have, you get \$17,000 more because now you can use search online. You don't have to go to the library, you don't have to go and look at different stores and go find what you need. And you could multiply this over and over again in terms of everything that we do.
- 8.104** Our quality of life, our standard of living, is so much higher. What is the value? What is the value, the monetary value for you, of arriving—here we are in Phoenix, Arizona—arriving in Phoenix, Arizona from wherever you are in the world and not having to use a map. Not even plugging in your phone—you don't have to plug in your phone in a rental a car anymore. It's all Bluetooth— and using CarPlay to use Google Maps to get wherever you want to go and not struggling with a map. How do we monetize that?
- 8.105** How do we monetize the fact that you can access every piece of music ever produced and load it up, put on a record at a marginal cost of zero? Pay a monthly fee and you get access to the entire library of all the music ever produced. Or the access on Netflix to the entire library of movies. There's no way for us to even monetize this, to put a real value on it, because what I pay Apple for Apple Music is a fraction, a tiny fraction of the value access to all that music has for me or for any human being.

- 8.106** So one of the problems in economics is we can't put a number on the value a consumer gets from the things that they purchase. All we can say is it's worth more than \$1,000 if you paid \$1,000 for it. We can't say how much more. And in this case, given how much we pay, very little for all the apps we get on the phone, in most cases, zero—how do we measure that value? And in that sense, life today in the United States is so much better than it's ever been before, from a material perspective. So much better.
- 8.107** We are so much wealthier in ways that are not captured just by income, but are captured by the quality of life, by what's available to us, by what we can access. On my phone today, I can access almost all the knowledge the human race has put together. I can access most artworks that are being produced in at least a photo image. It's all right there at my fingertips. How do I value that? We have no clue. And this is only going to get bigger. Our quality of life, our standard of living, and the intangibility of it, the fact that we're not really paying for it, is only going to get much greater with artificial intelligence, with AI.
- 8.108** AI has the potential to completely revolutionize all of these technologies. I mean, now I will have a travel agent. It'll be my AI. I'll say, "I need to get to Phoenix. You already know what hotels I like. You already know what flights I tend to have. Book it. Go and book it. Just let me know where I need to show up and make sure I have a rental a car, an Uber, or a Waymo—more likely a Waymo—waiting for me at the airport when I arrive so I can go to where I need to be. And make sure to track my flight as it comes so if I'm late, the Waymo comes exactly when I get there."
- 8.109** And it can track where my phone is. It can actually track, "Oh, you're leaving baggage claim now. There's the Waymo." That's not science fiction. Give it a couple of years. And that's just me who lacks imagination. Imagine what those entrepreneurs out there who have great imaginations about how to use this new technology to make our lives better. But then, are we going to be missing the days when we made our own flight arrangements? Are we going to be missing the days when we actually produced stuff with our hands?
- 8.110** We live in a time where technology, and I think the potential of the U.S. economy, is really unbelievable. Unfortunately, we're dominated by politics of nostalgia, politics of populism, of emotionalism, that does not appreciate what we have and what the potential is. Because one of the themes, I think, of this course is that there is this constant interaction between markets and government at the corporate level, at the business level.
- 8.111** The real steps forward, the constant steps forward, are steps forward created by industry, created by businesses, created by corporations, constantly being held back by government regulations, government controls, government ambitions for central planning or antitrust considerations, none of which ultimately have any basis in reality in terms of our well-being, and many of them admit that. One could only imagine what could be possible if we truly unleashed business in America. If we truly unleashed a

corporation focused on shareholder wealth maximization, focused on innovating and growing and producing values, i.e., producing profits. If we unleashed it in an environment that didn't have all the regulations and the controls and the antitrust. Antitrust has given us nothing other than a break on progress.

- 8.112** But imagine the amount of progress we could have if we took the break away, if we let the market, these entrepreneurs, these individual entrepreneurs, [if we] actually allowed them to pursue their vision all the way, to build, to create, to make, and to produce values, ultimately, to make our lives, all of our lives better. I think the story of Silicon Valley, what they managed to achieve over the last 45 years, should be an inspiration for how things should be. Notice that Silicon Valley faced few regulations, antitrust, but few regulations other than that during this period. Imagine if that were true of the automobile industry and every other industry out there. Imagine the progress and success we could have.
- 8.113** So hopefully that is the future we're heading. Given the technological advances that we have today, what we need is a government that protects property rights and evolves as property rights change. AI presents its own property rights challenges, but that other than that leaves people free. Our success and prosperity is really unlimited and unimagined if that were actually true. Thank you.
- 8.114** We live in amazing times. Our standard of living, quality of life, materially is the best ever. And we're unbelievably richer than pretty much any other generation that comes before us. We have an incredible abundance of stuff. I mean, this computer that is more powerful than supercomputers—the crazy supercomputers of just 40, 50 years ago. I mean, look at how many cameras we have straight at me right now. I mean, look this facility. This is all amazing and technology that really—maybe we could imagine if we were a science fiction writer, but most people could not imagine, and certainly not imagine that we would have it so soon. Today.
- 8.115** And yet all of this stuff that we have—I'm looking at the air conditioning vents and air conditioning itself—all of it is a product of a particular business form. All of this stuff is a product of corporations, of businesses. The world would not be as wealthy, as prosperous. We wouldn't have the material abundance that we indeed have without business. And the reality is that in America, as in most of the wealthy world, business has always—or over the last 150 years, when this wealth has been created—has manifested itself in a particular form, and that is the corporate form, this idea of corporations.
- 8.116** And yet, our culture resents businesses, generally. Big businesses, in particular. Notice that most of the stuff—this is the biggest company in the world. I'm not allowed to say what it is, but my computer over here—the biggest company in the world. The cameras are all a big company. The air conditioning is probably one of the big corporations out there. Big businesses—hate it—we despise it, we ridicule it, we go after it. People have very little understanding of what it takes to run a business, or what's involved in

creating a business and then running it and being successful at it. They have no real idea what a corporation is.

- 8.117** How is it different than a partnership? How is it different than a lot of other business forms? They don't know how they function. They don't know why they're successful and how they achieve success. What do they run for? There's a lot of confusion about what is the purpose of business. And of course, "what is the relationship between business and the state and government?" is a huge topic of confusion, of misunderstanding, and of evolution, because that relationship constantly is altering and changing.
- 8.118** Now, we're going to talk about all these things in this course. We're going to talk about what a corporation is, how it functions, why it's structured the way it is, how they became so successful, what their relationship with government is, what is the role of regulation, antitrust, industrial planning in America. We're going to be focused on the American Corporation. There's some differences between the way corporations have evolved and become successful here, than let's say, in Japan or in Germany or other places around Europe. We followed very much our own path for better and for worse in the United States relative to these other countries.
- 8.119** This is an issue that has constant relevance. There's constantly things going on in our world around us that makes this question about corporation, the world of government, how they function, what they are super important to policy. I was just reading this morning a story: about a year and a half ago, Nippon Steel made a bid to buy U.S. Steel. U.S. Steel is a very old American business. It's one of the first big businesses in America. It was put together by Carnegie in the early 20th century—Carnegie and J. P. Morgan. It's like a classic big American business.
- 8.120** And the steel industry has a history—we'll touch on it a little bit during the course—history of up and down over the decades. And generally, American Steel has not been particularly competitive, particularly not U.S. Steel, since the 1970s, since it really started facing competition from Japan and Europe. And indeed, U.S. Steel in the U.S. is struggling. So Nippon Steel, a very successful Japanese company, offered to buy them—\$14.6 billion.
- 8.121** Shareholders of U.S. Steel: "Yes, this is fantastic. We love it. They're giving us a premium over what our stocks earned. We're going to make some good money here." Even the union at U.S. Steel loved it because it's an infusion of capital. Nippon is really good at running steel companies. This is re-energizing for U.S. Steel, this injection.
- 8.122** But it's a foreign company buying an American company, and that sets off certain alarm bells in certain quarters. It's steel, which is like— we have a long history of steel as being American. That sets off certain alarm bells in other quarters. And it's national security. We need a lot of steel to build tanks and airplanes. And yet the amount of steel that goes into the defense industry is very small, and U.S. Steel is not a factor

there. Other companies take care of it. But everybody—national security—they raise that banner very, very quickly. And the Union of Steelworkers— not the Union Chapter at U.S. Steel, but the Union of Steelworkers— they don't want a foreign company in the U.S. Now, their workers are working for a foreign company. So, they're against.

- 8.123** So all this is a situation in which the government steps in and says, "Wait a minute, we need to authorize this." Generally, the US government authorizes big mergers. We don't have the freedom to merge whatever companies we want. And Biden decided, no. Even though there's a commission that analyzes the national security risks involved in this, the commission came back with neither here nor there. They didn't want to commit, which was politically very good for them.
- 8.124** Biden said, "No, I'm against this deal." On the campaign trail, Donald Trump said, "I'm against this deal." Why? It's a foreign company—unions— always national security. The government is always going to use the national security terminology. Anyway, just last week, it was announced that Nippon Steel is indeed buying U.S. Steel. The deal is going to go through. Trump has approved it. But there's a caveat.
- 8.125** Nippon Steel has agreed to give the US government something called a golden share. That is, they've given them veto power over most corporate decisions related to U.S. Steel. For really the first time in American history, the government has a golden share in a private company owned by a foreign company where it can now decide who are going to be the directors— at least the outside directors—the so-called independent directors are going to be basically approved and appointed by the US government, by the President.
- 8.126** They can decide on every single major business decision that affects U.S. Steel. This is something completely new. How does this fit into the history of the corporation? Now we have a U.S.—not owned, because they're not going to get any revenue from U.S. Steel—but controlled by the President. For a while, there was some thought, "Okay, this will phase out after three, four years." The U.S. government has in the past tried to nationalize industries or taking big stakes in companies, but usually they get rid of it very quickly. In the U.S., generally, our system is such that we believe in private property and the government shouldn't own businesses.
- 8.127** No, no. This golden share is there forever in perpetuity. This is a deal that's being negotiated. It's being closed. It's being done. This changes again the nature of corporate governance, the nature of the corporation in America today. Now we have this overhang of, "Well, what does the government want next?" This relationship between the corporate America—between business—and government has shifted just this week. And we will see all the shifts or many of the shifts that have happened over the last 150 or so years during this course.
- 8.128** So in our culture today, corporations are often the villains. You could check out Netflix or Prime or whatever, and go check out the movies, and you won't find a corporate

hero, but you'll find lots and lots and lot of corporate villains. Every dystopian story, almost every dystopian story, that we have out there—who ultimately controls the world? Some corporation. It's always a corporation. You remember WALL-E? You remember WALL-E, the Disney movie, where you had this dystopian future? Who did that? A corporation. This is true, generally, in movies. The villain is always some corporate boss pursuing greedy prophets. And in pursuit of greedy prophets, grows, grows, grows, and ultimately takes over the world.

- 8.129** And that's the motivation. That's what they want to do. They're the bad guys. Our culture is ripe with this. Again, you find me a movie or a novel where a corporation represents goodness—where a corporate leader is a hero. It almost doesn't exist. It's very, very rare. There's some exceptions, but it's very unusual and very rare.
- 8.130** And indeed, today—it used to be, at least the perception used to be—that at least I'd say in the latter part of the 20th century, the perception was Democrats hated corporations because they are generally on the side of socialism. They're generally perceived to be on the side of that the state should control business and capitalism, free markets, which corporations represent to a large extent. That is inefficient, ineffective. So we want much more regulations, controls.
- 8.131** And it's always been perceived as the Democrats are the ones who hate big business. And indeed, if you look at statistics in the past, Democrats have always resented big banks, big corporations, big business. But 40% of Democrats say they don't trust them. It used to be the Republicans actually had a relatively favorable view of big business and big banks. But that has really changed. Today, Democrats and Republicans agree: big banks, big business—something's very wrong with them. They have generally negative views of both. We now have a consensus across the political spectrum; about 60% of both Democrats and Republicans agree that there's something very wrong and we need more control.
- 8.132** They have a deep distrust of American business, particularly big business. Everybody loves small business. Everybody loves the little guy. Little guys are great until they grow big. The whole point of having a small business is to do what? For the most part, if you're ambitious, it's to make it grow big. We like them when they're small. It's like kids, right? They're cute when they're small. It's when they grow up, they become a problem.
- 8.133** In corporations, we despise them when they grow up, when they become big, when they become successful. How do you become big? Mostly by success. We don't trust them. We resent them. It's why antitrust is so popular. We want to break them up. We want to make them small again. Generally today, there's a bipartisan suspicion of big business across the board. It really goes deep.
- 8.134** There's a lot of people who question things like: aren't corporations just a creation of the state? Did government make corporations? And therefore, shouldn't we have a lot

more control over them if they are creations of the state? Don't they have to register—get a corporate charter—and therefore, they're a government-created entity? And if such, the government has a role in regulating it.

- 8.135** And you see that widely on the left. And again, you see that—certainly Lina Khan, who was the head of the FTC under Biden, who is important because she was not just a regulator, but she was an intellectual. She came from a law school. She wrote a lot of papers. She explained her positions. She really viewed corporations as creatures of the state. The state should have a lot of involvement and should dictate how they should be run and what size they should be. You saw that in the old days. Those of you might remember Ralph Nader who used to go after corporations constantly—we want them controlled, we want them broken up, want them regulated, we want them to go away.
- 8.136** Of course, over the last 10 years—and maybe this is some of the reason Conservatives have changed their views, or Republicans, at least, have changed their views vis-a-vis business—is you've seen, again, business accept this idea that they're creatures of, if you will, the state, society, and maybe they shouldn't just pursue profit. Maybe they shouldn't be so greedy. Maybe they should pursue something called social responsibility or the public interest or the common good. And as a consequence, they've adopted things like DEI or ESG. DEI being diversity, equity, inclusion, and ESG being environmentalism, social, and governance, which, again, are slogans typically associated with the left. Once business adopted them, as, "Yeah, we're socially responsible. These are the things that make us socially responsible." Now the right has turned against them.
- 8.137** But that's because they embrace this idea, as we'll see, that they are not just there, in a sense, to make money, and we'll get into what making money means, but that they're there to serve some social purpose. But the reality is that Conservatives have always had a tricky relationship with business. They've never been real fans. They've never gotten on board completely.
- 8.138** There was recently an article saying, this is the title of the article: "Big Business and Conservatives Are Headed for Divorce." So now the Conservatives, Republicans are completely against. From the article, "The new Conservatism Is hostile to globalization, immigration, big tech, media companies, and the idea that market forces should determine the common good." So they might defer on why they resent big business, but again, it's a universal trend now.
- 8.139** Here's Irving Kristol. I don't know if you remember—you guys are too young. Irving Kristol was a leading conservative in the 1990s. He was really the founder of neoconservatism, one branch of conservatism, not very popular right now, but in those days, very popular. This is what he wrote: "But one must concede that both the founding fathers and Adam Smith would have been perplexed by the kind of capitalism we have in 1978. They could not have interpreted the domination of economic activity

by large corporate bureaucracies as representing in any sense the working of a system of natural liberty."

- 8.140** So he is objecting, again, to the large corporate entity. He goes on, "The large publicly owned corporation of today, which strives for immortality, which is committed to no line of business, but rather, like an investment banker, seeks best return on investment, which is governed by an anonymous oligarchy, would have troubled and puzzled them, just as it troubles and puzzles me. And they would have asked themselves the same question we have been asking ourselves for almost a century now: Who owns this Leviathan? Who governs it? And by what right and according to what principle?"
- 8.141** Those are the kind of questions we're going to try to answer in this course. Who governs them? By what right? And what principle? He says, "No other institution in American history, not even slavery, has ever been so consistently unpopular, as has the large corporation with the American public." Not even slavery has been this unpopular. "It was controversial from the outset, and it has remained controversial to this day."
- 8.142** So he's right. Back in the 19th century, what did they call the successful businessmen of the 19th century? The big businesses. What were they called—the people who ran the big businesses of the 19th century? Robber barons. Why is that such a bad term? A robber? It means you don't create wealth, you steal it. And baron? A baron is an aristocrat, somebody who doesn't, again, create their status. Baron is very un-American. It's very European.
- 8.143** Now, there used to be robber barons. They were robbers and they were barons. Really, the whole aristocracy in Europe, how do you become an aristocrat? Five, six, seven, thousand years ago, how do you become an aristocrat? By stealing. There's no production. There's no actual production of values. There's no actual production of wealth. It's stolen wealth. In Europe, this association of wealth with stealing makes a lot of sense because they've never had really—up until about 150, 200 years ago—they never had a system in which wealth was created. Wealth was always redistributed or stolen. Robber barons reflects aristocracy of Europe.
- 8.144** And that is now attributed to the Carnegie's and the Mellon's and the Rockefellers and the J. P. Morgans. People who—I think as we'll see to some extent, but as if you know history—actually built, created, the wealth that they had. Very different than a baron, very different than the robber.
- 8.145** He goes on to say, "If large corporations are to avoid having government appointed directors on their boards"—golden share, government appointed directors on their boards; they're getting it now—"they will have to take the initiative and try to preempt that possibility by themselves appointing distinguished outside directors, directors from outside the business community. I love that. What you want running a company are directors who know nothing about business so that they can monitor and supervise

CEOs whose business is business. We want somebody completely ignorant of the business of business. That makes them good directors. We'll see, that is the dominant view today.

- 8.146** There's political hostility. Again, more so, I think, today than in the 1970s, although there's some similarities. Globalization has affected our views, our view on globalization, our views on immigration. There was just now Trump— what was it last week? Of course, this will be broadcast sometime in the future. So sometime in June of 2025—Trump said, "ICE is not going to go into farming and certain businesses, hotels and restaurants, in order to deport immigrants because those businesses really need those workers."
- 8.147** Huge backlash against that. "Wait a minute. Why are you discriminating? If they're illegal immigrants, they're illegal immigrants, why do we care if it's business or not?" Now, just on Monday, it was rescinded, and now they're going after all of them. There's a resentment towards immigration, and then there's a resentment towards business that employ immigrants because, "Hey, they're making a profit off of illegal immigrants. They should be paying employees a lot more money in order to hire Americans," is the view.
- 8.148** There's resentment against automation, technology. It always has been. It'll destroy jobs. Automation. What will we do once machines take all our jobs? We won't have anything to do. There's always been and always is, resentment against CEO pay. They make too much money. They make all these figures 200 and something times what the lowest employee in a company makes. And that's always a conflict that exists in America. It has its cycles depending on what makes the news. About 10, 12 years ago, it was the big thing CEO pay, how much money they make? Less so today.
- 8.149** A lot of people resent the idea—if you own stock in a company, how much say do you actually have in running the company? Are you really an owner if you have no control? It's this idea that the owners of the company don't really control the company. Who controls the company? The managers, the CEO, and yet there's this separation. So are we really doing stuff that's good for owners? CEO pay is a good example of that. Maybe the CEO is making a lot more money than he should. Shareholders shouldn't allow it. It's an interesting case.
- 8.150** If you remember, again, this is just not that long ago, Elon Musk, it's probably a year ago, Elon Musk got a pay package about five years ago from the board of Tesla, which was worth something like \$50 billion in stock. If he reached certain thresholds, he was going to get \$50 billion, like the biggest pay package, the biggest pay day anybody has ever had in the history of any place. And some minority shareholders sued and said, "Wait a minute. This board is a board appointed by Elon Musk. They don't represent our interests. We would never approve a \$50 billion payday. This is wrong."

- 8.151** It actually went to a Delaware Court—we'll get to Delaware later, why Delaware court —It went to a Delaware court where Tesla is incorporated. And the court said, "Yeah, that's nuts. That's unreasonable." And they voided the whole pay package. Later on, it went to shareholder vote, and shareholders actually voted for it. But there's still this issue of, majority shareholders versus minority shareholders? Is this okay? Again, the courts have ruled, "No, he can't get that pay package." So these issues around both CEO pay and around whether shareholders are properly represented is an issue that comes up when it comes to corporations.
- 8.152** And then there's the argument that corporations are inefficient, that they focus on profit at the expense of everybody else and everything else. This idea of small is beautiful. Small is beautiful, big is bad. And of course, tangled in all of that is envy, envy of people's success. Certainly, if you look at some of these very, very wealthy entrepreneurs in the tech industry, people hate their guts. They make their lives better. I think of somebody like Jeff Bezos. Changed the world. Who doesn't use Amazon? Everybody uses Amazon. Our life is dramatically better because of Amazon. How would we have survived COVID lockdowns, ridiculous lockdowns without Amazon? And we love to hate Jeff Bezos.
- 8.153** A lot of that is envy. It's also this moral notion that his self-interest, his pursuing his profits, his pursuing his own ideal, his pursuing a lot of money, his pursuing success, and these days, the good life—there's something wrong with that. He should be much more, a little bit of sacrifice, a little bit of sharing with the rest of humanity his wealth and success. And there's a lot of ignorance and unthinking that leads to resentment, these kinds of resentments.
- 8.154** All right, so we're going to try to cover a lot of these. We're going to cover a lot of these issues and bring them up and try to see how they should be addressed. They're really out there in the world, very little defense of business, very little defense of corporations. They keep chugging along because everybody knows we need them. Everybody knows our jobs depend on them. Everybody knows our standard of living depends on them. We love to hate them.
- 8.155** And then the government comes in once in a while and shakes them all up or breaks them up or just harasses them for a while and then goes away and we continue going because what's the alternative? None of us actually want to embrace the alternative. So we live with our resentment and our hate, and then we express it in our movies and our books. We cheer for the guy who's going after the corporate leader.
- 8.156** We need to remember, though, how successful they have been, how crucial they are to our lives. And in spite of the crises that have happened—I don't know, we'll talk about the Great Depression and World Wars and the inflation of the 1970s and the great financial crisis—and in spite of some legitimate horror stories of real fraud, of people doing really bad things—in spite of all that, somehow, here we are in pretty good shape. At least again, materially doing fantastic.

- 8.157** And even this idea of materially, I say materially, but even if you think about non-material spiritual stuff—let's take a simple example of something like music. Music is a spiritual experience. Our lives are better off for being able to enjoy music. Do you think you have more ability to enjoy music today than you did 20 years ago, 100 years ago, 400 years ago? I mean, it's not even close. If you have one of these, you basically have access to every piece of music ever produced at a marginal cost of zero.
- 8.158** You pay a monthly fee, listen to anything, anytime, anywhere. And that, again, is not an accident. That is something created by a business. That changed music forever. And it's a sequence of events. Ultimately, it's resulting in Apple creating iTunes, creating those, remember those iPods originally, and of course, today in an iPhone. And pursuing what, by the way? Why did it do all these things? In pursuit of innovation, success, vision, beauty, but ultimately in pursuit of money. Greedy Apple made it possible for us to elevate ourselves spiritually by having more access to better music all the time.
- 8.159** All right. So in this course, we're going to talk about what corporations are, how they function, why they function the way they do. We're going to talk a lot about the history—because this is about history—about the history of corporations. A lot about the interaction between corporations and the rest of the world, particularly government. So we'll talk about antitrust. We'll talk about regulations, some regulations. You could have a whole course on it easily. Antitrust, you could have a whole course on.
- 8.160** So we'll condense all these. We'll talk about some, about industrial planning, the attempts of government to try to control winners and losers and structures of businesses. We'll talk about how all this affected the evolution of businesses to where we are today. We'll talk about the Big Five. What are the Big Five? The Big Five tech companies that everybody's so obsessed about today. How did we get to the Big Five? How did we get here? We'll talk about some of the dynamics that brought us here. In doing so, we're going to cover a history that basically starts somewhere in the middle of the 19th century, but really in terms of the corporation gets going at the end of the 19th century, early 20th century. And the 20th century is just fascinating, I think, in terms of the ups and downs, the changes, the evolution that has happened to the corporation and to our economy more broadly, and the two are heavily related, as we will see.
- 8.161** So let's start. Let's start really by jumping into what these things are that we call corporations. What is a corporation? A corporation is a particular type of business organization. It's a particular type of business organization that is created in order to pursue a business goal. And behind that business goal is profit via economic activity. Now, not all corporations are business organizations. Not all corporations seek profits. You can have nonprofits. Nonprofits are also corporations. They're also structured, as we'll see, as corporations. Today, there's even something called a B Corp. A B Corp is a corporation that lets you know in advance before you invest money in it, that it is not

focused on making money as the predominant—it'll also make money, but it has a social goal, whatever that is. It's telling you up front.

- 8.162** Now, nonprofits is a big topic. We're not going to talk about it. It's just not relevant to what we're talking about in this course. We're going to be talking about business organizations. The B Corp, it's a small—it's not very popular, it's not very big. I suspect for a variety of reasons, I think you'll glean through the course, that it's never going to be very big. To a large extent, what drives business, what drives corporation is profit. It's a single-minded, if you will, pursuit of economic success, of business success. Once you pollute that with other goals, it's going to be a mess, and it's unlikely to be successful.
- 8.163** I'll give you one example. It's not exactly a B Corporation, but it's similar. Everybody know OpenAI? It's in the news, it's everywhere. This is one of the biggest successes. Recently, ChatGPT, this is huge. It's going to change the world. Well, the interesting thing about OpenAI, and there's a lawsuit going on right now between the founders of OpenAI, Elon Musk and the other guy's name just slipped my mind. You want to say it? Sam Altman. Thank you. Sam Altman and Elon Musk. And when it was set up, it was set up as a nonprofit. OpenAI was a nonprofit with the idea that whatever it generated would benefit humanity. And then nobody would disproportionately benefit from this—that it was shared by everybody. And that's how it was set up.
- 8.164** But interestingly enough, it actually got over time—and a lot of money came in. Elon Musk, I think a country put in \$10 million or something, a lot of money went in and it grew. And what was about two years ago, three years ago, they had this product launch, which was a real breakthrough, ChatGPT, which was a step forward in a dramatic way. Microsoft had invested money in a a weird situation. Microsoft was investing in a nonprofit under the idea, I guess, that they would get part of the technology. But what OpenAI has now done is it spun off the ChatGPT portion, which is really the whole company, into a for-profit entity. Now, originally, they said this for-profit entity would just fund the nonprofit, and the nonprofit entity would distribute the money for good causes, and it would be responsible for the governance of the for-profit in making sure that it served the general public interest, the common good, however you want to define it.
- 8.165** And a lot of people invested in the for-profit spinoff of venture capitalists, again, Microsoft. It was always a little suspect because these people are investing in this subsidiary as if it's really making money. But the not-for-profit expects it not to make money because it has this other goal. And there's a huge rift, governance-wise, contradiction between what the nonprofit wants and what the subsidiary wants.
- 8.166** Basically, it was six months ago or so, they decided to spin off the subsidiary as its own for-profit company and completely sever the relationship with the nonprofit. And the nonprofit will get the money. They will, in a sense, sell it, and they will get the money, and they will do their philanthropy with that. But from now on, ChatGPT will be a for-

profit. And this is what the lawsuit—Elon Musk is saying, "Wait a minute, that wasn't the original intent." Now, he, of course, has an incentive because he's created his own AI company, XAI, which competes with ChatGPT. So he would rather they stay a nonprofit, which he knows is not going to be as good a competitor as a for-profit OpenAI.

- 8.167** Anyway, you can see the tangled mess. But the reality is that at the end of the day, in order to be successful, in order for ChatGPT to actually win, if you will, it's got to be a for-profit. And the nonprofit knows it, and therefore they're spinning it up. So we're not going to talk too much about nonprofits. We're going to focus in on the full profit, on the typical business corporation that we love to hate and that we're familiar with in all of our daily lives.
- 8.168** So what is it that makes a corporation a corporation versus a more typical business setup? If you go back in history, the most typical business setup is, in a sense, a self-proprietaryship. Somebody just opening a business— shoe horses: you make the shoes from the metal, you do it all. It's your one-man operation. Maybe you have a couple of employees, but basically it's your business. All the income flows through you. You get to make all the decisions. You control it and you own it, and that's it.
- 8.169** Now, you can't grow a business like that very big. There's only so much one person can handle. There's no real access to capital. There are no outside investors. It's just you. A step in growing is to create, again, very common, a partnership. Bring in other people who can help manage, who can also bring in capital, and now you can have a larger business.
- 8.170** And indeed, typical business organizations historically were partnerships. Anything of size was a partnership. These were distinct legal entities, but where the owners, the partners, are all liable for everything that happens in the business, at least in a traditional partnership, that is the case. They're liable for it. If you sue the business, you're really suing the partners, you're not suing the business. If the partners sue, they're suing as individuals. There really isn't a business separate from the partners. It's one unit.
- 8.171** A corporation is something different. A corporation is a creation of distinct business entity. A business entity that, in a sense, has a life of its own. It can sue in the name of the entity. It is sued by other people directly. And it can bring in investors, large numbers of investors, from all kinds of places. Those investors are not now partners, so they're not expected to participate in the management and the running of the company. They're also not liable for the things that the company does. They are just investors in this legal entity. And as a consequence, it can grow much bigger, raise a lot more capital, and indeed can do somewhat riskier things because the investors can only lose what they invested in the company.

- 8.172** You can't go—this is called limited liability. Let's say a company takes on a huge amount of debt and then goes bankrupt. All the debt holders, all the people who are owed money can get is what is held under the corporate name, the assets that are held there by the corporation. If you own shares today in Apple, and Apple goes bankrupt, and it owes a huge amount of money, and they sell all the assets of Apple, and they still owe a huge amount of money, they can't come after you as a shareholder of Apple to take your money.
- 8.173** Now, if you're a partner, and the partnership gets sued, and they liquidate the partnership, and they pay it all out. If they still owe money, they'll come after you as individuals, at least in, again, a regular partnership. There are ways in which partnerships can shield themselves from that as well. So this idea of a limited liability is a huge benefit to the corporation. Huge benefit for a business. All you can lose is your initial investment. And that makes investing much more attractive.
- 8.174** It also makes it so, I don't have to do quite as much due diligence when I make an investment. If I'm making an investment and my entire net worth is at risk, I'm going to know you really, really well. I'm going to make sure this business is a good business, and I'm going to make sure you don't take too many risks because I don't want you to take risks because my entire net worth is—and that is very constraining and very limiting. Think of all the real breakthroughs that have happened in business. They always involve risk.
- 8.175** How many companies try to establish their personal computer and how many of them failed, and how many succeeded? How many railroads succeeded versus how many failed? How many in any industry succeeded versus fail? And yet the reason they can succeed is because people are willing to invest in them because they know their entire wealth is not affected by it, just their investment. So limited liability is a huge benefit that corporations have. Now, you can create that in partnerships, but on a smaller scale with limited partners versus general partners. There are kind of legal ways to do this.
- 8.176** Corporations also are, theoretically at least, immortal. There's no lifespan. The lifespan is as long as they're successful, as long as nobody's bought them, as long as they haven't gone bankrupt. Because the owners are constantly changing. And therefore, when the owners die, the business doesn't close. The shares pass on or they've sold the share, somebody else owns it. So you can have a business like U.S. Steel that we talked about earlier, which was founded 150 years ago. Less than that, 120 years ago. And here it is, still around. And if not for Donald Trump, maybe it wouldn't be around anymore. It would be called Nippon Steel. But here it is. It might be owned by a Japanese company. Tomorrow, it might be owned by somebody else.
- 8.177** So you can get this idea of companies that can plan and can pursue strategy that is not limited by anybody's lifetime. It's actually now indefinite. The founder of Apple, Steve Jobs, died a few years ago. He's no longer around, and yet Apple continues. And if they do well, they'll continue for a long, long time. And if they don't do well, they won't.

- 8.178** This idea of an indefinite lifespan is, again, an attribute of corporation. Again, you can do that in partnerships. You can create partnerships that have that effect. But a lot of partnerships, it's a partnership. The partners die. It's done. It's over. Now, there are partnerships that survived long periods of time. Some of the investment banks, like Goldman Sachs, used to be a partnership. It lasted for a long time, that partnership. So you can structure legally a partnership into a different form. But even Goldman Sachs ultimately became a corporation because there are multiple advantages to being a corporation. It's much cheaper, easier, smoother to structure yourself as a corporation and get this idea of an indefinite lifespan and limited liability and all these other features that you could create elsewhere but are not necessary for that creation.
- 8.179** Part of the consequence of this is that one of the attributes of a corporation is that you have a separation of ownership from control. In most businesses, the owner runs the business. The owner controls the business. And if you think about most corporations, at the beginning, when they're small, that is probably the case. The entrepreneur, the founder owns a big chunk of the business. He maybe sold a little bit off to investors, but he owns most of it and he's running it.
- 8.180** But as the business grows, and when ultimately they go public, owners typically hold a fraction of the shares in the company. Jeff Bezos doesn't own a big percentage of Amazon. He's the second the richest man in the world, even though he only owes a small percentage of Amazon because Amazon is so big. And at that point, he is the CEO — when he was CEO—he's running it even though he's not a big owner. So you get the separation of ownership from control. You get a division of labor. You get specialization.
- 8.181** I can now go out and hire a really good manager to run my corporation without me as an owner or as one of the shareholders being involved on a day-to-day basis. So you get a separation. Adam Smith talks about the huge benefits of specialization and division of labor. Here you get another way in which labor is divided. Ownership is divided from management, and that increases efficiencies for managers. You can now become an expert at being a manager. Go get an MBA. I don't know if I'd recommend that, but theoretically, at least, go get an MBA. That gives you the skill set, supposedly, of becoming a good manager. Or at least a diploma that says you have the skill set for being a good manager. I used to teach an MBA program, so I can criticize it.
- 8.182** Corporations have these huge advantages legally and structurally. They make things more efficient and more productive. They are better utilizers of this division of labor. They provide for this massive amount of ownership. If you think about us as owners of shares, I mean, all of us probably own shares in our retirement account, some of that account, brokerage account. We can now diversify our own holdings. But how many of us—you probably get proxy. You get a vote on stuff once a year. How many of you actually read those? I know what I do to them. In the trash as soon as I get them. We're

not thinking about how to run the company. We don't manage this. If we don't like how the companies run, what do we do? We sell our shares.

8.183 Now, we'll get to the most sophisticated effects of that and what some people can do in terms of when they're not happy with running a company. But for most of us, this is a way in which all of us participate in the enormous success of business without being a part of that business, by just providing capital, without actually getting involved in how to run it. Massive benefits to investors. And again, the creation of this class, if you will, are people who now specialize in management. That's what they do. That's their shtick, is how to run a great business.

8.184 For the corporation side as well, they get the benefit. And if you take in my finance course on this platform, we talk about this. The managers get the benefit of now having the corporation listed on an exchange valued constantly by investors like us. That gives them incredible price information about how well they're doing. Stock markets are signals. Stock prices are signals to managers. You're doing well, you're doing poorly, you're just doing okay. You can compare their stock to their competitors, see who's doing better. And managers are constantly looking at this stuff, not just out of a sense of vanity, but out of a sense of, they're important signals the market is sending me through the prices of my industry and my own company.

8.185 I like to say that sometime in the early part of the 20th century, the best company that made the best buggies in the world, probably started to see their stock price decline, and the managers went, "What's going on? We're super profitable. We're making really good buggies. Everybody's buying our buggies. What's going on?" And they popped their head up out of their business and looked around, and what did they see? An automobile industry over there, which the market had seen before them. The market had recognized, lowered their valuation because they had no future—they were all going to go out of business, and shifted the capital to the automobile industry. That's a signal.

8.186 Think about in our world of tech, the changing of the guard in terms of tech companies, and what's hot now, and what's hot yesterday, and how stock prices change, and how that is a signal for: what is the value? What is the prospects for the future?

8.187 These are the qualities or the way organizations or corporations are structured. There are lots of complications on top of this. There are lots of iterations that we can do. That's the basic idea. Separation of ownership and control. You get a new class of managers. You get this idea of indefinite time span, and you get limited liability, which is crucially important.

8.188 Now, a lot of people, a lot of critics of this form, worry a lot about separation of ownership control. We'll get to that. But before even that, the only reason this can exist, the only reason we have corporations, is because of the state, because government has endowed them with the ability to have this indefinite nature, endowed

them with limited liability. You can't have limited liability unless the state recognizes that you have limited liability. This is all just a creation of the state.

- 8.189** Ralph Nader, famous consumer advocate, supposedly, "A corporate charter"—which is what the state gives any corporation. If you have a corporation, you have to go and register and you get a charter from the state. "A corporate charter is in effect an agreement whereby a government gives the corporate entity existence, and that entity in return, agrees to serve the public interest."
- 8.190** If you accept that definition, if you accept that that's true, then it's the state that tells corporations what they should pursue. It's government that tells them what they should pursue. Maybe the government decides, "Pursue profits," which is in the public interest, but then they might change their mind and say, "No, don't pursue profits. Pursue some other cause, ESG, DEI, climate change, national defense."
- 8.191** It's the state that dictates what that should be. But is it? Is it the case that the corporation is a product, a creature, as they call it, of the state? And the theory behind this is called the concession theory. If you look back at history, you can see its roots, really, all over the place. I mean, it used to be before about 200 years ago, that if you wanted to start a business, any business, you had to get a concession. From whom? From the ruler, from the king. In reality, the king owned everything. Even in England, which is the most liberal of all kingdoms 200, 300, 400 years ago, the King owned all property. He owned all rights to trade, and ultimately, he owned all the profits.
- 8.192** So if you wanted to do something new, like in England, in the case, if you wanted to go and basically get a boat and sail to India and start a new spice trade with India, well, that is owned by the king, that trade. So if you want to do that, you have to get permission. You have to get a concession. You have to get agreement in order to do that. The king grants you the ability to do it. So the king grants concessions. He grants the right to create an entity that is engaged in that activity with perpetual existence. And this is the original corporate forms or some of the original corporate forms. And what you got originally were towns that were incorporated in this way. You had churches, guilds, universities, all created corporations.
- 8.193** And then you had trading organizations like the East India Company that got a concession from the king to engage and trade with India, with the East. And these trading companies, for example, the East India Company or the Virginia Company that established ultimately Virginia—Virginia was a commercial venture. Virginia was not a political venture. It was a commercial venture. The first settlers who came to Virginia came there as part of the Virginia Company in order to create economic activity to benefit this company, all granted to them by a concession from the king. And these are the first stock companies. People actually issued stock, just like today. They had limited liability, and they could raise capital from a lot of different people.

- 8.194** But it was at the approval of a king, and almost always, really always, it involved a monopoly. A monopoly not in the sense of today's monopoly—we'll get to what is meant by monopoly today, or is not meant, or the confusion about what monopoly means today—but what monopoly meant was the king will not allow anybody else to do what you're doing. If somebody else wants to trade with the East Indies, they go to jail. There's no competition, not allowed competition. We're going to use the state to prevent competition.
- 8.195** This was created, the first people to really embrace this with the Dutch East India Company, and then the British, and there were shares, and they were traded. The beginnings of the stock markets were a lot of these companies that were granted monopoly power over a particular segment, a particular industry, a particular aspect.
- 8.196** One of the first corporations in the U.S. was a corporation in Massachusetts, granted the right to build a bridge over the Charles River and to take a toll. And it got a monopoly. It was the only company allowed to build a bridge over the Charles River until people got upset by the fact that there was and they wanted more. And it actually ended up going to American Supreme Court, where they said, "No, you can't do that. You have to grant more—if other people want to build bridges, it's okay."
- 8.197** But initially, corporate charters were associated with monopoly. But if you go back in history, as early as the 13th century, there were already stocks, shares, different business entities that had a lot of the corporate characteristics. So in America, we didn't really have much economic policy driven from the federal government. So in the early part of the 19th century, the federal government was not very involved in the economy, but the state governments were very involved in the economy.
- 8.198** So one of the myths about the 19th century in America is we had laissez-faire: anything goes, we had real freedom, free markets. No, not at all. There were quite a bit of regulations and controls at the state level, not at the federal level. There was always a tension between the commerce clause that says you can't regulate trade between states and between the reality on the ground where there was a lot of regulation of trade between states at the state level.
- 8.199** And that tension continued most of the 19th century. And indeed, states would grant monopolies to build canals, to create businesses, all again in the corporate form and their ability to sell stock and have this indefinite life. They all demanded certain benefits to the state. The state would take basically money. A lot of the state revenues came from this granting of corporate monopolies, corporate rights to these different companies. This is how early on corporations were structured.
- 8.200** A change to that only came in the 19th century, in the middle of the 19th century, where businessmen didn't pay attention to states, managed to create business entities that were corporate-like without getting a grant from the state and without getting any special favors, but also coming in under the radar.

- 8.201** The first new corporate laws that were starting to approach modern corporate laws were in Connecticut in 1837. They created standardization and simple procedures in order to start a corporation. Britain passed similar laws in 1844 and 1862. But really, the big change happened in the late 19th century in New Jersey. New Jersey was the first state to basically grant corporate charters with almost no strings attached. You paid something to set up your corporation, and that was it. They left you alone.
- 8.202** One of the big innovations for New Jersey was the ability to have your company incorporated in New Jersey, but to own businesses all across the country. And then states started competing on who could be better for business because they wanted to attract businesses to come and incorporate there. And then you got a little bit of that competition going, and state chartering became a lot easier, simpler, and allowed for a lot more activities and monopoly aspect of it. And the "do what the state tells you to do," disappeared.
- 8.203** In Delaware, funnily enough, in 1913, before he became President, Woodrow Wilson was the governor of New Jersey, and he hated big business. He hated corporations, so they passed a law to undo a lot of the incorporation stuff that New Jersey had done before and to penalize businesses who actually incorporated New Jersey. Guess what happened to the businesses? They all left, and they all incorporated in a small state. To this day, it dominates incorporation in the United States, 65% of all Fortune 500 companies are incorporated in one state. That is Delaware.
- 8.204** And it makes sense for a small state because they get the fees on the incorporation. Given that they're a small state, that constitutes a huge amount of their revenue, and they become very friendly towards businesses, very friendly towards corporations. And what Delaware has done over the last 100 and some odd years is specialize in corporate law. Delaware It has a body of law with regard to corporations that is more developed than any other state. They have judges that are specialists in corporate law. And as a consequence, and they're generally friendly towards managers.
- 8.205** So managers like to incorporate in Delaware. But shareholders don't penalize managers who go to Delaware. So it must be pretty friendly to shareholders as well. So generally, Delaware is considered the best place to incorporate, at least until this case with Elon Musk. And now everybody is saying, "Oh, we should incorporate in Texas." So we'll see if that actually happens. Saying you're going to move out of Delaware and move to Texas is one thing. Actually doing it is something else. We'll see how many of those Fortune 500 actually move to Texas. But as of last year, 65% of Fortune 500 or S&P 500 were there. It's predictable. Everybody knows exactly where it is and so on.
- 8.206** So given this history, we have to ask ourselves: are indeed corporations a creation of the state? Is it the state that gave them, made it possible for them to have this limited liability? It seems that way. Kings and then states and now Delaware. It seems that the state has provided them with all these benefits of incorporation. And if so, if they're a creation of the state, what duties in a sense, do they owe the state? Because that's the

claim, they're creatures of the state, they must be socially responsible. They must take into account, quote, "the public interest." And with that question, we will start next class.